

Six Birds Foundations II: Admissibility, Meta-Theory, and the Exact-Six Program

Ioannis Tsiokos 

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Abstract

Six Birds Foundations I [28] described how a layer of description closes over what it tracks, and identified six roles that such closure uses: descent (whether an update respects a quotient), representability of a macro-level specification, enriched route and coherence mismatch, staged order and refinement, packaging into quotients, and audit. This paper gives the six roles precise meanings and asks whether exactly six are needed. For each role it fixes what counts as evidence for it, the level at which it acts, and where it meets the other roles, and it states the bookkeeping rules a description must follow before any role claim counts.

The question is posed on a precisely delimited domain, **FATCD**, of finite audited typed closure descriptions: finite collections of typed records in which every claim carries an honest audit. Without further hypotheses the paper proves three things. Each of the six roles occurs in some description of the domain. Every description has a decomposition record with one channel per role, each channel in one of five statuses (active, collapsed, below threshold, outside scope, absent), and every feature is accounted for by an active channel, an inactive channel-status entry, or one of eleven residual labels; the last two, *unresolved candidate* and *recorded seventh-role screen*, apply only when none of the first nine places the feature. The six channels are part of the record's definition, so the count is of channels, not of active roles, and the record need not be unique. Probability, causality and agency, the most natural candidates for a seventh role, are placed without a seventh role in the forms the paper specifies.

The upper bound is conditional. Its recognition hypothesis is assumed, not derived, and asks, without reference to any decomposition record, that every feature be content of one of the six roles, an annotation, data that asserts nothing, a checked restatement of such content, or content whose check is not finite over the description (placed as outside-domain structure); the upper-bound theorem shows that each of these ways yields one of the first nine labels. Under the hypothesis a feature that no channel accounts for receives one of labels (iii)–(ix), and no content record, annotation or leftover record is licensed the last two. The hypothesis is a genuine restriction and depends on presentation. A standalone formula stating that finitely many recorded weights are nonnegative and sum to one fails it, while the same weights recorded as the single row of a transition kernel, with a certificate that the row is a probability distribution, satisfy it. On the other hand, any closed formula of the paper's finite check language that does not mention the six witness predicates can be recorded, provided the completed description lies in **FATCD**, with its truth value, as a one-variable specification over two records of the same kind, which the hypothesis accepts; a false formula is recorded through its checked negation, and the rest of the description must still satisfy the hypothesis. The paper does not claim six roles for all of emergence, a full algebra of the six roles, their independence in every setting, or an empirical realization. Appendix B describes the accompanying Lean development and what it covers.

1 Introduction

Six Birds Foundations I [28] studied how a layer of description closes over what it tracks [3]. It worked with finite theory packages $\mathcal{T} = (Z, f, \Sigma_f, E, \mathcal{A})$: a finite space Z of micro-descriptions, a lens f recording what the layer can see, the definable content Σ_f that the lens induces, a completion operator E whose fixed points are the layer's theories [10], and an audit functional $\mathcal{A}$. From two assumptions, that constructions compose and that the layer sees its states only through a limited interface, it derived six primitive closure roles $P1, \dots, P6$.

This paper asks whether the list of six is complete. Could a description need a seventh role, for probability, say, or for causation or agency? Asked of all possible descriptions, the question is too loose to answer, so the paper makes it precise. It gives each role an exact meaning, states the rules a description must follow before a role claim counts, and fixes a domain of finite descriptions on which the question can be settled. On that domain it proves that each role occurs, classifies the stated forms of probability, causality and agency, and gives every description a record with exactly six role channels. Under an explicit recognition hypothesis, every residual feature is placed by one of nine criteria (labels (i)–(ix), listed in Section 1.4): membership in one or two active role clusters, role content not yet absorbed, a checked re-presentation or level translation, an annotation, bookkeeping or assertion-free data, or content outside the covered scope. Residual features are of two kinds: every content record and structural annotation, even one an active role uses (a *kind residual*), and every record a decomposition record leaves unassigned (a *record residual*). Under the hypothesis, therefore, no residual feature of either kind is left as an unresolved seventh-role candidate. The exact-six statement is the last step of this chain, not its whole content.

1.1 The three parts

The paper has three parts, each supplying structure the other two lack.

- *Role architecture* (Section 2). It fixes what each role means; the three levels at which a role can be present (behavior, verification, structure); where each pair of roles meets without one collapsing into the other; and how data moves between levels.
- *Admissibility* (Section 3). It states the bookkeeping discipline: a role claim counts only when its hypotheses, changes of level, translations and visibility assumptions are explicitly recorded. Under this discipline the roles stay distinct as types (typed non-collapse), with no claim that they are independent generators.
- *The exact-six chain* (Sections 4–9). It defines the domain FATCD of finite audited typed closure descriptions and proves lower bounds (each role occurs), an upper-bound classification under the recognition hypothesis, with separate closures for the recognized forms of probability, causality and agency, and a decomposition theorem (six channels, each with a status), a checklist of misreadings that these results rule out, and the terminal theorem (Section 9).

1.2 The terminal theorem, informally

In plain terms: every finite audited description can be sorted into six role channels, each marked active or given one of four reasons for being inactive, together with a set of leftover records; under a stated hypothesis every leftover is of a listed kind (a re-presentation or level shift of placed content, role content not placed in an active channel, an artifact of what an instrument can see, an empirical-bridge annotation, structure outside the domain, or bookkeeping and data without an assertion of its own), so no leftover calls for a seventh role.

A finite audited typed closure description D is a finitely presented collection of typed raw records and annotations subject to finiteness, closure and audit conditions (every claim it carries is referred

to by an honest audit annotation, which need not certify it). The role vocabulary P1–P6 enters D not as a constraint on admissibility but as a family of derived projections $\pi_i(D)$, each available when threshold, witness-schema, collapse-case, level, nonclaim and audit data are attached to raw features. Theorems 7.3 and 7.4 state that every $D \in \text{FATCD}$ has a well-formed decomposition record $\text{Dec}(D)$ with exactly the six role channels P1, $\dots$, P6, each in exactly one of five statuses (active, or one of four reasons for inactivity), and that each $r \in \text{Residual}(D)$ receives one of eleven labels. Under an explicit recognition hypothesis (Definition 6.7), every such label lies in (iii)–(ix), and every kind residual and every record residual is licensed only labels among (i)–(ix), so none is left as an unresolved candidate for a seventh role. The claim is about role channels, not active projections, and the decomposition is existential, not unique.

Where the content lies. The parts of the terminal theorem have different sources. That a decomposition record has six channels is part of its definition (Definition 7.2), which is indexed by the six roles. That every description has a well-formed record in which each channel carries one status and every raw feature is accounted for is proved by construction, for every description and without the recognition hypothesis (Theorem 7.3 and Theorem 7.4(i)–(ii)). That each role occurs somewhere is proved by explicit witnesses (Theorem 5.1). That the recognized forms of probability, causality and agency need no seventh role is proved case by case (Propositions 6.3–6.5). For a description satisfying the recognition hypothesis, no kind residual and no record residual is licensed label (x) or (xi) (Theorem 6.9). Theorem 9.1 states these parts together, under that hypothesis. The recognition hypothesis lists five ways a feature can be recognized; each is a version, independent of the decomposition record, of label criteria among (i)–(ix), and the proof of Theorem 6.9 checks the correspondence. That check uses no property of the particular witness rows or seed clauses (Remark 6.10); the number six enters through Definition 7.2, and the particular predicates through Theorem 5.1 and Propositions 6.3–6.5.

What the hypothesis adds. Criterion (xi) presupposes (x). Hence a description meets the condition of Definition 4.18 exactly when some well-formed decomposition record licenses label (x) for none of its kind residuals and none of its record residuals. The recognition hypothesis is a sufficient condition for this that refers to no decomposition record: Theorem 6.9 proves sufficiency, and Remark 6.10 shows it is not necessary. Recognized features are therefore placed by (i)–(ix) whatever decomposition record is chosen.

How restrictive the hypothesis is. The exclusion of a seventh role rests on the hypothesis, which some descriptions in the domain fail (Remark 6.8). That remark also calibrates the hypothesis: the finite normalization description D_{norm} fails it; the same weights presented as the row of a transition kernel satisfy it; and a closed check-language formula φ without atoms W_j , in a description with two distinct declared records of one kind, can be re-presented as a covered P2 seed recording its truth value in the extended description, provided that description stays in FATCD ; a false formula is represented by its checked negation. In the completed description channel P2 is active and channel P5 is not *absent_with_record*. When wrapping preserves the truth value of a true closed assertion, the seed records the same checked content, so the hypothesis constrains only presentation, and the original record can be dropped when nothing else depends on it (the removal conditions of Remark 6.8); if it must be kept, it is classified on its own and may still fail the hypothesis. Label (xi) is realized in FATCD by the description D_{scr} of the same remark.

1.3 What the paper does and does not claim

The terminal theorem is scoped exact-six, not unrestricted exact-six. It makes none of the following stronger claims:

- unrestricted exact-six, or any claim about all emergence phenomena;
- that every $D \in \text{FATCD}$ activates all six roles or carries six active nontrivial witnesses;
- that the decomposition/exhaustion record $\text{Dec}(D)$ is unique or canonical;
- a full six-primitives algebra or global primitive independence;
- empirical realization;
- internal instrument-family completeness;
- any statement about broader domains without further work.

These are scope boundaries, collected in Section 11 and enforced locally by the theorem-chain results that would otherwise overread them.

1.4 The objects at a glance

The paper works with a small set of objects, defined formally in Sections 4 and 7; this preview fixes their roles so that the earlier sections can be read with them in mind. A *description* D is a finite audited typed closure description (Definition 4.7). Its raw content $\text{Raw}(D)$ is a finite collection of typed raw records, such as objects, maps, relations, paths, transitions, traces and checks (Definition 4.1), together with annotations of level, threshold, lift, forgetting, visibility and empirical bridge (Definition 4.2). Its audit part $\text{Audit}(D) \subseteq \text{Raw}(D)$ holds the bookkeeping record of every claim it carries (Definition 4.6). A *role projection* $\pi_i(D)$ reads a cluster of raw features as the role P_i , and is admissible only when threshold, witness-schema, collapse-case, level, nonclaim and audit data are attached to the cluster (Definitions 4.9 and 4.11). A *decomposition record* $\text{Dec}(D)$ gives each of the six role channels exactly one of five statuses (Definition 7.1): an *active_projection*, when an admissible projection exists; *collapsed_boundary_case*, when otherwise a candidate cluster offered as the witness fails its witness schema; *below_threshold*, when otherwise a covered seed is present (so no admissible projection and no failed offer exists; the name refers to incomplete witness or attachment data, not necessarily a numerical threshold); *inapplicable_outside_scope*, when otherwise role content is declared outside the covered scope; and *absent_with_record*, with the absence recorded, in the remaining case. The raw features that no channel accounts for form the set $\text{Residual}(D)$ of *record residuals*, each classified under the scheme of Definition 4.14; that definition calls every content record and structural annotation a *residual feature* (*kind residual*), including records used by an active cluster. The scheme has eleven labels: (i)–(ii) membership in, or sharing between, active clusters; (iii)–(iv) presentation variants and level shifts; (v) role-aligned or covered-seed content not absorbed by an active projection, content checked in a failed offer, or an activation-threshold refinement; (vi)–(vii) visibility and empirical-bridge annotations; (viii) content outside the covered scope; (ix) bookkeeping, annotation and assertion-free data; (x) an unresolved candidate, placed by none of (i)–(ix); (xi) a recorded seventh-role screen.

Working vocabulary. Several words carry a technical meaning throughout. The entries below anticipate definitions of Sections 4–7 and are meant to be consulted when a term is first met.

- *Honest bookkeeping*: every claim a description carries, and every change of level, translation, lift, forgetting map, visibility assumption and empirical bridge that the claim uses, has an explicit record carried in $\text{Audit}(D)$ (Definitions 4.2 and 3.1).

- *Threshold annotation*: a declared record attached to a cluster of raw features before the cluster is read as a role; the reading also needs a witness schema, the collapse cases, a level annotation, a nonclaim annotation and an audit record (Definition 4.11).
- *Witness schema*: a statement of what data in the cluster would support that reading.
- *Instrument*: the declared means under which content is visible or suppressed (Definition 3.9).
- *Covered scope*: the records not satisfying the outside-scope criterion of Definition 4.14. A record lies outside it when an assertion quantifies over a sort with no recorded finite carrier or uses a predicate whose extension D does not record, the assertion is true in some expansion of D and false in another, and its other assertions are of the same kind or serve a role witness (exact criterion: paragraph *Terms used in the label criteria* after Definition 4.14). Such a record receives label (viii) and is certified for no role. Whether a record lies outside depends on presentation: a finite assertion written as a formula over recorded field values lies inside the covered scope, while the same assertion written with a declared predicate name whose extension D does not record lies outside (Remark 6.8). Not every finite, recorded feature inside the covered scope meets the recognition criteria: the normalization predicate of Remark 6.8 does not.
- *Covered*: in *covered seed* the word names seed clauses (1)–(6) of Definition 7.1; in *covered scope* it refers to the outside-scope criterion of Definition 4.14, under which a record with an assertion that is not finitely checkable lies inside the covered scope when another of its assertion fields is finitely checkable and serves no witness; in label (ix), *already-covered* bookkeeping is an annotation whose non-exempt fields are data or references. The three uses are unrelated.
- *Closure*: several senses occur. The *closure content* of a feature (Definition 6.7) is the subdescription reached by its references; a *closure operation* in criterion (xi) is an extensive, monotone, idempotent map on a finite order; *reference-closed* means $\text{Closed}(D)$. A *closure role* is one of the six roles, in the sense of Foundations I; the *closure condition* $\text{Closed}(D)$ requires the references and bookkeeping specified in Definition 4.5; and the *closures* in Section 6 classify the stated recognized forms of probability, causality and agency without the recognition hypothesis.
- *Residual feature*: in Definition 4.14, every content record and every structural annotation (a *kind residual*: the raw grammar fixes no role reading for any of them, so members of active clusters are included); in a decomposition record (Definition 7.2), $\text{Residual}(D)$ contains the raw records and annotations that no channel accounts for (the *record residuals*); Section 7 compares the two sets. The upper-bound classification and the recognition hypothesis cover both classes; the decomposition theorems apply them to $\text{Residual}(D)$.
- *Admissible*: the meaning depends on the object qualified. An admissible claim satisfies Definition 3.1; admissible forgetting and lifts, visibility records and empirical-bridge claims satisfy Definitions 3.6, 3.9 and 3.11; an admissible description is a member of FATCD (Definition 4.7); an admissible projection is a derived role projection (Definition 4.11), whose cluster satisfies its witness schema W_i , with the attachment data of Definition 4.9; and an admissible role witness carries the common witness data of Definition 5.2. In Section 8, a misreading is called excluded when it is not licensed by the results of Sections 4–7.
- *Closed unconditionally*: said of probability, causality and agency, it means closed without the recognition hypothesis, in the recognized forms of Propositions 6.3–6.5. Other finite, audited, in-domain forms remain subject to the recognition hypothesis; content outside the covered scope is classified as an outside-domain structure.
- *Recognition hypothesis* (Definition 6.7): every residual feature is recognized in one of five ways: (R1) it is role content: it contributes a required field to an aligned cluster (that is, removing the

record from the cluster falsifies its witness predicate; the phrase concerns membership, not a particular field) or carries a covered seed, and each of its assertion fields is required for a witness at an aligned cluster containing it, an input to that witness's check, or a named field of its seed; or it is a member of an honestly failed offer whose assertion fields are inputs to the recorded check; (R2) it is a structural annotation carrying only the fields its declared operation requires and no independent assertion, or a bookkeeping annotation whose non-exempt fields are data or references; (R3) it is role-neutral data without an assertion of its own, or a predicate record supplying no witness field whose only assertion field is a formula with free variables whose truth value varies under assignments respecting the declared sorts of the free variables, using only predicates whose extensions D records and quantifying only over sorts whose carriers D records, referenced by no claim annotation and by certificates only through $s(t)$ (this branch constrains presentation, not content; Remark 6.8); (R4) it is a checked re-presentation or level translation of such content; (R5) an assertion of it uses an unrecorded predicate or uncarried sort and is true in some expansion of D and false in another, its other assertions being of the same kind or serving a witness (an outside-domain structure). The upper bound, that no seventh role is needed, is proved under this hypothesis. It is a genuine restriction: some descriptions in the domain do not satisfy it. Recognition constrains presentation, not content: Remark 6.8 shows how a closed check-language formula without atoms W_j can be recorded, with its truth value, as a covered P2 seed; recognition alone does not show that the feature concerns representability.

Symbols used before their definition. Sections 2 and 3 use the following symbols, which are defined later.

Symbol	Meaning	Defined in
FATCD	the domain of finite, closed, audited descriptions	Definition 4.7
Raw(D)	all declared records of D , content records (“raw records”) and annotations alike; its elements are the raw features	Definition 4.2
Ann $_{\Theta}$	a threshold annotation	Definition 4.2
Audit(D)	the honest audit annotations of D ; claim, promotion, translation and similar annotations they reference are <i>carried in</i> Audit(D)	Definition 4.2
$\pi_i(D)$, W_i	a derived role projection and its witness predicate	W_i : <i>Witness types</i> table, Section 2 (paragraph <i>Witness predicates</i>); π_i : Definition 4.11
Status $_i(D)$	the status of channel i , one of five values	Definition 7.1
Dec(D), Residual(D)	a decomposition record and its record residuals	Definition 7.2
kind residual	every content record and structural annotation, including members of active clusters	Definition 4.14
Classify	the scheme assigning a label to each residual feature	Definition 4.14
covered seed	the core record of a role witness, which may be present without the rest of that witness: an update map with its source relation; a one-variable macro-specification whose recorded evaluations take both truth values, with a realization relation targeting it; a route non-reducibility certificate; a law-bearing stage, order, transition or refinement record; a quotient relation or packaging map; an event, provenance or trace record, or a trace check	Definition 7.1
Audit $_0(D)$	the honest audits whose certificates and conclusion contain no atom W_j	Definition 4.6
declared use	the typed field of a map, partial-map or relation record, with value update, packaging, quotient, realization or other	Definition 4.1
assertion field	a field of role predicate, comparison, invariant, law or claim	Definition 4.1
honest audit	an audit annotation meeting the honesty condition	Definition 4.6
offer, failed offer	a claim naming P_i and C with an honest audit recording $W_i(C)$ (for a failed offer, recording $W_i(C) = \text{false}$)	Definitions 4.6 and 7.1
well-typed interpretation	an assignment of values to the datum fields of D keeping its records, kinds, declared uses, distribution-valued markers, field roles, references and formula- and term-valued fields fixed, unrecorded predicate extensions and unlisted-sort carriers varying arbitrarily; required certificates must match their prescribed conditions in all of them	Definition 4.6(c1)

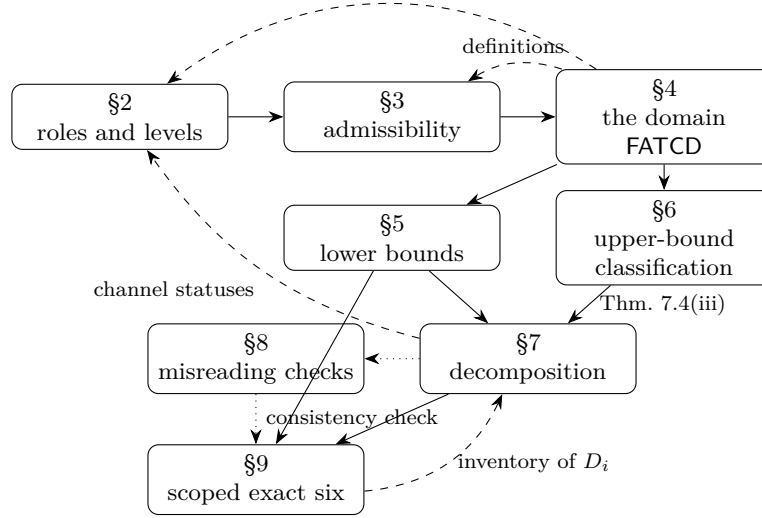


Figure 1. Dependencies between the sections that build the terminal theorem. Solid arrows show the order of the argument; dashed arrows mark forward uses: Sections 2 and 3 use definitions from Section 4, and Section 2 uses the channel statuses of Section 7. Propositions 7.5 and 7.6 also use the record-by-record inventory of $D_1, \dots, D_6$ in the proof of the last clause of Theorem 9.1, which uses only Theorem 5.1 and the definitions; this forward use creates no cycle. The solid arrow from Section 3 to Section 4 marks that Definition 4.11 uses Definitions 3.5 and 3.6, and label criteria (vi) and (vii) of Definition 4.14 use Definitions 3.9 and 3.11. Among the definitions carried by the dashed arrow back to Section 3, Definition 4.6 requires every claim of a description to be referred to by an honest audit, which Definition 3.1 further requires to certify the claim’s recorded statement or, for an offer, record the value of $W_i(C)$; the dotted arrow from Section 8 to Section 9 marks a consistency check that the proof of the terminal theorem does not use. The solid arrow from Section 5 to Section 7 marks Propositions 7.5 and 7.6, which reuse the descriptions D_i ; the existence argument in the proof of Theorem 7.3 does not use Section 5; only its closing non-vacuity remark cites Theorem 5.1.

1.5 Organization

Section 2 fixes the role meanings, the three levels, the boundaries between roles, and the selected-lift atlas. Section 3 states the admissibility rules. Section 4 defines **FATCD** and poses the exact-six question over it. Section 5 shows that each role occurs. Section 6 classifies the recognized forms of probability, causality and agency without the recognition hypothesis, and proves the general upper bound under it. Section 7 builds the six-channel decomposition record and shows that it exists but need not be unique. Section 8 checks the chain against misreadings, and Section 9 states and proves the terminal theorem. Section 10 discusses the result and places it beside earlier Six Birds work, and Sections 11 and 12 give its limits and open directions. Appendix A records development provenance and evidence hygiene, and Appendix B documents the Lean mechanization, with a note on the Lean counterpart of each result. Appendix C records the version history and is not needed for reading the paper. The Lean development checks the typed objects and rejects malformed cases; the proofs of the numbered results are the ones in the body. Figure 1 shows how the sections depend on one another. Sections 2 and 3 use the domain **FATCD** and the channel statuses before they are formally defined. The record, witness and audit definitions refer to one another. A reader checking proofs can use the following route, consulting forward references as needed:

- (1) Section 1.4;
- (2) Definitions 4.1 and 4.2;

- (3) the paragraphs of Section 2 from *Witness types* through *Route certificates*, with the witness table and the full Definition 4.6 at hand, using its clause (f) to order the evaluation of honesty and witness predicates;
- (4) Definitions 4.4–4.7, then Definitions 3.1, 3.5 and 3.6;
- (5) the rest of Section 4 and Definition 7.1;
- (6) the rest of Section 2;
- (7) Definitions 3.9 and 3.11 and Proposition 3.12;
- (8) Sections 5–7 and 9.

Propositions 7.5 and 7.6 also use the record inventory in the proof of Theorem 9.1 (p. 88). The other definitions and results of Section 3 are not cited in Sections 4–9, and Section 8 is a separate check against misreadings; the proof of Theorem 9.1 does not use it.

Example descriptions. The running example D_{ex} (Sections 2, 4 and 7); the lower-bound descriptions $D_1, \dots, D_6$ (Propositions 5.3–5.8); D_0 and the normalization description D_{norm} (Remark 4.8, analysed in Remark 6.8, paragraphs *The counterexample* and *Verification*), the latter failing the recognition hypothesis; D_{scr} , which realizes label (xi) (Remark 6.8, paragraph *A description passing the screen*); and D_c , D_o , D_T and D_{all} , which realize the five channel statuses (proof of Proposition 7.5).

2 Primitive role architecture

This section fixes the meanings of the six typed closure roles $P_1, \dots, P_6$, the behavioral, verification, and structural-categorical level pattern they share, the local boundary distinctions between them, and the selected-lift atlas vocabulary. These meanings are fixed for the remainder of the paper and govern every subsequent definition, lower bound, and upper-bound classification. Sections 2 and 3 use only the following from later sections: a description is a finite set of typed records and annotations (Definitions 4.1 and 4.2); a claim counts when an honest audit certifies it (Definition 4.6); an annotation is attached to a cluster when its target field lists exactly the cluster’s members (Definition 4.9); and a channel’s status is the first of five cases that applies (Definition 7.1). The paragraph *Symbols used before their definition* (Section 1.4) lists the rest. A reader checking proofs may read the paragraphs *Witness types* to *Route certificates* below, then Section 4 and Definition 7.1, before the rest of this section. The proofs of Sections 4–7 and 9 use from this section only the role names of Definition 2.1, the *Witness types* table with the check-language, certificate and law conventions before it and the paragraph *Route certificates* and the prescribed verification conditions after it, Remark 2.2, the level values of Definition 2.4 and Table 1; Proposition 2.6 is used by Proposition 3.3, and the other remarks and Definition 2.8 fix vocabulary.

2.1 Role meanings

Definition 2.1 (Primitive closure roles). The six primitive closure roles are

- P_1 — update-vs-quotient compatibility / descent obstruction [14, 16];
- P_2 — macro-specification representability [21]; when a satisfying realizer exists, the witness also requires a comparison with the classes of the kernel of a declared packaging map, whose declared domain contains the declared candidate class, certifying that some class of that kernel meets both the realizer set and its complement in the candidate class;

- P3 — enriched route/coherence mismatch [4, 25]; here “enriched” and “coherence” name only the recorded rewrite rules (and, in certificate (b), the move weights), not enriched-category or coherence-theorem structure; P3 is not a directionality certificate;
- P4 — stage/order/transition/refinement structure [5, 24];
- P5 — quotient packaging / quotienting [17];
- P6 — audit/bookkeeping/provenance/replay/checkability [13, 23].

The operative content of role P_i is the witness predicate W_i fixed by the *Witness types* table below (the table in the paragraph *Witness predicates*, p. 16, read with the paragraphs that paragraph lists); its record kinds and declared uses are those of Definition 4.1, and $\text{Audit}(D)$ and $\text{Audit}_0(D)$ are those of Definition 4.6. The phrases above name the roles.

Reading the roles. In plain terms, and with the lower-bound witnesses of Section 5 as instances:

- P1 asks whether an update respects a grouping of states. For an update F and a grouping q , F *descends* when $q(x) = q(y)$ implies $q(Fx) = q(Fy)$; two grouped states of the declared carrier obstruct descent when their defined images lie in different target classes, or when exactly one lies in the domain of a partial F . A W_1 witness of descent (alternative (a)) further requires the domain of F to be saturated under q (every q -class meeting it lies inside it; elements in no listed class are allowed) and some class to contain two of its elements. Trivial descent is not excluded: if some class of q has two elements, source and target are both the relation of q , that relation lists every class of q , and the update’s declared domain is the domain of q , then the identity update satisfies (a), as does every constant update.
- P2 asks whether a macro-level specification, a predicate s , has a realizer through a declared realization relation. In short, W_2 holds in two cases: no candidate in a declared class K , the full declared domain of a packaging or update map, satisfies s , although s is true at a declared record of the same kind outside K ; or some candidates satisfy s and some class of a declared packaging kernel whose domain contains the candidate class contains both a satisfying and a non-satisfying candidate. The specification s is, relative to a declared finite nonempty candidate class, *realizable* when some candidate satisfies it and *unrealizable* when none does, and *representable over E* when its realizer set is a union of the sets $[x]_E \cap K$ for the candidate class K , that is, when no class of E meets both the realizer set and its complement in K . The empty realizer set of an unrealizable s is such a union; unrealizability is counted as a separate witness, not as a failure of representability over E . The role is named for the question it settles; its witnesses are the two answers the packaging does not already supply: no candidate realizes s although s is satisfiable outside the candidate class, or some packaging class splits the realizers. A W_2 assertion of unrealizability requires the recorded finite absence certificate, and a satisfying realizer yields a W_2 witness only when the comparison certifies that the realizer set is not a union of E -classes. If it is such a union, as for the pullback of a predicate on packaged states, the comparison fails and there is no W_2 witness; the pullback alone does not establish a W_5 witness. “Macro” refers to the declared status of s , not to definability over the quotient: in the satisfying branch the witness certifies that realizability of s is not already decided by the packaging, so the P2 content is not P5 content. For example, with the states 0, 1, 2, 3 and the packaging q of the running example, the specification $s(r) \equiv (r = 0 \vee r = 2)$ has realizer set $\{0, 2\}$; the q -class $\{0, 1\}$ meets both $\{0, 2\}$ and its complement, so realizability of s is not decided by q (Proposition 5.4).
- P3 compares two routes with the same start and end. A route grammar has generating moves and rewrite rules, each rule relating two words with the same source and target; two routes

from u to v mismatch when they are distinct in the category presented by the moves and rules (a monoid when all moves are endomorphisms of one object), that is, when no chain of rule applications, in either direction and in any context, connects them, and a certificate of this, of one of the two types in the W_3 row, is a P3 witness. For example, let $a, b : u \rightarrow u$ be generating moves; if a certificate shows that no chain of rule applications, in either direction and in any context, connects the routes ab and ba , they give a P3 mismatch. With the single rule $aa \rightarrow a$, the monoid $\{e, \bar{a}, \bar{b}\}$, with $\varphi(a) = \bar{a}$ and $\varphi(b) = \bar{b}$, in which e is the identity and $xy = x$ for $x \neq e$, records the first move: it identifies aa with a and separates ab from ba .

- P4 is ordered structure: stages $i_0 < i_1 < i_2$, transitions between them, and refinements with stated laws. The W_4 row requires no order: one transition record with a certified law suffices, including the identity transition $T(z) = z$ on a one-point set (D_T , proof of Proposition 7.5).
- P5 packages distinct states into one object, for example $x \neq y$ with $\text{pack}(x) = \text{pack}(y)$, together with evidence that the resulting quotient is valid.
- P6 is checkable record-keeping: events, a trace, their provenance, and a replay or check of the trace.

A running example. Throughout the paper we follow one small description. Its states are $0, 1, 2, 3$, its update F sends $0 \mapsto 2$, $1 \mapsto 0$, $2 \mapsto 3$ and $3 \mapsto 1$, and its lens q groups the states into the classes $A = \{0, 1\}$ and $B = \{2, 3\}$; the grouping is recorded as a relation record E of declared use quotient whose classes are A and B , so $E = \ker q$. It also records a trace $0 \rightarrow 2 \rightarrow 3$ of the update, together with a replay and a check of that trace. The lens packages the states into a quotient object $\{A, B\}$ with packaging map q ; this is P5 content. Descent of F through that quotient fails: $q(0) = q(1) = A$, while $q(F0) = q(2) = B$ and $q(F1) = q(0) = A$. This split pair is P1 content, and the example shows the permitted interaction of the row P1/P5 of Table 1: P1 uses the P5 packaging when it checks descent, and neither is the other. The recorded trace, with its replay and check, is P6 content. The description also records F as a transition record with a certified law and an index record naming A and B . Its full record list is the table in the paragraph *The running example as a description* (Section 4), and its decomposition record is the paragraph *The running example, decomposed* (Section 7); both use these two records.

Witness predicates and the check language

Witness types. Here a description is a finite set of typed records (Definition 4.1); map and relation records declare a use (update, packaging, quotient, realization or other); $\text{Audit}(D)$ is the set of honest audit annotations (Definition 4.6); and a cluster is a finite set of records. The table below fixes, for each role, when a finite cluster C of records *supplies the witness type* of P_i , that is, when $W_i(C)$ holds; this table, with the paragraphs listed under *Witness predicates*, is the definition of W_i that Definition 4.11 uses. “Recorded” means given by finite typed fields of, or references to, declared records of the description. For $W_i(C)$, witness fields must be carried by members of C . A reference to a declared record outside C establishes its identity and type but does not import its fields as witness data. A certificate carried by a member of C is evaluated over the recorded values of D . A formula *reads the values of* a record d when d ’s identifier occurs in it as the record argument of a field-value term, as the record whose graph, table or relation an application or relation atom uses, in $t \in \text{dom } d$ or $t <_d t'$, as the record whose field a list operation or a quantifier bound uses, or as the predicate record of an abbreviation $d(t)$; it also reads d when a variable or an indexed list entry occurring as the record argument of a field-value term can denote d under the formula’s quantifier bounds and recorded lists (for a variable bounded by all declared records, or by those of

one kind, every such record). This is a syntactic condition, met even when the truth value does not depend on those values. A record whose values a certificate reads is an input to it and, when the row names that record, must belong to C . A row names the records it lists as constituents, together with the declared endpoint or index records of a W_4 law-bearing record; the map, partial-map or transition record of a W_6 replay certificate is named by the row and belongs to C ; other records a certificate reads, such as that map's declared domain, need not belong to C . The P6 audit entry is checked separately in $\text{Audit}_0(D) \subseteq \text{Audit}(D)$ and need not be a content member of C .

Check language. Checks are written in a fixed language: in summary, terms are record identifiers, rational constants, recorded field values, list lengths, entries and concatenations, applications of recorded maps and tables, finite sums and products, and word products $\varphi^{m,e}(w)$; atoms are equality, order, $t <_o t'$, $t \in \text{dom } F$, list membership and subword occurrence, relation atoms and $W_i(C)$ (defined at the end of the paragraph *Sorts*); formulas close atoms under $\neg, \wedge, \vee, \rightarrow$ and bounded quantifiers. In detail: terms are record identifiers, rational constants, recorded field values and applications of recorded finite maps, that is, of the graphs listed by map, partial-map and transition records, combined by finite sums and products; a recorded field value is written $f(d)$, with f the name of a datum or reference field and d a record-sorted term, its record argument (for instance $v(w_k)$, $A(E)$), and $f(d)_k$ is entry k of a list-valued field; the record argument may be an identifier, a variable or an indexed list entry, provided the denoted record carries that field, and a field name is not a record identifier, so $f(d)$ is never read as an application $F(t)$ of a map, partial-map, transition or table record; formulas are built from equality and order by $\neg, \wedge, \vee, \rightarrow$ (with $\leftrightarrow$ as an abbreviation, expanded before polarity is counted) and by quantifiers bounded by declared finite records or by the finite set of all declared records or of the declared records of one kind. An object record listing no members is not a declared finite record in this sense: a quantifier over it is a sort quantifier of the assertion language, and no certificate contains one.

Lists and maps. For every recorded finite list (a word in generating moves, the entry list of a trace, a list held in a datum field, or the identifier list of a reference field, such as the records an event or provenance record references or the members an object record lists) the language also has list length, indexing (τ_k for the entry of a trace τ at position k , positions counted from 0), membership, concatenation and contiguous-subword occurrence, with their standard finite-list meanings; the rewrite rules recorded in D , taken in a fixed order of their rule fields, form a finite list R whose k -th entry is a pair (λ_k, μ_k) , and λ_k , μ_k and $\lambda_{k,p}$ (the entry of λ_k at position p) are terms; an entry of a recorded finite list that is itself a finite list is again a recorded finite list, so indexing and the other list operations apply to it at every finite depth ($L_{k,p}$, $L_{k,p,q}$, as for $\lambda_{k,p}$); for a recorded finite list w and a term $u(x)$, $\sum_{k < |w|} u(w_k)$ is a term; and for a finite table field m recording a binary operation on a recorded list N of rational constants, an element e of N and a recorded finite map φ into N , the term $\varphi^{m,e}(w)$ is the left-to-right m -product of $\varphi(w_0), \dots, \varphi(w_{|w|-1})$, with $\varphi^{m,e}(\epsilon) = e$. In route certificate (a) the extension of φ to words is $\varphi^{m,e}$, and in (b) the weight of a word w is $\sum_{k < |w|} \text{wt}(w_k)$; quantifiers over rules, positions and list entries are bounded by declared finite lists. Applications of recorded finite maps include the graphs of refinement records and finite tables recorded in a field; for a datum field L holding a finite list of finite tables on a declared finite record X , $L_k(t)$ is the value at t of the k -th listed table, so that closure of L under composition is $\forall k, l < |L| \exists j < |L| \forall x \in X (L_j(x) = L_k(L_l(x)))$ and “ c differs from every listed map” is $\forall k < |L| \exists x \in X (c(x) \neq L_k(x))$.

Domains, rows and orders. For a map, partial-map or transition record F the language has the atom $t \in \text{dom } F$, true exactly when t is defined and its value occurs as the first coordinate of a pair in the graph of F (an interpretation assigns a map or transition record a graph defined at

every member of its declared domain), and an atomic formula containing an undefined map or table application is false; for a transition record t with codomain $\text{Dist}(S)$, $t(x)(s)$ is the entry of the row $t(x)$ at the position of s in the member list of S ; for a declared finite record X and a term $u(x)$, $\sum_{x \in X} u(x)$ is a term; for each order record o it has the atom $t <_o t'$.

Relations and predicates. For a recorded finite relation record R of arity k , the language also has atoms $R(t_1, \dots, t_k)$, true exactly when the tuple of values of $t_1, \dots, t_k$ is among the tuples R lists (for a relation of declared use quotient, when the values lie in one listed class); $x \sim y$ denotes such an atom. For a predicate record s given by an explicit check-language formula with one free variable, $s(t)$ in a certificate abbreviates that formula with t substituted for the variable; the abbreviation is unfolded at each recomputation using the formula then recorded in s , so a certificate containing $s(t)$ reads the formula field of s and every record the unfolded formula reads.

Certificates, assertion language and laws. A *certificate* is a record of a closed formula φ of this language, the inputs it names, and a truth value b ; it is correct when φ evaluates to b over the recorded values of the description, and *recomputation* is this evaluation. For an atom $W_j(C)$ the evaluation is classical: its value is fixed by its row over the description, but the free-form conjuncts allowed in $W_3(b)$ need only express their conditions in every well-typed interpretation, so recomputing a certificate that contains W_3 atoms is not in general an effective procedure; “finite” in the outside-scope criterion refers only to recorded extensions and carriers. Predicate, invariant and claim fields hold formulas of the *assertion language*, and a comparison or law field holds such a formula or a certificate, whose formula is in the check language. The assertion language is the check language extended by the predicate symbols declared by predicate records of the description, each with a declared arity, and by quantifiers over sorts declared by object records that list no members. Given an assignment to any free variables, such a formula has a truth value over D when every predicate extension it uses and every quantified sort’s carrier are recorded (Section 4); otherwise it is evaluated in *expansions* of D : an expansion keeps every recorded value, extension and carrier of D , assigns to each sort declared by an object record listing no members an arbitrary set (possibly empty), and assigns to each declared predicate whose extension D does not record an arbitrary relation of its declared arity on the carriers of its argument sorts ($\mathbb{Q}$ for a rational-sorted argument). Certificates, the rows of the table below and the seed clauses use the check language only; in particular the formula of a macro-specification in the W_2 row and in seed clause (2) of Definition 7.1 is a check-language formula containing no atomic W_j formula.

Sorts. Record identifiers and rational constants are distinct sorts. The declared types of fields, list entries and map or table arguments determine their sorts. A numeral or rational constant in record-sorted position (a record-typed argument of a recorded map, an argument of W_j , a term substituted for a record-sorted variable, as in $s(t)$, or compared with a record-valued term) denotes the record, with $=$ as identity of records and $<$ between record-sorted terms as the atom $t <_o t'$ of the order record o the formula names (in an order law, the order record carrying it; in a refinement law for $\rho : I \rightarrow I'$, $<$ and $<'$ are the atoms of the declared order records on I and on I'); a numeral in rational-sorted position (an argument of $+$ or $\cdot$, a rational-typed argument of a map or table, or compared with a value field) denotes the rational constant. In particular, monoid-table elements and arguments are rational-sorted, while the lists supplied to W_j contain record identifiers; equality compares terms of the same sort. The language also contains, for each $i \leq 6$ and each finite set C of declared record identifiers (written as a list of its members), an atomic formula $W_i(C)$; its value is determined by the P_i row of the table below, with the $\text{Audit}(D)$ conjunct of the W_6 row evaluated as in Definition 4.6. This is not circular: every certificate a row, law or seed clause reads contains no atom W_j (paragraph *Failure and placement*), and the $\text{Audit}(D)$ conjunct of W_6 ranges

over $\text{Audit}_0(D)$, whose certificates and conclusions contain no such atoms (Definition 4.6(f)); so the rows are evaluated before any W_j atom is. Record kinds, declared uses and field roles are recorded field values. These tag values, and elements of declared finite sets whose declared type is neither record-valued nor rational-valued, form a third sort, compared only by equality; record identifiers and rational values retain their respective sorts. Recorded field values are the values of datum and reference fields together with these; formula-valued fields and the formula of a certificate field are not terms, and a term-valued field enters only through its recorded term.

What a required certificate must express. Whenever a row of the table below, a law, or a covered seed requires a certificate of a property, the certificate's formula must express that property: over the named inputs it must be equivalent, in every well-typed interpretation of Definition 4.6, to the verification condition prescribed for that certificate by its row, law or seed clause. In addition, its formula must be obtained from that condition, with the named inputs substituted, using only the following steps: omitting conjuncts true in every well-typed interpretation (as described below), expanding a quantifier or sum bounded by a reference-field list into the finite conjunction, disjunction or sum over its members, and replacing an application of a term-valued field by its recorded term. No other conjunct may be added, even one true in every well-typed interpretation.

Exceptions. Two rows adjust this rule where they are displayed: the W_6 certificate has one of two fixed forms, and for $W_3(b)$ only the displayed conjuncts and the step checks are fixed in form, the other conjuncts needing only to express their conditions.

Further conventions. A well-typed interpretation is an interpretation in the sense of Definition 4.6(c1); in particular it keeps the records of D , their kinds, declared uses, field roles, references and formula- and term-valued fields, and lets every other datum field, such as a map graph or a multiplication table, take any value of its declared type. The prescribed conditions are displayed after the table: for W_6 the exact forms, and for the other rows and for laws the formulas in the list beginning "For the other rows and for laws". The certificate's truth value must be true, and it must be correct; an evaluation certificate $(s(t), b)$ recorded under the W_2 row or seed clause (2) records a value, not a required property, and may have $b = \text{false}$.

Named elements. When a row, law or seed clause prescribes that some element, pair or class has a property, the prescribed verification condition is that property for the element, pair or class the certificate names among its inputs, as in alternative (b) of the W_1 row, except where the displayed condition states the existence by a quantifier (the second conjunct of $W_1(a)$, the coverage conjunct $\forall z \in Q \exists u \in X (q(u) = z)$ of W_5 , the last conjunct of the order law, the formula Step in the step checks of $W_3(b)$, and the conditions of criterion (xi)(e)).

Omitted conjuncts. A conjunct that is true in every well-typed interpretation, such as $x \in X$ for a declared domain X or $k_0 \in K$ for a declared candidate class K (reference-field lists being fixed by Definition 4.6(c1)), may be omitted without affecting equivalence. The $W_1(b)$ certificates of D_{ex} and D_1 , the W_2 certificates of D_2 , and the comparison and realization-relation certificates of Remark 6.8 omit exactly these conjuncts. The certificate $(\neg s(c), \text{true})$ of the paragraph *One record suffices for recognition* instead expands $\forall k \in K$ over $K = \{c\}$. These conditions must imply the asserted property; for W_3 they are the monoid or rewriting checks, not an equivalence to route non-reducibility itself.

Failure and placement. A certificate that fails recomputation supplies no evidence for a witness, law or seed clause requiring that certificate; the record's other fields are assessed independently. A certificate used to establish a witness-table row, law or covered seed has a formula containing no atomic $W_j(C)$; such atoms may instead be certified in an audit after the witness predicates have been determined. A certificate may be carried in a comparison or check record, or in the fields of the relevant relation, map, order, transition or refinement record.

Laws. A *law* is an order, transition or refinement record (a binary relation record whose declared source and target are one object record I listing its members carries an order when it carries a certificate whose formula is the order law displayed after the table on the declared indices I , with $<$ read as its relation atom; it then counts as an order record in the W_4 row, in seed clause (4) of Definition 7.1 and for the atom $t <_o t'$, read as its relation atom) together with a certificate of a stated property of it: that an order relation on the declared indices is irreflexive and transitive and relates at least one pair, that a transition map is defined on its declared domain (for a transition record this holds in every well-typed interpretation, so that conjunct may be omitted) with values in its declared codomain, where the codomain is a declared record listing finitely many members, or the set $\text{Dist}(X)$ of probability distributions on such a listed set X , recorded by a codomain reference to X together with a datum field marking the codomain as distribution-valued (this marker, like a declared use, is kept fixed by every well-typed interpretation of Definition 4.6(c1)), membership in $\text{Dist}(X)$ being checked by nonnegativity and sum one, or that a refinement map between declared orders sends its declared source into its declared target and preserves the order.

Witness predicates. A role witness does not by itself supply the separate threshold, level, collapse-case and audit attachments of Definition 4.9. Operationally, $W_i(C)$ holds exactly when C records the data and relationships required by row P_i and its defining paragraphs: named constituent records belong to C ; certificates of required properties are carried by members of C , recorded true, recompute true over D , and satisfy the prescribed form requirements; for W_2 , recorded evaluations of s , including false evaluation certificates, are correct; and the W_6 audit entry satisfies that row's reference and conclusion requirements in $\text{Audit}_0(D)$. The predicates W_i are defined by the table together with the paragraphs *Check language* and *Certificates, assertion language and laws* before it (the latter fixes the laws of the W_4 row), the paragraph *Route certificates* (for W_3), the paragraph *What a required certificate must express* with its continuations *Exceptions* and *Failure and placement*, the verification conditions displayed after the table, the membership convention of the paragraph *Witness types*, and, for the $\text{Audit}(D)$ conjunct of W_6 , Definition 4.6(f); Proposition 6.1 derives from it which candidate features can support each role.

Role	$W_i(C)$ holds exactly when C records
P1	a map or partial-map record F whose declared use is <i>update</i> (Definition 4.1), source and target relation records of declared use quotient (the two may be the same record; they are the relation records that the row's certificate names as its source relation $\sim$ and target relation $\sim'$), and either (a) a finite certificate that the domain of F is saturated under the source relation, that some class of the source relation contains at least two elements of that domain, and that every related pair in that domain has related images, or (b) a comparison or check record of C naming a pair (x, y) of members of the declared domain of F and carrying a certificate that x and y are related by the source relation and that either exactly one of them lies in the domain of F or $F(x)$, $F(y)$ are defined, each lies in a class of the target relation, and they are not related by the target relation; in either case the certificate identifies its inputs and the update
P2	(1) a macro-specification predicate s whose formula has exactly one free variable r , occurring in it and ranging over records of the candidates' kind; (2) a relation record $\text{Real}(r, s)$ of declared use realization and, in fields of that realization relation or of comparison or check records of C , the evaluation of the formula of s at each member of the declared finite, nonempty class K of candidate realizers, every evaluation of s recorded in C being correct (equal to the value of the formula of s at that record in D); (3) a certificate either exhibiting a satisfying realizer or checking that none in that class exists. <i>No-realizer branch</i> : also (a) an evaluation, recorded in a field of the realization relation or of a comparison or check record of C , at which the formula of s is true at a declared same-kind record outside that class and (b) a candidate class that is the full declared domain of a map record of C of declared use packaging or update. <i>Satisfying branch</i> : also a comparison with the classes of the kernel E of a map record of C whose declared use is <i>packaging</i> and whose declared domain contains the declared candidate class, certifying that some class of E meets both the realizer set and its complement in the candidate class, that is, that the realizer set is not a union of the sets $[x]_E \cap K$, so that in the satisfying branch realizability is not reducible to quotient-class equality
P3	two defined (paragraph <i>Route certificates</i> , <i>Definedness</i>) path or derivation records with the same declared source and target, written as words in the generating moves (map or partial-map records; since definedness reads their graphs, they belong to C), the finite rewrite rules $\lambda \rightarrow \mu$ recorded with those routes or in referenced records that are members of C , these being all rewrite rules recorded in D , with the images $\varphi(m)$ in (a) and the weights in (b) given for every move occurring in the two routes or in those rules, and a comparison certificate that the two routes are not related by those rules, in one of the two forms of the paragraph <i>Route certificates</i> below
P4	at least one stage, order, transition or refinement record with declared endpoints or indices and a law for it in the sense above; an index label without such a law does not suffice
P5	distinct declared object or state records x, y (the micro-elements), a relation record E whose declared use is <i>quotient</i> , a map record q whose declared use is <i>packaging</i> to a declared quotient-object record Q (named by the row, so it belongs to C), and a check record of C carrying a certificate of quotient validity: E is the declared equivalence, q identifies x and y , and the fibers of q agree with E , with the quotient object covered by q ; a P1 split-pair certificate alone does not suffice
P6	a declared event e , a trace record τ , a record tied to e (one of the two references the other) that is τ itself or a provenance record also tied to τ , a replay or check record tied to τ carrying a certificate, in the sense above, whose formula states a replay property of the trace (each successive trace entry is the image of the preceding one under a declared map, partial-map or transition record whose codomain is not of the form $\text{Dist}(S)$) or a provenance property of it (each trace entry is a record referenced by a provenance record p of C tied to τ , and the trace begins at a record the event references), and an entry $a \in \text{Audit}_0(D)$ that refers to a claim annotation, to the event and trace records of C and to that replay or check record, and whose conclusion is the formula of that record's certificate

Witness types table. Witness predicates $W_1, \dots, W_6$: for a fixed description D , $W_i(C)$ holds exactly when the finite cluster C records the data in row P_i ; the W_3 row also refers to all rewrite rules recorded in D , and the W_6 row to $\text{Audit}_0(D)$. Every certificate a row requires must also meet the form requirements displayed after the table and in the paragraph *What a required certificate must express*.

Route certificates. (a) *Monoid certificate*: a finite monoid N given by its multiplication table (its elements recorded as rational constants in a list field and its multiplication as a finite table field) and the images $\varphi(m) \in N$ of the moves, such that the extension φ of these images to words satisfies $\varphi(\lambda) = \varphi(\mu)$ for every rule and $\varphi(\rho_1) \neq \varphi(\rho_2)$, together with the checks that $e \in N$, that N is closed under the recorded multiplication, and that N is associative with identity e . (b) *Rewriting certificate*: a certificate that, for a recorded rational weight at least 1 of each move (finitely many such weights generate a discrete set of word weights, so strict decrease terminates) extended additively to words, each rule strictly decreases the weight. It also records the list of all critical pairs (every overlap of left-hand sides, as in the prescribed condition below, including proper overlaps of a left-hand side with itself or with another, and inclusion overlaps of left-hand sides; non-equivalence in the free monoid on the moves implies non-equivalence under rule applications restricted to composable routes), with a check that the list is complete. For each critical pair and for each of ρ_1, ρ_2 it records a finite rewrite sequence, each step checked to replace one occurrence of some λ by its μ , words being recorded as finite lists of moves; the two sequences of each critical pair end in one word, and those from ρ_1, ρ_2 end in distinct words containing no left-hand side. By the critical-pair lemma the rewrite system on all words is locally confluent, and by Newman's lemma confluent. The distinct normal forms therefore exclude any sequence of applications of the recorded rules, in either direction, relating the two routes, even when intermediate words need not be composable. *Definedness*: a path or derivation record is *defined* when it lists the successive records it passes through and each move's map or partial-map record is recorded as defined at the preceding record, with value the next one. A record is *tied to* another when one of the two references the other. The recorded rule list may be empty; a W_3 witness then certifies only that two parallel words are distinct, as for two distinct moves $a, b : u \rightarrow v$, and in D_3 the rule $aa \rightarrow a$, applied in either direction, never identifies ab with ba , since it changes only the lengths of runs of a . Because the W_3 row and both certificates range over all rewrite rules recorded in D , a W_3 witness is in general not preserved when records carrying further rewrite rules are adjoined; Proposition 7.5 therefore forms disjoint unions in which D_3 alone records rules.

For W_6 , the certificate formula is exactly $\bigwedge_{k+1 < |\tau|} \tau_{k+1} = F(\tau_k)$, for $|\tau| \geq 2$ and a map, partial-map or transition record F whose codomain is not of the form $\text{Dist}(S)$, or exactly $\bigwedge_{k < |\tau|} \tau_k \in P \wedge \tau_0 \in R_e$, with P and R_e the concatenations of the reference-field identifier lists of p and of e , expanded over the entry list of τ ; it contains no additional assertion. The row admits degenerate instances: a trace (s, s) replayed by an identity map, and, since the provenance branch imposes no length condition, a one-entry trace whose entry the event and provenance record reference. In the provenance branch, the provenance record p and every record whose fields the check reads belong to C .

For the other rows and for laws, the prescribed verification condition is the following formula over the named inputs ($\sim, \sim'$ the source and target relations, X the declared domain, and $|\sim|, |E|$ the lists of members of the listed classes of the relation records $\sim, E$; $\forall u \in |E| \chi$ abbreviates $\forall u (E(u, u) \rightarrow \chi)$, u ranging over all declared records).

- $W_1(a)$: $\forall x \in X \forall y \in |\sim| ((x \sim y \wedge x \in \text{dom}F) \rightarrow y \in \text{dom}F) \wedge \exists x, y \in X (x \in \text{dom}F \wedge y \in \text{dom}F \wedge x \neq y \wedge x \sim y) \wedge \forall x, y \in X ((x \in \text{dom}F \wedge y \in \text{dom}F \wedge x \sim y) \rightarrow F(x) \sim' F(y))$;
- $W_1(b)$, for the named pair: $x \in X \wedge y \in X \wedge x \sim y \wedge ((x \in \text{dom}F \leftrightarrow y \notin \text{dom}F) \vee (x \in \text{dom}F \wedge y \in \text{dom}F \wedge F(x) \sim' F(y) \wedge F(y) \sim' F(x) \wedge \neg F(x) \sim' F(y)))$;
- W_2 , no-realizer branch: $\forall k \in K \neg s(k)$ for the non-realizer certificate, together with a separately recorded true evaluation $s(t)$ for a named same-kind record $t \notin K$; satisfying branch: $k_0 \in K \wedge s(k_0)$ for the named k_0 and, for the comparison, $x \in K \wedge y \in K \wedge q(x) = q(y) \wedge s(x) \wedge \neg s(y)$ for the named x, y ;

- W_5 : $\forall u, v \in X (E(u, v) \leftrightarrow q(u) = q(v)) \wedge x \neq y \wedge q(x) = q(y) \wedge \forall z \in Q \exists u \in X (q(u) = z) \wedge \forall u \in |E| (u \in X) \wedge \forall u \in X (q(u) \in Q)$;
- order law, on declared indices I : $\forall i \in I \neg(i < i) \wedge \forall i, j, k \in I ((i < j \wedge j < k) \rightarrow i < k) \wedge \exists i, j \in I (i < j)$;
- transition law, with declared source X and codomain Y : $\forall x \in X (x \in \text{dom } t) \wedge \forall x \in X (t(x) \in Y)$, where, when $Y = \text{Dist}(S)$ for a declared finite carrier S , membership means $\forall s \in S t(x)(s) \geq 0$ and $\sum_{s \in S} t(x)(s) = 1$ (a transition law imposes no nontriviality; the identity transition T of D_T in the proof of Proposition 7.5 satisfies it);
- refinement law, for a declared map $\rho : I \rightarrow I'$: $\forall i \in I (\rho(i) \in I') \wedge \forall i, j \in I (i < j \rightarrow \rho(i) <' \rho(j))$ (the order-preservation conjunct is vacuous when the order on I relates no pair; the range conjunct remains required; like the transition law it imposes no nontriviality).

The W_3 certificates are those of the paragraph *Route certificates*. For $W_3(a)$, with R the recorded rule list, M a list field of the comparison record, and w_1, w_2 the recorded words of ρ_1, ρ_2 , the prescribed condition is $e \in N \wedge \forall x, y \in N (m(x, y) \in N) \wedge \forall x, y, z \in N (m(m(x, y), z) = m(x, m(y, z))) \wedge \forall x \in N (m(e, x) = x \wedge m(x, e) = x) \wedge \forall a \in M (\varphi(a) \in N) \wedge \forall k < |R| (\varphi^{m,e}(\lambda_k) = \varphi^{m,e}(\mu_k)) \wedge \neg \varphi^{m,e}(w_1) = \varphi^{m,e}(w_2) \wedge \forall k < |R| \forall p < |\lambda_k| (\lambda_{k,p} \in M) \wedge \forall k < |R| \forall p < |\mu_k| (\mu_{k,p} \in M) \wedge \forall p < |w_1| (w_{1,p} \in M) \wedge \forall p < |w_2| (w_{2,p} \in M)$. For $W_3(b)$ it is the conjunction of the positivity of the weight of every move occurring in R , w_1 or w_2 , written $\forall k < |R| \forall p < |\lambda_k| (\text{wt}(\lambda_{k,p}) \geq 1) \wedge \forall k < |R| \forall p < |\mu_k| (\text{wt}(\mu_{k,p}) \geq 1) \wedge \forall p < |w_1| (\text{wt}(w_{1,p}) \geq 1) \wedge \forall p < |w_2| (\text{wt}(w_{2,p}) \geq 1)$, $\forall k < |R| (\text{wt}(\lambda_k) > \text{wt}(\mu_k))$, completeness and correctness of the critical-pair list (for all rule indices $k, l < |R|$, including $k = l$, and every overlap of λ_k with λ_l , whether an inclusion overlap (an occurrence of λ_k as a contiguous subword of λ_l for $k \neq l$, including one at position 0 even if $\lambda_k = \lambda_l$; for $k = l$ there is no inclusion overlap) found by contiguous-subword occurrence or a proper overlap (a nonempty proper suffix of λ_k equal to a prefix of λ_l) found by indexing, the list has an entry consisting of the overlap word and the two words obtained from it by the two overlapping rule applications, and every entry has this form), the step check of each recorded rewrite sequence, the condition that the two sequences of each entry start at its two words and end in one word, and that the sequences for ρ_1, ρ_2 start at w_1, w_2 , and irreducibility and distinctness of the normal forms of w_1, w_2 , each written with the list operations of the check language. The step check of a recorded sequence L is $\forall j < |L| (j + 1 < |L| \rightarrow \text{Step}(L_j, L_{j+1}))$, where $\text{Step}(u, v)$ abbreviates $\exists k < |R| \exists p < |u| (p + |\lambda_k| \leq |u| \wedge |v| + |\lambda_k| = |u| + |\mu_k| \wedge \forall q < |u| (q < p \rightarrow v_q = u_q) \wedge \forall q < |\lambda_k| (u_{p+q} = \lambda_{k,q}) \wedge \forall q < |\mu_k| (v_{p+q} = \mu_{k,q}) \wedge \forall q < |u| (p + |\lambda_k| + q < |u| \rightarrow v_{p+|\mu_k|+q} = u_{p+|\lambda_k|+q}))$. For $W_3(b)$ the exact-form requirement of the paragraph *What a required certificate must express* applies to the displayed conjuncts and to the step checks; each remaining conjunct (critical-pair completeness and correctness, the endpoint conditions, irreducibility and distinctness) may be any check-language formula that expresses its condition in every well-typed interpretation. Equivalence in every well-typed interpretation is not decidable in general (datum lists may change length across interpretations and the language has $+$ and $\cdot$ on list lengths, so solvability of Diophantine equations over $\mathbb{N}$ reduces to it), and neither are W_3 , the entailment requirement (c2) or criterion (viii); the paper uses these predicates classically and claims no decision procedure.

In the no-realizer branch of the W_2 row, condition (a) supplies a true evaluation of s outside the candidate class, so that the recorded evaluations of s take both truth values, as seed clause (2) of Definition 7.1 requires; condition (b) fixes the candidate class as the full declared domain of a packaging or update map, so that it is not chosen ad hoc. These conditions restrict the recorded evaluations of s . Given the map required by (b) and the evaluations and certificates of the row, $s(r) \equiv (r = t)$, for any declared same-kind record $t \notin K$, satisfies the no-realizer requirements on s .

In the satisfying branch, given a packaging map q of the cluster whose declared domain contains K and the evaluations and certificates of the row, $s(r) \equiv (r = x)$ suffices whenever $x \in K$ and its q -class contains a second member of K . A W_2 witness thus certifies the recorded form of a specification relative to the declared class (Remark 6.10).

The three levels of Definition 2.4 separate what happens (behavior), the certificate that it happens (verification), and the structure that explains it, such as a quotient object with its universal property (structure). The running example above shows P1, P5 and P6 in one small description.

Remark 2.2 (Excluded alignments). Three alignments that Definition 2.1 and the *Witness types* table rule out are used throughout later classification: generic transition data is not P3; generic path or derivation data is not P5; and generic semantic or interpretation data is not P1. These constraints determine which feature clusters can later be interpreted as active projections of each role. Precisely: a cluster without two defined path or derivation records, or without a comparison certificate of the W_3 row, fails W_3 ; a cluster without a relation record of declared use quotient, or without a map record of declared use packaging, fails W_5 ; and a cluster without a map or partial-map record of declared use update fails W_1 . Each is read off its row of the *Witness types* table, so these alignments add no hypothesis beyond that table.

Remark 2.3 (Drive face of P6). Foundations I uses a specialization $P6_{\text{drive}}$ of P6 for arguments about affinities and detailed balance [27]. This paper assigns it no witness type and no result uses it; the point retained here is that a P3 route/coherence-mismatch witness does not by itself establish directionality (Definition 2.1).

No proof in this paper uses the drive face; the terms of the remark are as follows. For a Markov chain with stationary law π , *detailed balance* means $\pi_x P_{xy} = \pi_y P_{yx}$ for every pair of states. For a cycle of states with positive stationary mass and positive transition probabilities in both directions, its *affinity* is the logarithm of the product of the forward probabilities divided by the product of the reverse ones; a nonzero affinity witnesses a failure of detailed balance. The remark records that a P3 mismatch alone does not establish a direction, the distinction also recorded in Definition 2.1.

2.2 Behavioral, verification, and structural-categorical levels

Each role admits a common three-level pattern.

Definition 2.4 (Level trichotomy). For each role $P_1, \dots, P_6$, we distinguish

- the *behavioral level*: status, occurrence, or observed behavior of the role in a description;
- the *verification level*: proof, witness, certificate, or checkable record supporting a behavioral claim [22];
- the *structural-categorical level*: reusable structure supporting the primitive role.

Forgetting proceeds from structural-categorical to verification to behavioral and is in general lossy. Lifting proceeds from behavioral to verification to structural-categorical only after adding explicitly recorded data, and selected lifts are not unique. Formally, a level is one of the three values recorded in the level field of a level annotation; the proofs of Sections 5–9 require that such a value be recorded and use only whether two recorded values coincide (for the level crossings of Definition 4.11), not which value is recorded.

Remark 2.5 (Level content per primitive). The level trichotomy instantiates as follows.

- P1: descent success or descent obstruction status; descent proof or obstruction witness; quotient data equipped with a descended map, or an explicit no-descended-map obstruction record.

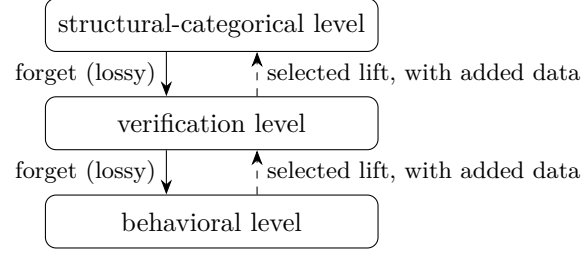


Figure 2. The three levels of Definition 2.4. Forgetting moves down and loses data; lifting moves up only with recorded added data, and the lift is a selection, not in general unique.

- P2: certified unrealizability or failure of representability over the declared packaging kernel; the certificate of that status; macro-specification layer together with a realization relation.
- P3: witness behavior of route disagreement; generated route terms and non-reducibility certificates; categorical route grammar or a reduced categorical universe.
- P4: stage status, transition occurrence, or refinement behavior; order/preorder, transition, or refinement certificates; stage/refinement architecture.
- P5: packaging or identification behavior; equivalence and quotient-validity certificate; quotient object with a packaging map and universal-property or tower/coarsening data.
- P6: audit-relevant event occurrence or record–event association; checkable audit record; audit schema with provenance, replay, and check relations.

The meta-theoretic properties of this trichotomy — lossy forgetting, selected non-unique lifts, and the absence of lower-to-higher determinacy without added data — are formalized in Section 3.

2.3 Local boundary distinctions

With role meanings and level pattern fixed, we turn to how the six roles remain distinct. Table 1 records the pairwise local rules. These are local distinctions under scoped bookkeeping, not a global primitive-independence theorem: two roles may interact legitimately in a single description, but neither collapses into the other. The rules below use the domain *FATCD* and the channel statuses before they are defined; a reader who prefers definitions first can read Sections 4.1–4.4 and Definition 7.1 first.

Table 1. Local boundary distinctions. Each row records the distinction rule between two roles, a permitted interaction, the collapse that is forbidden under scoped bookkeeping, and the bare partner datum X_b on which that collapse would credit a role.

pair	distinction rule	permitted interaction	forbidden collapse	bare datum X_b
P1/P5	P5 supplies quotient packaging; P1 tests selected update compatibility with quotient packaging	P1 uses P5 when checking descent or obstruction	packaging alone is not P1	the P5 packaging map record alone
P1/P4	P4 supplies stage/refinement structure; P1 may vary by stage	stage-indexed descent or obstruction	stage index alone is not P1	the P4 stage-index record alone
P1/P2	P1 is update compatibility; P2 is macro-specification representability	representability may be studied over quotient structures affected by descent	representability failure is not automatically descent obstruction	the P2 non-representability certificate alone

pair	distinction rule	permitted interaction	forbidden collapse	bare datum X_b
P1/P3	P1 concerns update maps, possibly partial, whose undefined values leave a route undefined; P3 compares already-defined routes	P1 may block a route before P3 comparison	route undefinedness is not route mismatch	the partial update map and the undefined route alone
P1/P6	P6 audits a P1 event; P1 is the event content	audited descent success or obstruction	audit record is not descent obstruction	the P6 audit record alone
P2/P5	P5 is quotient packaging; P2 requires a macro-specification, a realization relation and a finite existence or nonexistence certificate, plus a comparison with quotient classes when a realizer exists	a specification representable over E supplies no P2 witness in the satisfying branch; a separately witnessed P2 projection may accompany P5 packaging	packaging alone is not P2	the P5 packaging map record alone
P2/P4	P4 supplies stages; P2 representability may vary by stage	stage-indexed representability	stage variation alone is not P2	the P4 stage records alone
P2/P3	P2 concerns representability of specifications; P3 concerns route disagreement	route grammar may be built over represented macro structure	non-representability is not route mismatch	the P2 non-representability record alone
P2/P6	P6 audits representability claims; P2 is the representability content	audited representability or non-representability	audit trail is not representability	the P6 audit-trail records alone
P3/P4	P4 supplies stage/regime structure; P3 route mismatch may be stage-indexed	route systems vary over valid stages	stage-indexed route mismatch is not P4 itself	the P3 route-disagreement certificate alone
P3/P5	P5 supplies quotient objects; P3 compares defined routes over objects	routes over quotient objects	quotient object is not route mismatch	the P5 quotient-object record alone
P3/P6	P6 audits route comparisons; P3 is the route mismatch content	audited route mismatch	audit record is not route mismatch	the P6 audit record alone
P4/P5	P4 supplies stage/refinement structure; P5 supplies packaging at a stage	stage-indexed packaging	packaging is not stage structure	the P5 packaging map record alone
P4/P6	P4 supplies staged structure; P6 audits staged histories or transitions	audited stage transitions	audit record is not staging	the P6 audit record of staged histories alone
P5/P6	P5 supplies packaging; P6 audits packaging events	audited packaging	audit record is not packaging map	the P6 audit record of packaging events alone

Proposition 2.6 (Boundary distinctions under scoped bookkeeping). *In each row of Table 1, isolate the bare partner datum X_b listed in its last column, presented as a boundary record in the sense of Definition 4.2 (carrying exactly the fields of that datum, the typing fields of its kind, the datum and reference fields its certificates read, and reference fields to other members of its cluster). That datum alone lacks a record that the credited role's witness row requires a witness cluster to contain: $W_i(C)$ is false for a cluster C formed only from that datum, without importing fields of referenced records or adding independently supplied credited-role records. Consequently a claim that the credited P_i*

projection follows from that datum, without the further data required by W_i , is inadmissible (in the sense of Section 3) under the alignments of Remark 2.2 together with the honest bookkeeping required by a FATCD. A cluster carrying independently supplied witnesses for both roles may activate both channels; this is the permitted interaction, not their identification.

The proof uses the domain FATCD, the truth of honest audit conclusions and the certification condition for claims (Definitions 4.7, 4.6(c3) and 3.1); the row-by-row verdicts after it use the offers of Definition 4.6(d) and are contrasted there with the statuses of Definition 7.1; a first reading can skip it. The idea of the proof is short. Each forbidden collapse would credit one role with data that only the other role's witness type supplies. The borrowed content does not supply the credited role's witness, so the attempted identification receives an inactive local verdict and is inadmissible; independent data may still activate the description's channel. The proof checks this row by row; the three rows worked in full show the two shapes the argument takes. The local verdicts *no offer* and *failed offer* below describe two failures of a local attempt: respectively, no credited-role witness is offered, or an offered cluster fails its witness check. They are not channel statuses $\text{Status}_i(D)$, which Definition 7.1 assigns from the whole description. In the general argument, P_i is the role the collapse would credit and P_j the role whose content it borrows; in the enumeration of the rows, each row is written P_i/P_j in the order of Table 1, and each clause names the credited role, which may be either member of the pair.

Proof. In short, each credited row of the witness table requires a record that the row's bare datum lacks: an update map or a source relation record of declared use quotient for W_1 (rows P1/P5, P1/P4, P1/P2, P1/P6); a macro-specification with a realization relation for W_2 (rows P2/P5, P2/P4, P2/P6); two defined path or derivation records for W_3 (rows P1/P3, P2/P3, P3/P5, P3/P6); a law-bearing order, transition or refinement record for W_4 (rows P3/P4, P4/P5, P4/P6); and a quotient relation with a packaging map for W_5 (row P5/P6). Hence $W_i(C) = \text{false}$. A claim that the credited P_i projection follows from the datum has statement $W_i(C)$. By Definition 4.6(c3), an outright conclusion of an honest audit that uses no unrecorded predicate symbol and quantifies only over recorded carriers is true in D , and $W_i(C)$ is such a formula; so no honest audit certifies $W_i(C)$, the certification condition of Definition 3.1 fails, and the claim is inadmissible. This proves the proposition. $\square$

Row-by-row local verdicts. The local attempt verdicts below are illustrative and are not used by later results. The enumeration and table also identify each row's credited role and missing witness field, used in Proposition 3.3. A reader may continue at Remark 2.7 (p. 25). The verdict is not derived from the row: in every row it is failed offer if the attempt is recorded as an offer and no offer otherwise, and only $W_i(C) = \text{false}$ is proved. No later result uses these verdicts. Each row of Table 1 names an ordered pair (P_i, P_j) , a permitted interaction, and a forbidden collapse. The forbidden collapse always has the same logical form: it would license treating witness or structural data carried under the meaning of one role as if it were, by itself, witness or structural data for the other role — that is, it would identify the two roles' admissible data types. The following enumeration records, row by row, the credited role and the missing witness field that the proof of Proposition 2.6 summarizes, and maps each row's forbidden collapse to a common schema. The enumeration discharges all fifteen rows of Table 1, mapping each row's forbidden collapse to the schema, with the role-specific data types read off Definition 2.1 and the per-primitive level content of Remark 2.5. Three rows are then worked in full as illustrations.

Schema. Fix a row and isolate the particular partner-role datum named in its forbidden-collapse entry. Compare that datum with the fields required by the credited role's witness row. If a required

field is absent, the bare datum cannot certify W_i ; hence the implication from that datum alone to an active P_i projection is invalid. An offered cluster containing only the bare datum may honestly record a failed W_i check. Adding the missing P_i data may make W_i true and may activate both channels, without identifying their meanings; this is the permitted interaction of the row.

Application to the rows. Every entry in the forbidden-collapse column has one of two shapes. In the *carried form*, “ X alone is not P_i ” or “ X is not P_i ”, the datum X is content named by the distinction-rule column as the meaning of the partner role; the credited-role field absent from the bare datum is listed below. In the *non-implication form*, “ P_j -status S is not automatically P_i -status S' ”, the partner’s status or certificate is not the credited role’s required certificate. These are checks of the stated bare datum, not assertions that every P_j -bearing cluster fails W_i . The fifteen clauses record the missing fields.

Enumeration of all fifteen rows. We list each row as P_i/P_j , name the credited channel, the borrowed content, and the distinct witness type the credited channel requires but does not receive, so that the schema applies with no residue. Throughout, the local attempt verdict of a clause describes the attempted identification; it is not the channel status $\text{Status}_i(D)$ that Definition 7.1 assigns from a whole description; a description may carry such an attempt and, independently, an active witness for the same role. In every clause the attempt may be recorded as an offer, with check result $W_i(C) = \text{false}$ (verdict failed offer), or not recorded (verdict no offer); the clauses below illustrate the first case in (3), (4), (8), (10), (11) and (13) and the second in the others. These stipulations concern the listed attempts, not the status $\text{Status}_i(D)$ of a full description; the two verdicts describe the attempts as listed, and the statement uses only $W_i(C) = \text{false}$, which holds in every clause.

- (1) P_1/P_5 , “packaging alone is not P_1 ”: credits P_1 on the P_5 packaging map record alone; the update or partial-map record and the certificate or related-pair witness required by W_1 are absent; the displayed local attempt verdict for P_1 is no offer.
- (2) P_1/P_4 , “stage index alone is not P_1 ”: credits P_1 on P_4 stage-index content; no descent or obstruction witness is thereby present; the displayed local attempt verdict for P_1 is no offer.
- (3) P_1/P_2 , “representability failure is not automatically descent obstruction”: credits P_1 on a P_2 non-representability status; a non-representability certificate is not a descended-map obstruction record and does not instantiate the P_1 obstruction witness; the displayed local attempt verdict for P_1 is failed offer.
- (4) P_1/P_3 , “route undefinedness is not route mismatch”: credits P_3 on the undefinedness of a route through a partial update map, where the map is undefined at a value the route passes through; a non-reducibility certificate between *already-defined* routes is required, and an undefined route supplies no such certificate, so the candidate cluster does not instantiate the P_3 witness schema; recorded as an offer, the attempt has local verdict failed offer.
- (5) P_1/P_6 , “audit record is not descent obstruction”: credits P_1 on a P_6 audit record; an audit record is not a descended-map or obstruction witness; the displayed local attempt verdict for P_1 is no offer.
- (6) P_2/P_5 , “packaging alone is not P_2 ”: credits P_2 on P_5 packaging content alone. The packaging map supplies neither the realization relation nor its finite existence or nonexistence certificate; in the satisfying-realizer branch it also supplies no required comparison. The local attempt verdict for P_2 is no offer. A separately witnessed P_2 projection may still accompany the packaging.

- (7) P2/P4, “stage variation alone is not P2”: credits P2 on P4 stage content; a representability certificate with realization relation is required, and stage variation supplies none, so the local attempt verdict for P2 is no offer.
- (8) P2/P3, “non-representability is not route mismatch”: credits P3 on a P2 non-representability status; a non-representability record is not a non-reducibility certificate between defined routes and does not instantiate the P3 witness schema, so the local attempt verdict for P3 is failed offer.
- (9) P2/P6, “audit trail is not representability”: credits P2 on a P6 audit trail; an audit record is not a representability certificate; the displayed local attempt verdict for P2 is no offer.
- (10) P3/P4, “stage-indexed route mismatch is not P4 itself”: credits P4 on the P3 route-disagreement certificate alone, without the stage law that indexes it; a non-reducibility certificate carries no order, transition, or refinement structure and does not instantiate the P4 witness schema; if the certificate is carried in the fields of an order, transition or refinement record, that record still carries no law, since its only certificate expresses non-reducibility of two routes, so W_4 fails and the local attempt verdict for P4 is failed offer.
- (11) P3/P5, “quotient object is not route mismatch”: credits P3 on a P5 quotient object; a quotient object is not a non-reducibility certificate between defined routes and does not instantiate the P3 witness schema; the displayed local attempt verdict for P3 is failed offer.
- (12) P3/P6, “audit record is not route mismatch”: credits P3 on a P6 audit record; an audit record is not a non-reducibility certificate; the displayed local attempt verdict for P3 is no offer.
- (13) P4/P5, “packaging is not stage structure”: credits P4 on the P5 packaging map alone; a packaging map is not an order/transition/refinement certificate and does not instantiate the P4 witness schema; the displayed local attempt verdict for P4 is failed offer.
- (14) P4/P6, “audit record is not staging”: credits P4 on a P6 audit record of staged histories; an audit record is not an order/transition/refinement certificate; the displayed local attempt verdict for P4 is no offer.
- (15) P5/P6, “audit record is not packaging map”: credits P5 on a P6 audit record of packaging events; an audit record is not a quotient object with a packaging map; the displayed local attempt verdict for P5 is no offer.

Clauses (1)–(15) exhaust the rows of Table 1: each names the credited channel, the borrowed partner-role content, and the distinct witness type the credited channel requires and does not receive, and each clause shows that the named bare datum lacks a field required by the credited witness; the displayed local verdict describes an absent or failed offer, not a status of the full description.

We now work three of these rows in full; they illustrate the carried and non-implication shapes.

Row P1/P5 (packaging alone is not P1), clause (1). By Definition 2.1, P5 is quotient packaging and P1 is update-vs-quotient compatibility, i.e. descent. By the W_5 row of the witness table, a P5-witness is a quotient object with a packaging map and quotient-validity evidence, whereas a P1-witness (the W_1 row) is an update map or partial map with source and target relation data and either its compatibility certificate or a related pair witnessing failed domain saturation or separated defined images. Possession of the packaging map is precisely the data that says *what* the quotient is; it says nothing about *whether* a selected update descends through it. In terms of the *Witness types* table, W_1 requires a map record whose declared use is *update*, and the P5 cluster of the packaging datum contains only the packaging map, whose declared use is *packaging*; so W_1 fails on that cluster. The collapse “packaging alone is P1” would credit the P1 channel on the strength of the packaging

map alone, but no descended-map witness and no obstruction record is thereby present, so by the schema the verdict on this attempted identification is no offer; an independently witnessed P1 channel may still be active. The descent question has not been answered by the packaging map. The permitted interaction — P1 using P5 when checking descent or obstruction — is untouched, since there P1 consumes the quotient packaging as the object *over which* a descended map or obstruction is then exhibited, and the descent witness remains the P1 channel’s own content.

Row P3/P4 (stage-indexed route mismatch is not P4 itself), clause (10). By Definition 2.1, P4 is stage/order/transition/refinement structure and P3 is enriched route/coherence mismatch; by Remark 2.2 generic transition data is *not* P3 and, symmetrically, a route-mismatch witness is not P4. A P4-witness is an order/transition/refinement certificate or a stage architecture; a P3-witness is a non-reducibility certificate exhibiting disagreement between already-defined routes. When route mismatch is indexed by stages, the stage index is P4-content and the route disagreement at each index is P3-content; these are recorded as data of different roles. The collapse “stage-indexed route mismatch is P4” would credit the P4 channel on the strength of the route-disagreement certificate considered without its stage-law record, but a non-reducibility certificate is not an order/transition/refinement certificate — it carries no stage, order, or refinement structure of its own — so by the schema the local attempt verdict for P4 is failed offer: the candidate cluster does not instantiate the required P4 witness schema. The permitted interaction — route systems varying over valid stages — is untouched, since there the P4 stage structure is supplied independently and the P3 mismatch is merely indexed by it, each channel keeping its own witness.

Row P2/P5, clause (6). P2 concerns a macro-specification and its realization relation; P5 concerns quotient packaging. In the satisfying-realizer branch of W_2 , the recorded comparison must show that some class of the kernel of the declared packaging map meets both the realizer set and its complement in the candidate class. In the no-realizer branch, recorded satisfaction data, a finite certificate that no candidate satisfies the specification, and a recorded true evaluation at a declared same-kind record outside the class, and a candidate class that is the full declared domain of a declared packaging or update map, suffice; no quotient comparison is required. A packaging map alone supplies neither branch’s realization relation and certificate. Thus the proposed credit of P2 from P5 packaging alone makes no W_2 offer; the local attempt verdict for P2 is no offer. A P2 witness may still be carried alongside P5 quotient data when its own required records and attachments are present.

The enumeration (1)–(15) covers every row of Table 1, and the three worked rows confirm both shapes in detail. The argument is local to each listed pair under the covered scope and the audit condition of Definition 4.6; it establishes pairwise non-collapse, not global independence of the six roles (cf. Remark 2.7).

The clauses of the proof follow the rows of Table 1; the credited role of each, and the verdict its clause illustrates, are as follows; in every row the verdict is *failed offer* if the attempt is recorded as an offer and *no offer* otherwise, and only $W_i(C) = \text{false}$ is proved:

	pair	credited	local verdict		pair	credited	local verdict
(1)	P1/P5	P1	no offer	(9)	P2/P6	P2	no offer
(2)	P1/P4	P1	no offer	(10)	P3/P4	P4	failed offer
(3)	P1/P2	P1	failed offer	(11)	P3/P5	P3	failed offer
(4)	P1/P3	P3	failed offer	(12)	P3/P6	P3	no offer
(5)	P1/P6	P1	no offer	(13)	P4/P5	P4	failed offer
(6)	P2/P5	P2	no offer	(14)	P4/P6	P4	no offer
(7)	P2/P4	P2	no offer	(15)	P5/P6	P5	no offer
(8)	P2/P3	P3	failed offer				

Remark 2.7 (Not a global independence theorem). The boundary matrix asserts local distinctions, not global independence. Primitives interact legitimately in a single description: P1 descent is audited by P6; P3 route terms may be indexed by P4 stages; in the satisfying branch a P2 witness is carried over P5 packaging data. No claim is made that the six roles are independent generators of a global algebra. The typed-non-collapse counterpart of this observation appears in Section 3.

2.4 Selected-lift atlas and global nonclaims

The remaining architectural data are the selected lifts between levels and the nonclaims that accompany them. Upgrading from behavioral or verification content to the structural-categorical level requires explicitly recorded added data. The resulting lifts are selected, not unique, and the atlas carries structural content that is lost under forgetting.

Definition 2.8 (Selected-lift atlas). The *selected-lift atlas* for the six roles records, for each P1, ..., P6:

- P1: proof or witness together with quotient structure, carrying either a descended map or an explicit no-map obstruction;
- P2: certificates of unrealizability or of failure of representability over the declared packaging kernel, together with the macro-specification layer and realization relation;
- P3: the selected lift data, carrying generated route terms, non-reducibility certificates, and categorical route grammar or a reduced categorical universe;
- P4: validity certificates together with structural stage/refinement architecture;
- P5: quotient-validity certificates together with the structural quotient data, a packaging map, and universal-property or tower/coarsening data;
- P6: record-event or check evidence together with the audit schema and its provenance and replay/check relations.

Each lift is selected, and not unique in general; where uniqueness holds, as for the descended map $\bar{F}([x]) = [F(x)]$ of a descending update, it is recorded with its proof. Forgetting from a higher level loses structural content that is not recoverable without re-adding data, unless, as for the descended map, the higher datum is determined and this is recorded with its proof. Formally, the selected-lift atlas of a description D consists of those lift annotations of D that satisfy Definition 3.6, are attached to a cluster C such that some witness-schema annotation targeting exactly C names P_i , and carry the corresponding added data listed above; such a lift is indexed by each such P_i . Other admissible lifts need not belong to this role-indexed atlas.

Remark 2.9 (Global nonclaims for the role architecture). The role architecture preserves the following nonclaims, all carried forward into the admissibility regime of Section 3 and the theorem chain of Sections 5–7:

- no full six-primitives algebra is claimed;
- no global primitive-independence theorem is claimed;
- no empirical realization is claimed;
- no uniqueness of primitive presentations is claimed;
- no uniqueness of selected lifts is claimed;
- no lower-level-to-higher-level determinacy is claimed;
- no instrument-free formulation is claimed;

- direct cross-level comparison remains inadmissible without translation.

3 Admissibility and meta-theory

This section states when a claim about a description counts. In Definitions 3.1–3.11 and Propositions 3.3 and 3.12, a claim is a claim annotation of some $D \in \text{FATCD}$, and “inadmissible” means that condition (a) or (b) of Definition 3.1 fails, or that a record required by Definition 3.5, 3.6, 3.9 or 3.11 is missing; in particular a claim other than an offer that no honest audit annotation of D certifies is inadmissible; an offer satisfies condition (a) of Definition 3.1 when an honest audit records the value of $W_i(C)$, even when that value is false. The central rule is honest bookkeeping (Definition 3.1): every claim, and every change of level, translation, visibility assumption or empirical bridge it relies on, must be explicitly recorded. Together with the role meanings and boundary distinctions of Section 2, that rule governs the claims developed below.

- The six roles are kept apart as types (typed non-collapse); the paper does not claim that they are independent generators.
- A claim moves between levels only through recorded forgetting, which loses data, and recorded selected lifts, which add it.
- A role counts as active only with the required threshold, witness, level, collapse, nonclaim and audit data.
- An instrument decides what is visible and what is suppressed.
- An empirical claim enters only through a recorded bridge.

The section uses the domain FATCD , the audit part $\text{Audit}(D)$ and threshold attachment, which are defined in Section 4; a reader checking proofs can follow the reading order at the end of Section 1.5, keeping Definition 4.6 at hand while reading Definition 3.1. Several claim-level rules below use records and annotations defined in Section 4. The main correspondences are:

- Definition 3.1 and Definition 4.6: Definition 4.6 requires every claim of a description to be referred to by an honest audit; Definition 3.1 further requires that audit to certify the claim’s recorded statement (for an offer, the value of $W_i(C)$);
- Remark 3.8 and Definition 4.9: when a role reading of a cluster is licensed;
- Definition 3.6 and the lift and forgetting annotations of Definition 4.2: how a claim changes level;
- Definition 3.9 and the visibility annotations: what an instrument shows and suppresses;
- Definition 3.11 and the empirical-bridge annotations: how an empirical claim enters.

The rules keep the global nonclaims of Remark 2.9 in force. Of this section, only Definitions 3.1, 3.5, 3.6, 3.9 and 3.11 and Proposition 3.12 are cited in Sections 4–9, the last only in Section 8. The other results of this section fix how claims are stated; several return in Sections 10–12.

3.1 Honest bookkeeping as an admissibility regime

The audit condition of Definition 4.6 records enough data for each claim, promotion, translation, lift, forgetting, visibility, and empirical-bridge assertion to be interrogated on its own terms. The following definition stipulates that these record requirements jointly constitute the admissibility regime, in a spirit analogous to the axiomatic specification discipline of Hoare logic [15].

Definition 3.1 (Admissible claim under honest bookkeeping). Honest bookkeeping is the admissibility regime for the claim annotations carried by a description $D \in \text{FATCD}$. A claim annotation

of D is *admissible* if and only if (a) some honest audit annotation of D refers to it and certifies the formula recorded as its statement (for an offer, $W_i(C)$ or $\neg W_i(C)$ according to the recorded result, Definition 4.6), and (b) every assertion below has an explicit record carried in $\text{Audit}(D)$ in the sense of Definition 4.2:

- the claim itself, with its statement, hypotheses, and scope;
- every promotion, with its source level, target level, added data, losses, and preserved nonclaims;
- every translation, with its source context, target context, preserved data, suppressed data, and claim strength;
- every selected lift, marked selected and non-unique unless uniqueness for its specified source data and target class is proved and recorded;
- every forgetting map, with its record of losses, empty only when losslessness is proved and recorded;
- every visibility claim, with both its visible and its suppressed data;
- every empirical-bridge assertion, with its substrate, observable, instrument, level map, threshold-witness relation, and suppressed-structure nonclaims;
- every cross-level claim, with a level annotation targeting it and translation annotations for each pair of levels recorded on the records it names, as in Definition 3.5.

Accordingly, unrecorded claims, implicit promotions, silent translations, asserted-but-unselected lifts, lossless forgetting without proof, visibility overreads, and bare empirical-realization assertions are all inadmissible.

Remark 3.2 (Why the regime is necessary). The bookkeeping requirement is a structural check, not a stylistic one. Without explicit records, passive observation can be misread as P6, mere packaging as P1 descent, and route failure as P3 mismatch. The regime prevents those misreadings from entering the theorem chain.

3.2 Typed non-collapse

The regime does not include flat primitive independence — the claim that P1, ..., P6 are independent generators of a global algebra. Its meta-theoretic counterpart here is typed non-collapse. Proposition 2.6 excludes each forbidden collapse of Table 1; Proposition 3.3 states the same exclusion as a rule of the admissibility regime of Definition 3.1 and adds that each row's permitted interaction is not thereby excluded. Each row is checked in one direction, for the role it credits (listed in the table after the proof of Proposition 2.6); the row label names the pair without order.

Proposition 3.3 (Typed non-collapse). *Under honest bookkeeping (Definition 3.1) and the alignments of Remark 2.2, consider each forbidden-collapse row of Table 1. Write P_c for the role that row would credit and X_b for the bare partner datum it borrows, as listed in the last column of Table 1 and presented as a boundary record in the sense of Definition 4.2. For every $D \in \text{FATCD}$, no honest audit annotation of D certifies $W_c(C)$ for a cluster C formed only from X_b ; the “permitted interaction” of the same row is not excluded by this proposition. More generally, for each $k \leq 6$, no honest audit annotation of D certifies $W_k(C)$ for a cluster C lacking any record that the W_k row requires a witness cluster to contain (among them, for $k = 1$: a map or partial-map record of declared use update, or a relation record of declared use quotient; $k = 2$: a macro-specification predicate record or a realization relation; $k = 3$: two defined path or derivation records; $k = 4$: a law-bearing stage, order, transition or refinement record; $k = 5$: a relation record of declared use*

quotient, a packaging map record or a check record; $k = 6$: an event record, a trace record or a replay or check record); in particular each row's exclusion holds with its two roles exchanged whenever the exchanged datum lacks that record.

Proof. Fix a forbidden-collapse row and its credited role P_c and borrowed datum X_b . The row records a “forbidden collapse”: an identification asserting that the content of its two roles is the same closure role, so that one projection may be read off the other without further data. To be admissible under Definition 3.1, such an identification would have to enter the theorem chain as a claim with a recorded statement and scope. The role meanings are distinct, and Proposition 2.6 checks, for each forbidden-collapse row, a field required by the credited witness that the named partner datum alone does not supply. An audit certifying the credited projection solely from that datum would therefore certify a conclusion unsupported by its recorded check and would not be honest under Definition 4.6: by that definition the check-language conclusion $W_c(C)$ of an honest audit would be true in D , whereas $W_c(C)$ is false by Proposition 2.6. Independently supplied data may satisfy both witness conditions; the proposition excludes the claimed automatic identification, not that joint occurrence. Hence the collapse is inadmissible. More generally, if C lacks a record required by the W_k row, that row gives $W_k(C) = \text{false}$. Definition 4.6(c3) therefore excludes an honest audit certifying $W_k(C)$; this also proves the stated exchanged-role exclusions.

It remains to check that ruling out the collapse does not also rule out the row's “permitted interaction”. The permitted interaction does not identify P_i with P_j ; it records a legitimate joint occurrence of the two distinct roles in a single description — for instance, P1 descent audited by P6, or P3 route terms indexed by P4 stages, as in Remark 2.7. Such an interaction carries its own honest-bookkeeping record for each role separately and asserts no identification. Nothing in the argument above touches it: the inadmissibility derived for the collapse rests on the unsupported inference from the bare partner datum to the credited witness, which the permitted interaction never makes. Therefore the permitted interaction remains admissible whenever its own bookkeeping conditions are met. $\square$

Remark 3.4 (Typed non-collapse versus global independence). Typed non-collapse is not a global primitive-independence statement. It does not assert that the six roles are logically or categorically independent in a global algebra; it asserts only that the pairwise identifications listed as forbidden collapses in Table 1 are inadmissible under honest bookkeeping, leaving the corresponding typed interactions available.

3.3 Level-profile architecture

Typed non-collapse fixes which identifications are inadmissible; cross-level claims also require recorded levels and translations. The level trichotomy of Definition 2.4 therefore supplies the profile for such claims.

Definition 3.5 (Level profile of an admissible claim). A claim annotation is *cross-level* when records it names are listed in the target fields of level annotations recording different levels; it is a *direct comparison* when its statement contains an equality or order atom, or an implication, whose two sides read records listed under different recorded levels. Its *level profile* is a level annotation targeting the claim, together with, for each such pair of levels, a translation annotation carried in $\text{Audit}(D)$ whose source and target contexts are those levels. Every admissible cross-level claim about a description $D \in \text{FATCD}$ carries a *level profile*: it locates the claim on the behavioral, verification, or structural-categorical level, and each cross-level assertion is accompanied by a translation record in the sense of Definition 3.1. Direct comparison of claims at different levels without a translation record is inadmissible.

3.4 Forgetting and selected lifts

Level transitions are controlled by forgetting and selected lifts.

Definition 3.6 (Admissible forgetting and selected lifts). Forgetting from a higher to a lower level (structural-categorical to verification, verification to behavioral, or structural-categorical directly to behavioral) is admissible if and only if it carries a record of its losses carried in $\text{Audit}(D)$; the loss record may be empty only when losslessness is proved and recorded. Lifting from a lower level to a higher level is admissible only when accompanied by

- the added data required to support the higher-level statement;
- the mark that the lift is selected and non-unique unless uniqueness for its specified source data and target class is proved and recorded;
- explicit preservation of lower-level invariants and explicit losses where the lift is partial.

Lossless forgetting and unique lifts are inadmissible unless the losslessness or uniqueness is itself proved and recorded.

Remark 3.7 (Lower-to-higher non-determinacy). Definition 3.6 formalizes the non-determinacy of lower-level data: given behavioral or verification content alone, the admissibility regime forces no choice of structural-categorical content, and any asserted choice must be marked as a selected lift. Where the lower-level data do determine the higher-level datum, as the descended map of a descending update is determined, that determinacy is admissible with its recorded proof.

3.5 Activation thresholds

The regime must also state when a projected role is active. The attachment protocol of Definition 4.9 is lifted to a regime-level principle.

Remark 3.8 (Activation thresholds). At the regime level, the threshold-attachment discipline of Definition 4.9 reads as follows. An admissible active derived role projection $\pi_i(D)$ exists only when D carries the threshold, witness, level, collapse, nonclaim and audit data required by Definition 4.9. A channel with no admissible projection receives *collapsed_boundary_case* if a failed offer is recorded, and otherwise *below_threshold* if a covered seed is present (Definition 7.1); these are statuses of channels, not of clusters, and a channel with an admissible projection is *active_projection* whatever other clusters do. Arbitrary enrichment, adding structure solely to force a role projection, is inadmissible unless the additions are themselves honestly bookkept as raw features or annotations. This restates the attachment protocol at regime level: the sentences before the enrichment sentence restate Definitions 4.9 and 7.1, and the enrichment sentence follows from Definition 4.7, since every added record of a member of FATCD is subject to Definitions 4.4–4.6.

3.6 Instrument-relative visibility and suppression

Threshold-governed activation still leaves open what an instrument makes visible or suppresses. Admissibility of visibility claims is therefore instrument-relative.

Definition 3.9 (Admissible visibility record). Every admissible visibility claim carries an explicit *visibility record* comprising its visible content, its suppressed content, the instruments under which the visibility and suppression are asserted, and the resulting instrument-relative claim strength. Overreading a suppressed claim as visible under a different instrument is inadmissible. Partial visibility, in which a claim is visible only up to some coarsening or translation, is admitted when the visibility record itself makes the partiality explicit. An *instrument* is a declared record of the

description named in an instrument-visibility annotation's instruments field or as the measuring instrument of an empirical-bridge claim carried in $\text{Audit}(D)$; the *claim strength* is a value of a declared finite ordered set recorded in that annotation; a visibility claim *asserts instrument-free truth* when it asserts visible content without naming an instrument. The *instrument family* of a description is the set of its instrument records.

Remark 3.10 (No instrument-free truth). Definition 3.9 does not claim that instruments are free of distortion or that any single instrument is canonical. It asserts only that admissible visibility claims carry an instrument-relative visibility record. Instrument completeness is deferred to Section 12.

3.7 Empirical-bridge admissibility

Any claim crossing from the closure-algebraic side of the framework to a substrate-level observable must pass through the empirical-bridge discipline, related to but narrower than the empirical-adequacy view of theories [31].

Definition 3.11 (Admissible empirical-bridge claim). An empirical-bridge claim about $D \in \text{FATCD}$ is admissible only when it has, carried in $\text{Audit}(D)$,

- the substrate and observable to which the bridge relates;
- the instrument under which the observable is measured;
- the level map between the closure-algebraic content and the substrate-level content;
- a threshold-witness relation recording when the bridge is crossed;
- the nonclaims suppressed by the bridge record, in particular any structural content of D that is not tracked by the substrate or instrument.

Empirical realization — the claim that D is itself a substrate observation — is stronger than empirical-bridge admissibility and is not asserted by any empirical-bridge claim.

3.8 Nonclaim closure across the admissibility regime

The rules of this section, from honest bookkeeping (Definition 3.1) to admissible empirical-bridge claims (Definition 3.11), keep the global nonclaims of Remark 2.9 in force. The next proposition lists the overclaims they block.

Proposition 3.12 (Admissibility-regime nonclaim closure). *Under the admissibility regime, each of the following claims is inadmissible; the second kind of claim in the first item cannot occur, since no member of FATCD carries it:*

- a claim identifying two of the six roles along a forbidden collapse of Table 1, or a claim annotation whose statement applies a recorded finite map with domain the role names $P_1, \dots, P_6$ (or tuples of them), or an independence predicate on them, whose referenced records are only boundary and witness records, and whose statement is not represented by their fields in the sense of Definition 4.2 (by Definition 4.6(e));
- an unconditional empirical-realization claim, that D is itself a substrate observation, where D records no extension of that predicate (criterion after Definition 4.14), whether or not an empirical-bridge record is present;
- a lift asserted to be unique without a recorded proof of uniqueness;
- a claim that lower-level content determines higher-level content, without a recorded proof of that determinacy;

- *a visibility claim asserting instrument-free truth;*
- *a direct cross-level comparison without a translation record;*
- *a forgetting asserted to be lossless without a recorded proof of losslessness;*
- *an empirical-bridge claim lacking any of: a threshold-witness relation, a level map, the instrument under which the observable is measured, or suppressed-structure nonclaims.*

Proof. Each requirement of the regime blocks a specific class of overclaims; we check that it blocks the listed claims.

Honest bookkeeping (Definition 3.1) declares inadmissible every unrecorded claim, implicit promotion, silent translation, asserted-but-unselected lift, lossless forgetting without proof, visibility overread, and bare empirical-realization assertion; these are requirements of Definition 3.1, beyond the description-level audit condition of Definition 4.6.

Typed non-collapse (Proposition 3.3) blocks the forbidden identifications of Table 1. A claim annotation asserting a full algebra (a map applied to role names) or a global independence theorem (an independence predicate on role names) must represent that substantive assertion in fields of its referenced records (Definition 4.2). Boundary and W_i -witness records carry exactly the fields of their declared checks, and those checks supply neither a map on role names nor a global-independence certificate. An assertion not represented by the fields of the cited boundary and witness records requires an additional declared content record. A claim citing only those records and failing the representation condition cannot be carried by a member of FATCD. This does not prove that no full algebra or global independence theorem can exist. Both broader statements remain nonclaims of Remark 2.9.

Level-profile architecture (Definition 3.5) blocks direct cross-level comparison without a translation record, which is exactly the non-claim that cross-level comparison remains inadmissible without translation.

Admissible forgetting and selected lifts (Definition 3.6) block lossless forgetting, unique lifts, and lower-to-higher determinacy: forgetting is lossy unless losslessness is proved and recorded, no lift is admitted as unique unless uniqueness is proved and recorded, and, by Remark 3.7, no structural-categorical content is asserted to be forced by behavioral or verification content without a recorded proof. These are the non-claims that forgetting remains lossy, that no unique-lift theorem is claimed, and that no lower-to-higher determinacy is claimed.

Activation thresholds (Remark 3.8) block arbitrary enrichment and threshold-free nontriviality claims, since an active projection is admissible only with the threshold, witness, level, collapse, nonclaim and audit data of Definition 4.9.

Instrument-relative visibility (Definition 3.9) blocks instrument-free visibility claims and visible-to-suppressed overreads; it does not impose instrument indexing on every claim kind.

Empirical-bridge admissibility (Definition 3.11) blocks empirical-realization overclaims and any bridge claim lacking a threshold-witness relation, a level map, the instrument under which the observable is measured, or suppressed-structure nonclaims. An empirical-realization claim asserts a predicate, being a substrate observation, whose extension D does not record; its truth does not follow from any finite check of D , so by Definition 4.6 no honest audit certifies it outright, and it is inadmissible whether or not a bridge record is present. An unrecorded realization predicate may occur as a hypothesis of an audited conditional consequence; no audit certifies realization outright or as a consequent. This yields the non-claim that no empirical realization is claimed and that empirical-bridge claims carry the required data.

Collecting these residues gives each of the listed non-claims. Those that restate a global nonclaim of Remark 2.9 are thereby carried forward into the regime; the two further items — that

forgetting remains lossy absent a recorded proof of losslessness, and that empirical-bridge claims carry their threshold-witness relation, level map, instrument and suppressed-structure nonclaims — are additional disciplines the regime imposes at this level. $\square$

4 The FATCD domain and the exact-six question

This section defines the class of descriptions over which the exact-six theorem is stated. A *finite audited typed closure description*, abbreviated FATCD, is a finitely presented collection of typed raw records and annotations that is finite, closed under its own references, and audited in the sense of Definition 4.6; the setting is that of finite model theory [19]. Roles are not part of the definition. A role enters only when a threshold annotation is attached to a cluster of raw features, and the role projections $\pi_1(D), \dots, \pi_6(D)$ are readings of the raw material, not conditions for membership in the domain. Every content record and structural annotation is a residual feature in the sense of Definition 4.14 and is licensed some of the eleven labels of Definition 4.14; the unresolved label (x) is licensed exactly when none of labels (i)–(ix) is; the features that no channel of a chosen decomposition record accounts for are its record residuals. So a seventh role is not ruled out by definition. The section closes with non-circularity and non-overbreadth schemas and formulates the exact-six question precisely over FATCD: whether every $D \in \text{FATCD}$ admits a decomposition/exhaustion record with exactly six role channels relative to which no kind residual and no record residual is licensed label (x) or (xi) of Definition 4.14. A six-channel record always exists (Theorem 7.3); the question is decided by the residual condition. The theorem chain answering that question appears in Sections 5–7.

The domain is framed in the finite theory-package setting of Foundations I [28], where one works with tuples $\mathcal{T} = (Z, f, \Sigma_f, E, \mathcal{A})$ carrying a lens, a completion operator, and an audit functional. FATCD does not identify a description with such a tuple; it extracts from that setting the structural requirements relevant to the scoped exact-six question: finite typed raw data, closure under references, sources, targets, endpoints, and annotations, and the audit condition of Definition 4.6, on which the regime of Section 3 builds. Optimization, objective, and richer semantic material are not excluded at the outset; they belong in the domain when finitely represented and honestly recorded as raw or annotated data.

4.1 Raw grammar

We begin with the neutral typed raw grammar from which every element of FATCD is built. The alphabet is fixed in advance of the role vocabulary and follows standard typed-record conventions [8].

How to read this section. A description is a finite collection of records. Each record is either a raw record of one of the kinds of Definition 4.1 or an annotation of one of the six structural or seven bookkeeping kinds of Definition 4.2; it carries finitely many typed fields and may refer to other records of the same description. The conditions of Section 4.2 then require three things: everything is finite (Definition 4.4); every reference, endpoint and annotation target is a record of the description, and every claim it carries has a bookkeeping record (Definition 4.5); and those bookkeeping records are complete (Definition 4.6). A *claim* is a claim annotation (Definition 4.6(a)), an assertion the description makes about its own records, such as that a cluster is read as a role; a field with field role claim on another record is not a claim. Its bookkeeping record is an audit annotation in $\text{Audit}(D)$ that refers to the claim, lists the certificates it uses and records its conclusion (Definition 4.6(a)); the claim annotation records the statement, hypotheses and scope. For a claim admissible under Definition 3.1, the promotions, translations, lifts, forgetting maps, visibility and empirical bridges it

uses have further annotations carried in $\text{Audit}(D)$. Appendix B describes the typed Lean mirror of these definitions.

In set terms, a description can be written as a finite set R of records with a kind map from R to the kinds of Definitions 4.1 and 4.2, finitely many typed fields on each record, a reference relation on R (an annotation refers to its target), a finite set of claims, and a map sending each claim to its bookkeeping record in $\text{Audit}(D) \subseteq R$. Definition 4.4 makes all of these finite, Definition 4.5 requires every reference to land in R and every claim to have a bookkeeping record, and Definition 4.6 lists the entries each bookkeeping record must carry. Here R is $\text{Raw}(D)$. A *typed field* is a triple (n, ρ, v) : a name n , a field role ρ (datum, reference, predicate, comparison, invariant, law or claim; Definition 4.1), and a value v of the type fixed by the record kind and ρ . For a datum field this is a rational, an element of a declared finite set, or a finite list or table of such values and record identifiers, such as the graph of a map record; for a reference field an identifier or a finite list of identifiers (the member list of an object record, the candidate class of a realization relation, the entry list of a trace, the list of successive records and the word of a path or derivation record, and every declared domain, codomain, source, target, endpoint and index field are reference fields); for a term- or formula-valued field a check- or assertion-language expression (a term-valued field, such as the row $v(w_1), \dots, v(w_n)$ of the transition record of Remark 6.8, has field role datum; a formula-valued field has an assertion role); and for a certificate field a triple $(\varphi, \text{inputs}, b)$. A field may carry a value-type tag. Deleting a field removes one triple; changing its value replaces v by another value of the same type. A *declared record* of D is an element of R , and a *raw feature* is any declared record. A *content record* is a declared record of a kind of Definition 4.1, and an *annotation* one of a kind of Definition 4.2. Thus $\text{Raw}(D)$ contains annotations, while “raw record” means a content record.

Definition 4.1 (Raw record). A *raw record* is a finite typed syntactic or relational feature of a description. The admissible record kinds are

- *object* and *state* records;
- *relation* records, *map* records, and *partial-map* records;
- *predicate* records and *comparison* records;
- *path* or *derivation* records;
- *index*, *order*, *transition*, and *refinement* records; a *stage* is an index record together with an order record ranging over it; in the W_4 row and seed clause (4) of Definition 7.1 a stage counts through its order record;
- *trace*, *event*, and *provenance* records;
- *replay* and *check* records.

The rest of the definition fixes, in order: declared uses of map and relation records (including realization relations), the sort declarations of predicate records, rule fields, field roles, and value-type tags. A map or partial-map record declares its use in a typed field: *update*, for a map from a declared state set to itself, *packaging*, for a map to a declared quotient object (surjectivity is certified by the W_5 check, not assumed), or *other*; a relation record likewise declares its use as *quotient*, for an equivalence relation presented as the classes of a quotient, or *other*. The graph of a map or transition record lists exactly one value at each member of its declared domain; only a partial-map record may leave members of its declared domain without a value. The listed classes of a relation record of declared use quotient are nonempty and pairwise disjoint, so its relation atom is an equivalence relation on the members of its classes. These conditions are part of the record’s type, so the recorded values of D form a well-typed interpretation in the sense of Definition 4.6(c1). A relation record may

instead declare its use as *realization*: it declares a finite nonempty candidate class K of declared records and, as its target, a predicate record s declared as a macro-specification whose formula has exactly one free variable, which occurs in the formula and ranges over records of the kind of the members of K , and whose formula is a formula of the check language, so that it has a truth value over D at every member of K ; the relation lists exactly those $r \in K$ at which that formula evaluates to true in D (a condition on the recorded values of D , required for membership in FATCD; unlike the graph and class conditions above, it is not imposed on other well-typed interpretations of Definition 4.6(c1)); it may also record the value of that formula at further declared records of the same kind. A macro-specification's formula may be closed; the one-free-variable condition is imposed only by realization relations, seed clause (2) and the W_2 row. A predicate record declares in a typed field whether it is a *macro-specification*. It also declares, in a datum field, for each free variable of its formula and, for a predicate record without a formula field, for each argument place of its declared arity, the record kind over whose declared records that variable or argument ranges, the rational sort when it is applied to rational-valued terms, or a sort declared by an object record listing no members; for a macro-specification targeted by a realization relation this is the common kind of the members of K , which must all be of one kind. A path, derivation or comparison record may carry *rule fields*: datum fields whose declared type, fixed by the record kind, is a pair (λ, μ) of finite lists of identifiers of map or partial-map records; the rewrite rules recorded in D are the values of all rule fields of records of D . Each typed field of a record declares one of the field roles *datum*, *reference*, *predicate*, *comparison*, *invariant*, *law* or *claim*; a field recording a checked equality, isomorphism or level translation is a comparison field. A field whose value is a proposition about recorded values (a formula, constraint or asserted property) declares one of the assertion roles predicate, comparison, invariant, law or claim; datum and reference fields record values and identities only. “Predicate field”, “comparison field” and the like refer to these declared roles. A field holding a certificate has an assertion role: law when it is the law field of an order, transition or refinement record, or of a relation record carrying an order in the sense of the paragraph *Laws* (Section 2), and comparison otherwise; the certificate of an equality, isomorphism or level translation is part of the comparison field it checks. A typed field may also carry a declared value-type tag from a finite list recorded in D (for instance probability, weight, kernel, confidence, error, intervention, counterfactual, causal dependence, goal, objective, plan, policy or choice); tags make no assertion and enter no witness or residual-label criterion; Section 6 uses them to delimit its threat classes. A raw record is not, by itself, a P1–P6 role.

Definition 4.2 (Annotations). In addition to raw records, a description admits *annotations* of six declared structural kinds:

- level annotations Ann_L ;
- threshold annotations Ann_Θ ;
- lift annotations Ann_{Lift} ;
- forgetting annotations $\text{Ann}_{\text{Forget}}$;
- instrument-visibility annotations Ann_{Vis} ;
- empirical-bridge annotations $\text{Ann}_{\text{Bridge}}$.

A description also admits *bookkeeping annotations* of kinds audit, claim, witness-schema, collapse-case, promotion, translation and nonclaim. A *promotion* annotation records a claim's move from a lower to a higher level of Definition 3.5, with the fields of Definition 4.6(g); the selected higher-level datum the move adds is recorded by a lift annotation. They belong to $\text{Raw}(D)$, have finite typed fields and declared targets, and are subject to the same reference-closure condition as the other

annotations. $\text{Audit}(D)$ is the collection of its honest audit annotations (as in Definition 4.6(e)); a record is *carried in* $\text{Audit}(D)$ when it is such an annotation or a claim, promotion, translation, collapse-case, witness-schema, nonclaim, lift, forgetting, visibility or bridge annotation referenced by one.

Claim annotations and the representation condition. A claim annotation records a statement, its referenced records, its hypotheses and its scope; the scope field is a reference field listing the records and clusters the claim concerns, and makes no assertion. It may additionally carry a *witness-cluster field*: a typed reference field listing exactly the members of a cluster proposed for a named role. In every construction in this paper, a claim described as naming P_i and a cluster C for a witness check or derived projection lists C in this field. Merely naming a content record does not supply a witness-cluster field. A claim annotation is bookkeeping about the declared content records it names. It satisfies the *representation condition* when every substantive assertion in its statement, that is, any predicate, comparison, invariant or law content of the statement other than naming its referenced records, hypotheses and scope, is represented by fields of those records, and an assertion introducing independent content requires an additional declared content record. The audit identifies the fields and checks the asserted consequence. Precisely, an assertion of the statement is *represented by fields* of the named records when it is the formula of one of their predicate, comparison, invariant, law or certificate fields, a conjunction of such formulas, or a formula logically entailed by a specified finite list of signed certificate formulas carried in those fields (true in every well-typed interpretation, under every assignment of truth values to the atomic W_j formulas, in which those signed formulas are true; this is the entailment of Definition 4.6(c2) with the W_j atoms uninterpreted). For this representation test, (φ, true) supplies φ and (φ, false) supplies $\neg\varphi$, and atomic W_j formulas are treated as propositional atoms. Certificate correctness, completeness and audit honesty are checked separately under Definition 4.6. The statement $W_i(C)$ or $\neg W_i(C)$ of an offer, and the statement $W_i(C)$ of a role claim, is represented by the members of C listed in its witness-cluster field, the $\text{Audit}(D)$ conjunct of the W_6 row being handled as in the next sentence. For a $W_6(C)$ claim, its event, trace and check content is represented by the named content records; its separate $\text{Audit}(D)$ -entry conjunct is evidenced by the honest event-or-trace audit entry required by the W_6 row.

Citation roles. Two citation roles, (1) and (2), are fixed as follows; (3) is a condition common to both.

- (1) A W_i -witness record is a content record of a cluster offered or used as a P_i witness that meets condition (3).
- (2) A *boundary record* is a record meeting condition (3) of a cluster on which a row of the boundary matrix (Table 1) is checked: a content record or, for the rows whose bare datum is a P6 audit record, that audit annotation; a bare datum consisting of several records is presented as the set of their boundary records.
- (3) A W_i -witness record carries exactly the fields specified by its declared witness check together with the datum and reference fields that check uses and reference fields to other members of the same cluster; a boundary record carries exactly the fields constituting the bare datum of its row of Table 1, the typing fields of its kind (kind and declared use), the datum and reference fields its certificates read, and reference fields to other members of the same cluster. A record carrying further fields is not a W_i -witness or boundary record; this is a condition on which records a claim may cite as such, not a membership condition of FATCD.

Independent operations on the six primitives, axioms for those operations, and global-independence certificates require separate content records.

Witness-schema annotations. When a witness-schema annotation names P_i , it carries a target field listing the cluster to be checked. A witness-schema annotation either names P_i and the corresponding row of the *Witness types* table of Section 2, or names a candidate role outside $P_1, \dots, P_6$ and a check-language formula schema $\psi(C)$ evaluated on declared clusters. A *formula schema* is a check-language formula without atomic W_j formulas, with a cluster parameter C occurring only in finite disjunctions $\bigvee_{d \in C} \chi(d)$ and conjunctions $\bigwedge_{d \in C} \chi(d)$, d record-sorted; its *instance* $\psi(C')$ at a finite set C' of declared identifiers expands each over the members of C' (the empty disjunction being false and the empty conjunction true). We write $t \in C$ for $\bigvee_{d \in C} t = d$. The latter annotation supplies no W_i witness or derived projection. An assertion that $W_i(C)$ holds additionally requires the finite evidence specified in its table row. These annotation kinds belong to the raw grammar; the finiteness and closure conditions on how they attach to declared records are imposed in the next subsection. For a description D , we write $\text{Raw}(D)$ for the finite collection of raw records and annotations of D .

Remark 4.3 (Independence of the feature alphabet). The record and annotation kinds in Definitions 4.1 and 4.2 are syntactic or relational entities. No kind is named $P_1, \dots, P_6$. Consequently, membership in FATCD does not require any derived role projection, does not require an exact-six decomposition/exhaustion record, and does not exclude candidate seventh-role features by definition. An optional witness-schema annotation may name a role, but no content kind is itself a role, and membership requires no witness schema or derived projection.

4.2 Finiteness, closure, and audit conditions

Three further restrictions turn the raw grammar into a domain of admissible descriptions: finiteness, closure, and audit.

Definition 4.4 (Finite description). A description D is *finite* when

- $\text{Raw}(D)$ contains finitely many records and annotations;
- every record or annotation carries finite typed fields;
- every relation, map, path, trace, or annotation is finitely represented;
- every reference in a record points to a declared record of D ;
- in particular, semantic, probabilistic, topological, geometric, causal, optimization, or resource structure enters D only through such finitely represented records, by the preceding conditions.

Definition 4.5 (Closed description). A finite description D satisfies the *closure condition* $\text{Closed}(D)$ when

- every source and target of every relation or map record is a declared record of D ;
- every endpoint of every path or derivation record is a declared record of D ;
- every index, transition, order, or refinement reference is a declared record of D ;
- every trace, event, provenance, replay, or check record references declared records of D ;
- every level, threshold, lift, forgetting, visibility, or empirical-bridge annotation references declared records of D ;
- every claim annotation carried by D is referred to by an audit annotation of D (its bookkeeping record).

Definition 4.6 (Audited description). Summary, made precise in (a)–(g) below. An audit is honest exactly when its references resolve (b1); its certificates recompute correctly (b2); it lists every check its claim requires (b3); predicates with unrecorded extensions occur in its conclusion only negatively (b4); its conclusion follows from the signed certificates in every well-typed interpretation (c1)–(c2); and unverified hypotheses occur only as antecedents of a conditional conclusion, every hypothesis used in an outright conclusion being recorded and checked true, and the conclusion uses no field it lists as non-certified (c4). These conditions are evaluated in the order fixed by (f). Consequently (c3), every conclusion, outright or conditional, using only recorded predicates and carriers is true in D .

(a) *Claims and audits.* A *claim* is a claim annotation with a statement, referenced records, hypotheses and scope. An audit annotation for a claim records references to the claim and to the records it names, a finite list of certificates (Section 2) with their recomputed truth values, and a conclusion field holding a closed formula of the assertion language; it *certifies* exactly the formula in its conclusion field.

(b) *Honesty.* It is *honest* exactly when (b1)–(b4) below, the entailment requirement (c2) and the requirements of (c4) all hold, evaluated in the order fixed by (f); (c1) defines the interpretations used in (c2), and (c3) is a consequence: (b1) its references resolve in D ; (b2) every listed certificate is correct; (b3) the list contains every check the claim requires, namely: (b3-i) for a claim naming P_i and listing a cluster C in a witness-cluster field (the claim of an offer in the sense of (d)), every certificate that the W_i row prescribes and a member of C carries (for W_2 , including every recorded evaluation certificate $(s(t), b)$), each listed as (φ, b') with b' its recomputed truth value, together with the certificate $(W_i(C), b)$ recording the result, or, when C lacks a record that every alternative of the row requires, that certificate alone, with $b = \text{false}$; (b3-ii) for $W_6(C)$, the content checks of the W_6 row; (b3-iii) for an assertion represented by fields of content records, the certificates in those fields; and (b4) the conclusion field contains a closed formula of the assertion language in which every atom whose predicate symbol has an unrecorded extension occurs only negatively (within an odd number of implication antecedents and negations). Here D records the extension of a declared predicate p when at least one complete presentation of p is supplied, either by a relation record that references the declaring record of p in a field declaring it the extension of p , its listed tuples being by definition the true tuples of p , provided the argument sorts of p have recorded finite carriers, or by its declaring predicate record giving an explicit formula of the fixed check language over recorded inputs and declared finite ranges (arguments of rational sort being then admitted and evaluated by that formula), all such presentations agreeing on every tuple; the criterion is repeated, with the definition of recorded carriers, in the paragraph *Terms used in the label criteria* after Definition 4.14.

(c) *Premises.* (c1) *Interpretations.* An *interpretation* (or *well-typed interpretation*) of D does three things. First, it keeps the records of D , their kinds, declared uses, the markers declaring a transition codomain distribution-valued, field roles and references; so reference-field carrier lists and declared map and transition domains are fixed across interpretations, while quantifier bounds supplied by datum fields may vary. Second, it keeps every formula-valued field (predicate, comparison, invariant, law, claim, certificate and conclusion fields) and every term-valued field syntactically fixed. A term-valued field is a field explicitly recorded as a symbolic check-language term or a finite list of such terms, including a transition row recorded as $v(w_1), \dots, v(w_n)$; these expressions are evaluated using the interpretation's datum values, componentwise for lists; ordinary numeric datum fields are not term-valued merely because their values are rational constants. Third, it assigns each remaining datum field a value of its declared type: the type of a map, partial-map or transition graph, or of a table field, fixes the sorts of its values but not their membership in a declared codomain or list, and a row into $\text{Dist}(S)$ has exactly one coordinate for each member of the fixed reference-field list S , with arbitrary rational entries. Unrecorded predicate extensions and

unlisted-sort carriers vary arbitrarily, and $W_j(C)$ is evaluated by its row, the $\text{Audit}(D)$ conjunct of the W_6 row ranging over $\text{Audit}_0(D)$ of D itself. (c2) *Entailment*. For a correct certificate (φ, b) , an audit may use φ as a premise if b is true and $\neg\varphi$ if b is false. Honesty requires the signed premises to entail the conclusion in every interpretation; the entailment requirement is therefore a validity condition over these interpretations; certificate correctness is checked separately by recomputation. When an audit conclusion is one of its signed premises, entailment holds by syntactic identity, but reference closure, certificate correctness, completeness and the remaining honesty requirements must also hold. (c3) *Truth of conclusions*. In particular, every conclusion of an honest audit, outright or conditional, that uses no unrecorded predicate symbol and quantifies only over recorded carriers is true in D : the recorded datum values of D , with any choice of unrecorded extensions and carriers, form an interpretation in which every correct signed premise is true, so the conclusion is true there, and its value does not depend on extensions it does not use. (c4) *Hypotheses and non-certification*. An unverified stated hypothesis may occur only as an antecedent of a conditional conclusion; an outright conclusion requires every hypothesis used to be recorded and checked true. A failed check may thus be honestly recorded without supporting a positive witness claim. An audit annotation may carry a *non-certification field*, a reference field listing fields of the records it names whose asserted property it does not certify; it is honest only if its conclusion does not use those fields.

(d) *Offers*. In particular, an *offer* of a cluster C as a P_i witness is a claim annotation naming P_i and listing C in a witness-cluster field, together with an honest audit annotation recording the check of $W_i(C)$ and its result; an offer requires no threshold, level or collapse-case annotation; its statement is $W_i(C)$ when the recorded result is true and $\neg W_i(C)$ when it is false, and its audit concludes that statement from the certificate $(W_i(C), b)$. A witness-schema annotation records the predicate to be checked, not a claim that every referenced cluster satisfies it.

(e) *Audited descriptions*. A description D is *audited* when every audit annotation it carries is honest and every claim it carries is referred to by at least one such annotation and satisfies the representation condition of Definition 4.2; the honest audit annotations form $\text{Audit}(D) \subseteq \text{Raw}(D)$.

(f) *Order of evaluation*. Honesty is determined in two stages. Let $\text{Audit}_0(D)$ be the honest audit annotations whose certificates and conclusion contain no atomic W_j formula; their honesty does not depend on any W_j . The $\text{Audit}(D)$ conjunct of the W_6 row ranges over $\text{Audit}_0(D)$, so an audit of $W_6(C)$ cannot itself discharge it. Every atom $W_j(C')$ is then determined, since the rows of $W_1, \dots, W_5$ refer to no audit annotation and so do not depend on which audits are honest, and W_6 depends on audits only through $\text{Audit}_0(D)$; this determines the honesty of the remaining audit annotations.

(g) *Required fields*. Promotions, translations, lifts, forgetting, visibility and bridge assertions must also carry the following fields, with their stated losses and limits preserved:

- every promotion records its source level, target level, added data, losses, and nonclaims;
- every translation records its source context, target context, preserved data, suppressed data, and claim strength;
- every lift is marked as selected and non-unique unless uniqueness for its specified source data and target class is proved and recorded;
- every forgetting map records its losses;
- every visibility claim records both visible and suppressed data;
- every empirical-bridge claim records substrate, observable, instrument, level map, threshold witness relation, and suppressed-structure nonclaims.

4.3 Finite audited typed closure descriptions

Combining the grammar with the three admissibility conditions yields the domain over which the exact-six question is posed.

Definition 4.7 (Finite audited typed closure description). A *finite audited typed closure description* is a finitely presented, closed, audited collection of typed raw records and annotations: every claim it carries is referred to by an honest audit annotation, and its promotions, translations, lifts, forgetting, visibility and empirical-bridge assertions carry the fields listed in Definition 4.6. Membership does not require every claim to be certified: a member may carry a claim whose honest audit does not certify its statement, making that claim inadmissible under Definition 3.1. Precisely, it is a description D whose raw content $\text{Raw}(D)$ is drawn from the kinds declared in Definitions 4.1 and 4.2, such that D is finite in the sense of Definition 4.4, satisfies the closure condition $\text{Closed}(D)$ of Definition 4.5, is audited in the sense of Definition 4.6, and each realization relation of D lists exactly the members of its candidate class at which its target's formula is true in D (Definition 4.1). We write **FATCD** for the class of all such descriptions, and $D \in \text{FATCD}$ for membership.

Remark 4.8 (No minimum richness). **FATCD** imposes no lower bound on description size. A description D_0 consisting of one declared object record and no claims is finite, closed and audited. Its record has no witness, predicate, comparison, invariant, law or claim field, so its record satisfies (R3) of Definition 6.7 and D_0 satisfies the recognition hypothesis. Richer descriptions carrying transitions, paths, derivations or annotations of every kind may also belong to **FATCD**; whether they satisfy the recognition hypothesis is checked record by record. The domain is defined by structural discipline, not by the presence of any particular feature kind. A second small member, used in Proposition 4.16, is the following description D_{norm} , analysed in Remark 6.8: state records $w_1, \dots, w_n$ ($n \geq 1$) with nonnegative rational value fields $v(w_k)$ summing to one (for instance $n = 2$ and $v(w_1) = v(w_2) = 1/2$), a predicate record N (unrelated to the monoid N of route certificate (a)) declared a macro-specification whose formula is $\bigwedge_k v(w_k) \geq 0 \wedge \sum_k v(w_k) = 1$, a claim annotation naming N whose statement is the formula of N , and an honest audit annotation listing the certificate $(\bigwedge_k v(w_k) \geq 0 \wedge \sum_k v(w_k) = 1, \text{true})$ and certifying that formula.

4.4 Threshold attachment

Membership in **FATCD** is fixed; the next step is to specify how role-theoretic claims may attach to the raw data of an admissible description. Role projections in **FATCD** enter only through a disciplined attachment of threshold and witness data to raw features. The condition below is the structural precondition for the role vocabulary introduced in the next subsection.

Definition 4.9 (Threshold attachment). Let $C \subseteq \text{Raw}(D)$ be a finite cluster of raw features of D , possibly a single feature; attachment of an annotation to C is defined below. A derived role projection on C is admissible only when all of the following are present in $\text{Raw}(D)$:

- the raw features of C themselves;
- a threshold annotation Ann_Θ attached to C ;
- a witness schema, in the sense of Definition 4.11, for the attempted projection, recorded by a witness-schema annotation whose target field lists exactly the members of C ;
- a collapse-case annotation targeting exactly C , a record of the collapse cases that would invalidate the witness; like a threshold label, its content is recorded and is not checked by the results of this paper;
- a level annotation Ann_L attached to C ;

- a nonclaim annotation targeting exactly C ;
- an audit record in $\text{Audit}(D)$ honestly bookkeeping a claim annotation that names the attempted role and lists C in its witness-cluster field.

These are necessary conditions; the complete condition is the admissibility clause of Definition 4.11. An annotation is *attached to C* when its declared target field lists exactly the members of C . A threshold annotation may carry a *threshold predicate* on recorded fields of C ; when it does, the projection on C is attempted only if $\text{Audit}(D)$ records that predicate checked true, that is, only if some audit annotation in $\text{Audit}(D)$ lists a correct certificate (θ, true) whose formula is the threshold predicate θ , a closed check-language formula. A threshold annotation may also carry a label field naming the attempted reading, and a level annotation carries a level field. Label and level fields are operation data in the sense of (R2) of Definition 6.7 and of criterion (ix-b), and make no assertion. A threshold predicate is operation data under (R2) and (ix-b) only when it reads the value of at least one record, every record whose values it reads is a member of C , and each such record satisfies (R1); otherwise it is an independent assertion.

Remark 4.10 (Pre-projection discipline). The items of Definition 4.9 form a structural precondition, not a process narrative. What matters is that each item is present explicitly in $\text{Raw}(D)$ before a role projection is asserted, so that no threshold, witness, collapse case, or level is supplied tacitly by the act of projection itself. The lower-bound examples use threshold annotations as recorded tags; Definition 4.9 does not require a numerical cut-off. By the ordered rule of Definition 7.1, *below_threshold* means that a covered seed is present but neither an admissible active projection nor a failed offer takes precedence. The missing data may be witness fields or attachments.

4.5 Derived role projections

The role vocabulary P1–P6 now enters FATCD, not as a constraint on membership but as a derived interpretation of raw features conditioned on the attachment data of §4.4.

Definition 4.11 (Derived role projection). A *derived role projection* for a description $D \in \text{FATCD}$ at role index $i \in \{1, \dots, 6\}$ is a tuple $\pi_i(D, C)$, written $\pi_i(D)$ when the cluster C is fixed by context (a description may carry admissible projections on several clusters for the same i), specified as follows, interpreting raw features of D as the role P_i , supported by threshold data, witness schema, level assignment, and honest bookkeeping. Concretely, $\pi_i(D)$ is specified by

- the proposed role name, one of P1, P2, P3, P4, P5, P6;
- a raw feature cluster $C \subseteq \text{Raw}(D)$ on which the role is interpreted;
- the threshold data Θ_i attached to C in the sense of Definition 4.9;
- a witness schema W_i : the predicate on pairs (D, C) of a finite description and a finite $C \subseteq \text{Raw}(D)$, not necessarily with $D \in \text{FATCD}$, given by the P_i row of the *Witness types* table in Section 2 (depending on annotations only through the $\text{Audit}(D)$ conjunct of the W_6 row, which is evaluated over $\text{Audit}_0(D)$, not over C), fixed by a witness-schema annotation $w_i \in \text{Raw}(D)$ naming P_i and that row, which the tuple also records; we write $W_i(C)$ when D is fixed;
- a level assignment Level_i ;
- an evidence or certificate type;
- a record of collapse cases avoided or accepted;
- loss and lift annotations for any level crossings involved;
- the explicit nonclaims preserved by the projection.

The supporting data for $\pi_i(D)$ are carried by $\text{Raw}(D)$ and certified by $\text{Audit}(D)$ in the sense of Definition 4.6. Where the cluster matters we write $\pi_i(D, C)$. The projection $\pi_i(D, C)$ makes a *level crossing* when the level recorded in its level annotation differs from the level recorded in some other level annotation whose target field lists a member of C . The projection $\pi_i(D, C)$ is *admissible* if and only if $W_i(C)$ holds with its check recorded by an honest audit annotation; the threshold, witness-schema, collapse-case and level annotations target exactly C ; every threshold predicate θ they carry is recorded in $\text{Audit}(D)$ as checked true in the sense of Definition 4.9; a claim naming P_i and C is admissible in the sense of Definition 3.1, some honest audit annotation referring to it and certifying $W_i(C)$; a nonclaim annotation targets exactly C ; if the projection makes a level crossing, it carries the translation and the lift or forgetting annotation required for that crossing, with the audited losses or selected-lift data required by Definitions 3.5 and 3.6; and every lift or forgetting annotation whose targets meet C satisfies Definition 3.6. The evidence or certificate is then an applicable alternative of the W_i row, since $W_i(C)$ holds, and no lift or forgetting annotation is required when no crossing occurs. An admissible derived role projection is called *active*; a channel is active when its status is *active_projection* (Definition 7.1).

Remark 4.12 (Fixed role meanings). The role names P1–P6 are defined by their meanings in Section 2: P1 concerns update-vs-quotient compatibility and descent obstruction; P2 concerns macro-specification representability; P3 concerns enriched route and coherence mismatch, not a directionality certificate; P4 concerns stage, order, transition, and refinement structure; P5 concerns quotient packaging; and P6 concerns audit, bookkeeping, provenance, replay, and checkability. The witness schema W_i in Definition 4.11 must be aligned with these meanings; a cluster whose raw features do not support the meaning of the proposed role contributes no $\pi_i(D)$.

What a witness type is. Proposition 6.1 lists, for each role, the data a candidate cluster must carry (tabulated under *Witness types* in Section 2): for P1, for instance, a map or partial-map record of declared use update, source and target relation records of declared use quotient, and a certificate of alternative (a) or (b) of the W_1 row. A cluster or feature *supplies the witness type* of P_i when it carries the data listed there for P_i . The witness schema W_i , recorded in $\text{Raw}(D)$ under Definition 4.9, specifies what in the candidate cluster would support the projection.

Remark 4.13 (Projection and membership are independent). Whether a description D supports a given derived role projection $\pi_i(D)$ is not part of the admissibility conditions of Definitions 4.4–4.6. A description $D \in \text{FATCD}$ may carry no derived role projection, projections for some roles, or projections for all six; its membership in FATCD is unaffected. The decomposition/exhaustion record of Section 7 accordingly assigns a status to every role channel of D , only some of which need be active.

4.6 Residual classification scheme

Some raw features remain role-theoretically unsettled at the raw-grammar level. FATCD provides an explicit scheme for classifying them.

The scheme is stated in four pieces: the notions it borrows from later sections (next paragraph); the list of labels (Definition 4.14); the criterion for each label (the table after it, part of the definition); and the terms those criteria use (the paragraph after the table). Section 6 restates the criteria in record-independent form as (R1)–(R5).

Forward notions. The label criteria below use six notions defined later; we state them here, and the cited definitions are the official source. A cluster is *aligned* with P_i when W_i holds of it, and the

required fields are defined after the label table. A *covered seed* for P_i (Definition 7.1, p. 77, which also defines failed offers and outside-scope declarations) is a record of one of the kinds (1)–(6) of that definition, with the named seed fields listed there. Briefly: (1) an update map with its source relation; (2) a one-free-variable macro-specification taking both truth values, with a realization relation; (3) a comparison certificate for two routes with the same endpoints; (4) a law-bearing stage, order, transition or refinement record; (5) a quotient relation or packaging map; (6) an event, provenance or trace record, or a replay or check record certifying a property of its trace. $A(\text{Dec}(D))$ and $S(\text{Dec}(D))$ are the records of the active projections and of the inactive-status entries of a decomposition record (Definition 7.2), fixed before its residual features are labelled. An $(R4)$ *chain* from r is a finite sequence of distinct declared records $r = r_0, \dots, r_n, n \geq 1$, in which, for each $j < n$, r_j records a checked equality or isomorphism of presentation data (a bijection between specified datum fields that are not term-valued, or member lists of object records, of r_j and r_{j+1} , with a certificate that each value equals its image; a level-translation step carries such a bijection and certificate as well; no structural isomorphism is meant) with, or a checked level translation to, r_{j+1} and, apart from one comparison field recording the equality, isomorphism or level translation with r_{j+1} , and the datum and reference fields it compares, has no predicate, comparison, invariant, law or claim field (Definition 6.7). Two further notions come from Definition 6.7. A record satisfies $(R1)$ when it contributes a required field to a cluster aligned with some P_i or carries a covered seed, and each of its predicate, comparison, invariant, law or claim fields is required for, or an input to the check of, W_{i_φ} at some cluster $C_\varphi \ni r$ aligned with P_{i_φ} , with the index and cluster chosen separately for each field φ , or is a named field of a seed clause it carries (so r carries no assertion beyond these fields; content beyond them must be recorded in a separate declared content record, which is classified on its own); or when it is a member of the cluster of an honestly recorded failed offer and each such field is an input to the recorded check. The *exempt fields of $(R2)$* are the statement and hypothesis fields of a claim annotation, the certificate field of an audit annotation and its conclusion field when its conclusion is true in every well-typed interpretation or is represented, in the sense of Definition 4.2, by fields of the content records named by the claim it audits, and the formula schema of a witness-schema annotation; a conclusion field that is not exempt is a formula-valued field, so an audit annotation carrying one does not have only datum and reference fields besides its exempt fields.

Definition 4.14 (Residual feature; residual classification scheme). Let $D \in \text{FATCD}$. A *residual feature*, or *kind residual*, of D is any content record of Definition 4.1 or any of the six structural annotations of Definition 4.2 in $\text{Raw}(D)$. The seven bookkeeping annotations are excluded from this kind-residual set; they may still be record residuals of a particular decomposition. Raw-grammar kind alone assigns none of these features a role; “residual” means residual to the raw kinds, not left over by a decomposition, and every content record is a kind residual, including the constituents of active clusters. The records left over by a particular decomposition record form the set $\text{Residual}(D)$ of Section 7. The *residual classification scheme* specifies, relative to the channel entries of a decomposition record (its six statuses and the clusters, offers, seeds and declarations they name, which fix $A(\text{Dec}(D))$, $S(\text{Dec}(D))$ and $\text{Residual}(D)$ before any label is chosen), which labels are licensed for each residual feature; a record chooses one licensed label $\text{Classify}(r)$ for each $r \in \text{Residual}(D)$ (Definition 7.2), drawn from the list

- (i) a declared constituent of the cluster that the decomposition record names for an active P1–P6 projection $\pi_i(D)$, whose cluster satisfies W_i ;
- (ii) shared constituent of two such projections;
- (iii) presentation variant of an already classified feature;

- (iv) level shift of an already classified feature;
- (v) unabsorbed role-aligned content, covered-seed content with no certified witness W_i , content checked in an honestly failed offer, or activation-threshold refinement;
- (vi) instrument-visibility artifact;
- (vii) empirical-bridge annotation;
- (viii) outside-domain structure;
- (ix) already-covered bookkeeping or annotation data, or role-neutral data without an assertion of its own;
- (x) unresolved candidate seventh-role threat: an in-scope feature placed by none of (i)–(ix), which need be neither true nor a proposed role (Remark 6.8);
- (xi) recorded seventh-role screen.

The same criteria apply to every member of $\text{Residual}(D)$ for a chosen decomposition record, including a bookkeeping annotation not assigned to a channel. They are evaluated in the same way for every bookkeeping annotation of D , assigned to a channel or not; Theorems 6.9 and 9.1 use them so. A claim annotation's bookkeeping fields may receive (ix); its substantive assertion is classified through the content records representing it, including any additional record required for independent content. For a feature with no claim about it, “honestly audited” in criterion (x) means that every claim concerning it, if any, has an honest audit; it does not require an audit annotation for an unclaimed raw record. A decomposition record fixes one such assignment; where the label criteria license more than one label for r , different records may choose differently (Proposition 7.6). A label applies to r under the criterion given for it in the table after this definition, which is part of the definition. In particular, (x) applies when r has finite, honestly audited content in the covered scope and no criterion (i)–(ix) holds for it, so that it remains an unresolved candidate; and (xi) applies under the criterion in the last row of the table. In words, (xi) holds for an unresolved feature r when a recorded formula ψ , attached to a named candidate role outside $P_1, \dots, P_6$, holds at some cluster containing r at which no W_i holds, fails at some other cluster and at every cluster at which some W_i holds, and r records a closure operator c on a declared finite order lying outside the monoid generated by id_X and the restrictions to X of the graphs of those map, partial-map, transition and refinement records of D that are defined at every member of X with values in X (a map recorded only in a field of a record of another kind is not counted). There the separation certificate is the one whose formula is the finite conjunction over all $C' \subseteq \text{Raw}(D)$ of $(W_1(C') \vee \dots \vee W_6(C')) \rightarrow \neg\psi(C')$, with $\psi(C')$ the instance of the schema at the list C' ; this separation certificate establishes no witness row, law or seed, so its formula may contain W atoms, although ψ itself contains none, and “ ψ false at some other cluster” is the certificate $\neg\psi(C_0)$ for a named $C_0 \neq C$. Criterion (xi) is a recorded screen, not the identification of a new closure role.

Example (illustrative; not part of the definition). For $X = \{0, 1\}$ with a declared order record for $0 < 1$ and a certificate of its order law, a relation record of declared use other may carry a finite table field c with $c(0) = 1$, $c(1) = 1$, and a datum field listing (id_X) , with correct certificates, recorded true, of extensivity, monotonicity, $c^2 = c$, closure of this list under composition, and $c \neq \text{id}_X$. If no listed generating-record kind records a total map on X , its generated monoid is $\{\text{id}_X\}$. These records establish only the closure-operation part of (xi). The full criterion additionally requires (x), a witness-schema annotation, and the certified ψ -separation specified above. For that full criterion, add a check record carrying the stated recomputable certificates, and a claim annotation referencing both r and that check record whose statement is exactly the certified screen conditions. An honest audit of the claim cites those records and certifies that statement. This ends the example.

The scheme is declarative. At the domain-definition level, FATCD records the available labels but neither proves that every residual feature lands in categories (i)–(ix) nor proves that categories (x) and (xi) are empty. Those claims are deferred to Section 6. The terms aligned, contributes a required field, required for W_i at C , input to the check and outside the covered scope are defined in the paragraph *Terms used in the label criteria* after the table below.

When each label applies. Labels are assigned relative to the channel entries of a fixed decomposition record $\text{Dec}(D)$ (Definition 7.2); well-formedness of $\text{Dec}(D)$ then requires each chosen label to be licensed in this sense. Labels (i)–(v) use the named channel entries where their criteria require them, together with the records and clusters specified in those criteria. The residual scheme allows the following labels for a residual feature r in the sense of this definition; the criteria are given in the table below, and Theorem 6.9 uses them to exclude (x) and (xi) under the recognition hypothesis. Labels (i) and (ii) do not occur in the set $\text{Residual}(D)$ of a decomposition record (Section 7).

Label	The residual feature r
(i)	belongs to the cluster that the decomposition record names for an active projection satisfying W_i , and every predicate, comparison, invariant, law or claim field of r is required for W_i at the cluster C of the clause, or is an input to the check of W_i at C , or is a named field of a seed clause that r carries (so r carries no assertion beyond these fields; content beyond them must be recorded in a separate declared content record, which is classified on its own)
(ii)	there are indices $i \neq j$ and clusters C_i, C_j named by the decomposition record for active projections such that r contributes a required field to both, and every predicate, comparison, invariant, law or claim field of r is required for, or an input to the check of, W_i at C_i or W_j at C_j ; the relation record E of the running example, which lies in the P1 and P5 clusters, is an example, as is the map record for F , which lies in C_1 and C_6 (the clusters of the paragraph <i>The running example as a description</i> below)
(iii)	re-presents a classified feature: $r = r_0$ begins a finite chain $r_0, \dots, r_n$ as in (R4), with $n \geq 1$, distinct records and only checked equality or isomorphism steps, whose terminal r_n lies in $A(\text{Dec}(D)) \cup S(\text{Dec}(D))$ or satisfies a criterion among (i), (ii) and (v)–(ix)
(iv)	restates a classified feature at a different level: $r = r_0$ begins a finite chain $r_0, \dots, r_n$ as in (R4), with $n \geq 1$, distinct records and at least one checked level-translation step, whose terminal r_n lies in $A(\text{Dec}(D)) \cup S(\text{Dec}(D))$ or satisfies a criterion among (i), (ii) and (v)–(ix)
(v)	one of: (v-a) r contributes a required field to a six-role-aligned cluster, or itself carries a covered seed for one of the six roles, but criterion (i) does not hold for r ; this includes a cluster missing an attachment and a seed lacking a complete witness; and every predicate, comparison, invariant, law or claim field φ of r is, for some index i_φ and some finite cluster $C_\varphi \ni r$ aligned with P_{i_φ} , required for W_{i_φ} at C_φ or an input to its check, or is named in a seed clause that r carries (so r carries no assertion beyond these fields; content beyond them must be recorded in a separate declared content record, which is classified on its own); (v-b) r is a threshold annotation targeting a cluster named for an active projection, which carries a threshold predicate and no field other than its typing, target and label fields and that predicate, in the sense of Definition 4.9, and the threshold predicate reads at least one record, every record whose values it reads is listed in the annotation's target field, and each such record satisfies (R1); (v-c) r is a member of the cluster of an honestly recorded failed offer for some P_i , each of whose predicate, comparison, invariant, law or claim fields is an input to the recorded check of W_i on that cluster
(vi)	is an instrument-visibility annotation whose fields other than its typing and target fields are its visible data, suppressed data, instruments and claim strength (Definition 3.9), each formula-valued one of which reads at least one record, reads only records listed in its target field, and reads only records satisfying (R1)

Label	The residual feature r
(vii)	is an empirical-bridge annotation identifying an admissible bridge claim whose associated entry in $\text{Audit}(D)$ carries the five items of Definition 3.11, and whose own fields other than its typing and target fields only identify that bridge claim and those items
(viii)	lies outside the covered scope: a field of r asserts a property whose check is not finite over the records of D : it quantifies over a declared domain for which D supplies no finite checking range, or uses a declared predicate whose extension D does not record, and its truth value is not fixed by the recorded values of D (it is true in some expansion of D and false in another, in the sense of the paragraph <i>Certificates, assertion language and laws</i> of Section 2; for a formula with free variables: for some assignment α of its free variables and two expansions of D whose carriers contain the values α gives to variables of unlisted sorts, it is true under α in one and false under α in the other). In addition, every other predicate, comparison, invariant, law or claim field of r either has this property or is required for W_i at a cluster C aligned with P_i , an input to the check of W_i at such a cluster, or named in a seed clause that r carries
(ix)	is one of: (ix-a) a bookkeeping annotation whose fields other than the exempt fields of (R2) are datum or reference fields; (ix-b) a structural annotation whose fields contain only its declared operation, its targets, and the finite typing, checking, preserved or suppressed data, losses and nonclaims required for that operation, with no independent assertion, when (v), (vi) and (vii) do not apply. Here a formula-valued field of a structural annotation (a threshold predicate, a preserved or lost invariant) is part of its declared operation only when it reads at least one record, every record whose values it reads is listed in the annotation's target field, and each such record satisfies (R1); otherwise it is an independent assertion and (ix-b) does not apply; (ix-c) role-neutral data without an assertion of its own: it supplies no witness field and has no predicate, comparison, invariant, law or claim field, such as an input used in checking a required or seed field, or a declared object or state that no role datum uses; also a predicate record supplying no witness field whose only assertion field is a formula with at least one free variable whose truth value is not the same under all assignments respecting the declared sorts of its free variables, using only predicates whose extensions D records, and quantifying only over sorts whose carriers D records, each free variable ranging over a record kind or the rational sort, referenced by no claim annotation, and referenced by certificates only through the abbreviation $s(t)$ of the paragraph <i>Check language</i> (a definition, which has no truth value over D)
(x)	has finite, honestly audited content in the covered scope, and no criterion (i)–(ix) holds for it; for $D \in \text{FATCD}$ the first two conditions hold for every record (Definitions 4.4 and 4.6(e)), so (x) applies exactly when r lies in the covered scope and no criterion (i)–(ix) holds

Label	The residual feature r
(xi)	satisfies (x), and: (a) D carries a witness-schema annotation naming a role label outside $P_1, \dots, P_6$ with a formula schema $\psi(C)$ (Definition 4.2) each of whose instances $\psi(C')$, at the identifier list of a cluster C', is a closed formula of the fixed check language containing no atomic W_j formula; (b) for some cluster $C \ni r$, an honest audit records $\psi(C)$ true, $W_1(C), \dots, W_6(C)$ all false, and ψ false at some other cluster of D; (c) the same audit records $\psi(C')$ false for every cluster $C' \subseteq \text{Raw}(D)$ at which some $W_i(C')$ holds, by the separation certificate specified in the body of this definition; (d) r records a closure operation c on a declared finite order $(X, \leq)$, with $\leq$ the reflexive closure of the relation of a declared order record whose law is certified, that is, a total map $c : X \rightarrow X$ recorded in a field of r with correct certificates, recorded true, whose formulas are obtained, by the steps permitted in the paragraph <i>What a required certificate must express</i> of Section 2, from the prescribed conditions $\forall x \in X (x <_o c(x) \vee x = c(x))$, $\forall x, y \in X ((x <_o y \vee x = y) \rightarrow (c(x) <_o c(y) \vee c(x) = c(y)))$ and $\forall x \in X (c(c(x)) = c(x))$, o being that order record; (e) a datum field of r lists finitely many total maps $X \rightarrow X$, among them id_X and, for every map, partial-map, transition or refinement record of D whose graph is defined at every member of X with values in X, the restriction of that graph to X, and correct certificates, recorded true, whose formulas are obtained in the same way from the prescribed conditions $\forall k, l < L \exists j < L \forall x \in X (L_j(x) = L_k(L_l(x)))$ and $\forall k < L \exists x \in X (c(x) \neq L_k(x))$, L being that datum field, stating that the list is closed under composition and that c differs from every listed map; the list then contains the monoid these maps generate, so c lies outside it

Criterion (viii) states the same condition as the paragraph *Covered scope* below and as (R5); criterion (ix-c) states the same condition as (R3); and criterion (v-a) is the first branch of (R1) together with the failure of (i).

Terms used in the label criteria. *Declared operations.* The *declared operation* of a structural annotation of Definition 4.2 is the operation of its kind (a level assignment, a threshold, a lift, a forgetting, an instrument restriction or an empirical bridge) together with the fields that the definition of that kind prescribes for it; an *independent assertion* of such an annotation is a predicate, comparison, invariant, law or claim field other than those prescribed fields, or a formula-valued prescribed field that is not part of its declared operation under the condition of criterion (ix-b).

Alignment and required members. A cluster $C \subseteq \text{Raw}(D)$ is *aligned* with P_i , or *six-role-aligned*, when $W_i(C)$ holds; it may lack the attachment data of Definition 4.9. A record $r \in C$ is a *required member* of such a cluster, and is said to *contribute a required field* to it, when $W_i(C \setminus \{r\})$ fails (the condition concerns membership, so a record without fields, such as a micro-element of the W_5 row, can contribute a required field), and r *supplies a witness field* when it contributes a required field to some aligned cluster or carries a covered seed.

Required fields and inputs. For $r \in C$ and a field φ of r , let $D[r \setminus \varphi]$ be the description with φ deleted from r , all else unchanged, and $C[r \setminus \varphi]$ the corresponding cluster; the rows of the witness table define W_i on every such pair, whether or not it lies in FATCD. The field φ is *required for* W_i at C when $W_i(C)$ holds and $W_i(D[r \setminus \varphi], C[r \setminus \varphi])$ fails, and is an *input to the check* of W_i at C when changing only its value (for a datum field its recorded datum; for a formula-valued or certificate field its recorded formula together with, for a certificate, its named inputs; the recorded truth value of a certificate is not read by recomputation) can change the recomputed result of some certificate κ carried by a member of C whose recorded formula, before the change, expresses, in the sense of the paragraph *What a required certificate must express* (Section 2), the conditions that

the W_i row prescribes for that certificate (for W_6 , a property of a trace in C tied to an event in C ; for W_3 , a non-reducibility certificate for two routes in C ; for W_4 , a law of an order, transition or refinement record in C ; κ may be φ itself only when φ 's recorded formula already expresses such a condition). For a failed offer on C , a field φ of $r \in C$ is an *input to the recorded check* when the offer's audit lists a certificate carried by a member of C , other than $(W_i(C), \text{false})$, whose formula expresses a condition prescribed by the W_i row and whose recomputed result changes when only the value of φ changes; when the audit lists $(W_i(C), \text{false})$ alone, no field is such an input. This field-level notion differs from the record-level phrase of the paragraph *Witness types*, which governs only cluster membership; a reference field is never an input to the check. For a covered seed, the required fields are the fields named in its seed clause of Definition 7.1.

Covered scope. A record r of D lies *outside the covered scope* when a field of r asserts a property whose check is not finite over the records of D : it quantifies over a declared domain for which D supplies no finite checking range, or uses a declared predicate whose extension D does not record, and its truth value is not fixed by the recorded values of D (it is true in some expansion of D and false in another, in the sense of the paragraph *Certificates, assertion language and laws* of Section 2; for a formula with free variables: for some assignment α of its free variables and two expansions of D whose carriers contain the values α gives to variables of unlisted sorts, it is true under α in one and false under α in the other). In addition, every other predicate, comparison, invariant, law or claim field of r either has this property or is required for W_i at a cluster C aligned with P_i , an input to the check of W_i at such a cluster, or named in a seed clause that r carries; the *covered scope* of D consists of its other records.

Recorded extensions and carriers. Here D *records the extension* of a declared predicate p when at least one complete presentation of p is supplied, either by a relation record that references the declaring record of p in a field declaring it the extension of p , its listed tuples being by definition the true tuples of p , provided the argument sorts of p have recorded finite carriers, or by its declaring predicate record (possibly r itself) giving an explicit formula of the fixed check language of Section 2 over recorded inputs and declared finite ranges (arguments of rational sort being then admitted and evaluated by that formula); all such presentations of p must agree on every tuple, and otherwise D does not record its extension. This depends only on D and p .

Finite checking ranges. D supplies a finite checking range for, and records the carrier of, a declared domain σ when σ is declared by an object record listing its members or is the set of declared records of one kind; a sort declared by an object record listing no members has neither. Both conditions are decided by inspecting the fields of r and the declarations of D . The record and its references remain finite and declared.

Outside-scope declarations; examples. An outside-scope declaration (Definition 7.1) additionally requires the honest audit of its claim to list that field in its non-certification field; the outside-scope criterion itself imposes no condition on audits. A finite, audited feature is not outside scope merely because it supplies none of the six role witnesses. Label (viii) applies to a record outside the covered scope: the record itself is finite and belongs to D , but the check its assertion needs is not finite over the records of D . Examples are an audited metaphysical-causation assertion whose declared predicate has no recorded extension (Proposition 6.4) and a proposed intentional or teleological predicate whose extension is unrecorded (Proposition 6.5), in each case provided the asserted property is true in some expansion of D and false in another.

Remark 4.15 (Optimization and objective data are not excluded). Raw features expressing optimization, objective, or selection criteria pass through the scheme of Definition 4.14; they are not dismissed by definition. Under the fixed role meanings (Remark 4.12), typical classifications include: a predicate or specification record possibly supporting a P2 projection; a map or transition-selection

record possibly supporting a P1 or P4 projection; a route-comparison or route-choice record possibly supporting a P3 projection; an audit or check record possibly supporting a P6 projection; an empirical-bridge annotation; or an outside-domain structure when an assertion of the feature uses an unrecorded predicate or uncarried sort and is true in some expansion of D and false in another, and every other assertion field meets the additional conditions of criterion (viii). A feature that fits none of these receives whichever of criteria (iii)–(ix) it meets (for instance (ix-c) for a record with only datum fields, such as a value tagged objective), and category (x) only when none holds.

The running example as a description. The example of Section 2 is a description $D_{\text{ex}} \in \text{FATCD}$ whose raw records are listed below, with the references the text of the example fixes and the projection that uses each record. Every reference points to a declared record, and the audit part carries a bookkeeping record for each claim, so D_{ex} is finite, closed and audited. Threshold, witness, level, collapse and audit annotations are attached to the three clusters C_1 , C_5 and C_6 described after the table.

Record	Relates to	Used by
four state records 0, 1, 2, 3		
state-set object record S_0 listing 0, 1, 2, 3	state records	declared domain and codomain of F and of its transition record, domain of q ; P1 and P5 projections
object records A , B and a quotient-object record $\{A, B\}$ listing them		P5 projection
map record for F , declared use <i>update</i>	S_0	P1 projection
relation record E , of declared use quotient, whose datum fields list the classes $\{0, 1\}$ and $\{2, 3\}$ under class-name fields A and B		P1 and P5 projections
map record for q , declared use <i>packaging</i> , declared domain S_0 , codomain $\{A, B\}$	S_0 , state records, quotient-object record	P5 projection
quotient-validity check record for q and E , checking the two fibers, their equality with the E -classes (so $E = \ker q$), and coverage of $\{A, B\}$, and that every member of a class of E lies in S_0	state, object, q -map and relation records	P5 projection
comparison record for the split pair (0, 1)	map and relation records	P1 projection
trace record $0 \rightarrow 2 \rightarrow 3$		P6 projection
a replay record carrying only reference fields to the trace and to the map record for F , and a check record carrying reference fields to the trace and to the map record for F together with the certificate $\tau_1 = F(\tau_0) \wedge \tau_2 = F(\tau_1)$, where $\tau = (0, 2, 3)$	the trace and the map record for F	P6 projection
event and provenance records	the trace	P6 projection
transition record for F , whose graph lists the same four pairs as the map record for F , carrying in its law field the certificate $(\forall x \in S_0 (t(x) \in S_0), \text{true})$, t being this record (the transition law of Section 2 with its definedness conjunct omitted), where $S_0 = \{0, 1, 2, 3\}$ (no separate check record)	map record for F , S_0 and state records	
index record idx with two datum fields named A and B (field names, distinct from the identifiers of the object records A and B , as are the class-name fields of E) holding the lists (0, 1) and (2, 3), with one comparison field recording the bijection β from these two fields to the class fields A and B of E , together with its certificate that each recorded value equals that of its β -image, written entrywise with indexing ($A(\cdot)_k$ denoting entry k of the field A): $\bigwedge_{k < 2} (A(\text{idx})_k = A(E)_k \wedge B(\text{idx})_k = B(E)_k)$	relation record E	

The split-pair comparison carries the certificate $(E(0, 1) \wedge ((0 \in \text{dom}F \leftrightarrow 1 \notin \text{dom}F) \vee (0 \in \text{dom}F \wedge 1 \in \text{dom}F \wedge E(F(0), F(0)) \wedge E(F(1), F(1)) \wedge \neg E(F(0), F(1))))), \text{true})$; $F(0)$ and $F(1)$ are defined because F is a total map record on S_0 , so this is alternative (b) of the W_1 row. The quotient-validity check carries the certificate, recomputed true, $\forall u, v \in S_0 (E(u, v) \leftrightarrow q(u) = q(v)) \wedge 0 \neq 1 \wedge q(0) = q(1) \wedge \forall z \in \{A, B\} \exists u \in S_0 (q(u) = z) \wedge \forall u \in |E| (u \in S_0) \wedge \forall u \in S_0 (q(u) \in \{A, B\})$, the W_5 formula with $X = S_0$, $Q = \{A, B\}$, $x = 0$, $y = 1$ (the numerals in $0 \neq 1$ replace the record-sorted variables x, y and denote records). Both certificates name their declared inputs. No threshold annotation targets a cluster containing the transition record. The P1 cluster C_1 contains F , E , S_0 , the split-pair comparison and their referenced state records. The P5 cluster C_5 contains S_0 , the four state records, the object records A and B , the quotient-object record $\{A, B\}$, the map record for q , the relation $E = \ker q$ and the finite quotient-validity check. The P6 cluster C_6 contains the event, trace, provenance, replay and check records and the map record for F . For each $i \in \{1, 5, 6\}$, D_{ex} carries a threshold, witness-schema, level, collapse-case, claim, honest audit and nonclaim annotation assigned to the displayed active projection, the threshold, witness-schema, collapse-case, level and nonclaim annotations each targeting exactly C_i , each of the three level annotations recording the verification level, so that no projection makes a level crossing (Definition 4.11); for $i = 6$ these annotations also include a claim annotation c_τ referencing the event, trace and check records, whose statement is the formula of the check record's certificate (so c_τ satisfies the representation condition of Definition 4.2), with an honest audit a_τ referencing c_τ , the event, trace, F and check records, and certifying precisely those equations from the check record, both assigned to the active P6 projection, which discharges the $\text{Audit}(D)$ conjunct of $W_6(C_6)$ certified by the P6 audit. It carries no lift or forgetting annotation and no other annotation. Each threshold annotation carries only its typing, target and label, each level annotation only its typing, target and level, and each claim, collapse-case and nonclaim annotation only datum and reference fields besides its statement and hypothesis fields. For each $i \in \{1, 5, 6\}$, the role claim has statement $W_i(C_i)$. Its audit references that claim and its named records, lists the row's content certificates, recomputed true, together with $(W_i(C_i), \text{true})$, and concludes $W_i(C_i)$. For $i = 6$, the latter certificate is recomputed after a_τ is established honest in $\text{Audit}_0(D_{\text{ex}})$. The P4 status names the law-bearing transition record as its seed. The index record references E ; its equality-check field records that its two lists equal, entry by entry, the class lists of E . Recomputing that equality verifies its presentation-variant label. The table and these annotations exhaust $\text{Raw}(D_{\text{ex}})$. Consequently, for the decomposition record that assigns all these annotations to their projections and names the transition record as the P4 seed, $A(\text{Dec}(D_{\text{ex}})) \cup S(\text{Dec}(D_{\text{ex}}))$ contains every feature except the index record, and $\text{Residual}(D_{\text{ex}})$ consists of the index record alone, classified as a presentation variant of the relation record; a well-formed record leaving c_τ and a_τ unassigned has them as further record residuals, each licensed (ix-a).

4.7 Non-circularity and non-overbreadth

Two proposition schemas record the definition-theoretic facts used by the later theorem chain: the domain does not presuppose the answer to the exact-six question, and it is structurally restrictive enough for that question to remain substantive.

Proposition 4.16 (Non-circularity schema for FATCD). *Membership in FATCD is not defined by requiring a decomposition/exhaustion record or by requiring any derived role projection to exist. Concretely, the description D_0 of Remark 4.8 lies in FATCD and has no derived role projection, and the description D_{norm} of Remark 4.8 (analysed in Remark 6.8) lies in FATCD, and its raw record N is licensed only label (x) in every well-formed decomposition record.*

Proof. We inspect the clauses that govern membership, namely those of Definitions 4.1–4.7, and the residual classification scheme of Definition 4.14.

- (a) The raw grammar permits a witness-schema annotation to name P_i , and honesty checks any role claim a description actually makes. Neither such an annotation nor a role claim is required for membership in FATCD. Its membership conditions require finite fields, closed references, and honest audits for claims present; they require no derived π_i , role-channel assignment, or decomposition record. Thus optional role-naming data do not presuppose a role projection.
- (b) No membership clause excludes a candidate seventh-role feature by definition. A candidate seventh-role feature is, at the raw-grammar level, a record or annotation of one of the declared kinds; nothing in Definitions 4.4–4.6 filters a feature out on the ground that it might resist absorption into a P1–P6 projection. Such a feature is therefore admissible whenever its containing description satisfies all membership conditions of FATCD, and is carried forward to the residual classification scheme rather than barred at the door.
- (c) The residual classification scheme (Definition 4.14) is declarative and carries the categories (x) and (xi)—an unresolved candidate seventh-role threat, and a recorded seventh-role screen—as explicit failure options. At the domain-definition level the scheme records the available labels but does not assert that those two categories are empty, so it cannot smuggle the exact-six answer into the definition of the domain.

Since none of (a)–(c) ties admissibility to the existence of a role projection or a six-channel record, membership in FATCD is independent of the exact-six question, which is what was claimed. For the concrete clause, D_0 carries no annotation, so no witness-schema annotation exists and no tuple $\pi_i(D_0)$ can be formed (Definition 4.11); D_{norm} lies in FATCD and its record N receives label (x) in every well-formed decomposition record: N has a predicate field containing a closed formula and is referenced by a claim annotation, so both branches of (ix-c) fail; D_{norm} has no map, relation, path, order, transition, refinement, event, provenance, trace, replay or check record, so no cluster satisfies any W_i , no record is a covered seed, every channel is *absent_with_record*, and (i), (ii) and (v) fail; N has a predicate field and records no checked equality, isomorphism or level translation, so it begins no (R4) chain and (iii), (iv) fail; N is not an annotation, so (vi), (vii), (ix-a) and (ix-b) fail; N gives its predicate by an explicit formula over recorded values, so (viii) fails; and D_{norm} has no witness-schema annotation naming a candidate outside $P_1, \dots, P_6$, so (xi) fails (Remark 6.8). $\square$

Proposition 4.17 (Non-overbreadth schema for FATCD). *FATCD is structurally restricted and does not admit arbitrary syntactic material: every record of a description in FATCD is finite, typed and reference-closed, every claim it carries is referred to by an honest audit annotation, unclaimed records may occur but support no projection without the attachment data of Definition 4.9, and mentioning a role name without the supporting threshold, witness, level, collapse, nonclaim and bookkeeping data does not by itself produce a derived role projection. In particular, if no witness-schema annotation of D names P_i , then no $\pi_i(D, C)$ is admissible.*

Proof. Let $D \in \text{FATCD}$. We inspect the admissibility conditions of Definitions 4.4–4.9 in turn.

- (i) By Definition 4.4, $\text{Raw}(D)$ contains finitely many records and annotations, each record or annotation carries finite typed fields, and no infinite semantic, probabilistic, topological, geometric, causal, optimization, or resource structure is admitted unless finitely represented and bookkept as a raw feature. Thus unbounded or unrepresented material is excluded at the level of finiteness.
- (ii) By Definition 4.5, every source, target, endpoint, index, transition, order, refinement, trace, event, provenance, replay, check, and annotation reference is bound to a declared record of D .

Hence no dangling or externally defined reference can occur, and the syntactic material of D is self-contained.

- (iii) By Definition 4.6(e), every claim carried by D is referred to by an honest audit annotation. Promotions, translations, lifts, forgetting, visibility and empirical-bridge assertions carry the fields required by Definition 4.6(g). Thus a claim lacking an honest audit, or one of these annotations lacking its required fields, cannot occur in a member of **FATCD**; an assertion field of an unclaimed content record needs no audit.
- (iv) By Definition 4.9, a role projection cannot be asserted on a cluster C until the raw features of C , a threshold annotation, a witness schema, a record of collapse cases, a level annotation, a nonclaim annotation targeting exactly C , and an audit record are all present in $\text{Raw}(D)$. Hence a role name supplied without this attachment data does not, by the act of naming alone, produce a projection.

Combining (i)–(iii), every record of D is finite, typed and reference-closed, and every claim of D is honestly audited; a record about which no claim is made needs no audit (Definition 4.6), and by (iv) it supports no projection unless the attachment data are present. By (iv), mentioning a role name without the supporting threshold, witness, level, collapse, nonclaim and bookkeeping data does not by itself produce a derived role projection. This establishes both halves of the claim. For the last clause, admissibility of $\pi_i(D, C)$ requires a witness-schema annotation naming P_i and targeting exactly C (Definition 4.11). $\square$

4.8 The scoped exact-six question over **FATCD**

We can now state the scoped exact-six question precisely.

Definition 4.18 (Scoped exact-six question). The *scoped exact-six question* over **FATCD** asks whether, for every $D \in \mathbf{FATCD}$, there exists at least one decomposition/exhaustion record $\text{Dec}(D)$ whose role-channel structure consists of exactly the six typed closure-role channels $P1, P2, P3, P4, P5, P6$, each assigned exactly one of the five statuses later formalized in Section 7, such that no kind residual of D and no element of $\text{Residual}(D)$ is licensed label (x) or (xi) of Definition 4.14 relative to $\text{Dec}(D)$. No uniqueness is built into this question. For a class $\mathcal{D} \subseteq \mathbf{FATCD}$, the affirmative answer for every $D \in \mathcal{D}$ is denoted $\text{ScopedExactSix}(\mathcal{D})$. Section 9 establishes $\text{ScopedExactSix}(\mathbf{FATCD}_{\text{RH}})$, where $\mathbf{FATCD}_{\text{RH}} \subseteq \mathbf{FATCD}$ is the class of descriptions satisfying the recognition hypothesis introduced in Section 6 (Definition 6.7); the recognized forms of probability, causality and agency (Propositions 6.3–6.5) are classified without that hypothesis. Label (x) is licensed for every in-scope assertion-bearing record placed by none of (i)–(ix), for instance an unclaimed predicate record whose closed formula is $0 + 0 = 1$, or the true formula $0 + 0 = 0$, between rational constants (Remark 6.8); one object record together with such a predicate record already lies in **FATCD** but not in $\mathbf{FATCD}_{\text{RH}}$ (it has no claims, so it is audited; the predicate record has a closed assertion field, carries no seed, contributes to no aligned cluster, begins no (R4) chain, is not an annotation and lies in the covered scope, so (R1)–(R5) fail as for N in Remark 6.8); a negative answer for D therefore need not exhibit any candidate role.

The unrestricted assertion $\text{ScopedExactSix}(\mathbf{FATCD})$ fails: the finite audited normalization description of Remark 6.8 lies in **FATCD** and has a residual feature labelled (x) in every well-formed decomposition record (verified in Remark 6.8; see also Proposition 8.7(d)). Label (x) there records that no criterion (i)–(ix) places the normalization predicate; it does not exhibit a seventh role.

Remark 4.19 (Channels, not active projections). An affirmative answer to the scoped exact-six question asserts the existence of six role channels per description, not six active derived projections

per description. The distinction introduced in Remark 4.13 and formalized in Section 7 is preserved throughout.

Remark 4.20 (Relation to the Foundations theorem). Proving $\text{ScopedExactSix}(\text{FATCD}_{\text{RH}})$ is not a re-derivation of the canonical unavoidability result of Foundations I [28], and $\text{ScopedExactSix}(\text{FATCD}_{\text{RH}})$ does not follow from that result alone. The Foundations theorem derives the P1–P6 vocabulary under minimal hypotheses of composability with limited interface access; the scoped exact-six claim asserts that, within the finite audited typed closure regime of FATCD and under the recognition hypothesis of Section 6, every admissible description admits a six-channel decomposition/exhaustion record without a seventh irreducible typed role in the covered scope. The two results share vocabulary but address different questions.

5 Six-role lower bounds

The exact-six question of §4.8 has two directions. The lower-bound direction asserts that each of P1, ..., P6 is instantiated in FATCD; the upper-bound classification of Section 6 will assert, under a recognition hypothesis, that no seventh role is needed in the covered scope. This section establishes the lower-bound direction by constructing one admissible witness for each role. The content is occurrence, not exhaustion: lower bounds alone do not answer the exact-six question, but they rule out the trivialization in which some role has no admissible witness in the domain.

5.1 Lower-bound theorem and common witness requirements

Theorem 5.1 (Six-role lower bounds). *For each role index $i \in \{1, \dots, 6\}$, there exists a description $D_i \in \text{FATCD}$ and a derived active projection $\pi_i(D_i)$ witnessing the role channel P_i in the sense of Definition 4.11. In particular, each of P1, P2, P3, P4, P5, P6 is instantiated by at least one admissible description in FATCD. This is an occurrence theorem, not an exhaustion theorem.*

A witness for role P_i consists of the data specified below, following the role-projection discipline of §4.5 and the threshold-attachment discipline of §4.4.

Definition 5.2 (Common witness data for role P_i). *A witness of role P_i in FATCD is a tuple*

$$(D_i, C_i, \Theta_i, W_i, \text{Level}_i, v_i^{\text{fin}}, v_i^{\text{cl}}, v_i^{\text{aud}}, n_i, \pi_i(D_i))$$

where

- $D_i \in \text{FATCD}$ is the ambient admissible description;
- $C_i \subseteq \text{Raw}(D_i)$ is the raw feature cluster on which the projection is interpreted;
- Θ_i is the threshold annotation attached to C_i ;
- W_i is the witness schema for the role meaning of P_i fixed in the *Witness types* table of Section 2;
- Level_i is the level assignment in the sense of Definition 2.4;
- $v_i^{\text{fin}}, v_i^{\text{cl}}, v_i^{\text{aud}}$ are the statements that D_i satisfies Definitions 4.4, 4.5 and 4.6 respectively, each with its proof;
- n_i is the explicit nonclaim record carried by the witness;
- $\pi_i(D_i)$ is the derived role projection of Definition 4.11.

The witness is *admissible* when $v_i^{\text{fin}}, v_i^{\text{cl}}$ and v_i^{aud} show $D_i \in \text{FATCD}$ and $\pi_i(D_i, C_i)$ is admissible in the sense of Definition 4.11, with Θ_i , the W_i witness-schema annotation, Level_i and n_i as its threshold, witness-schema, level and nonclaim annotations; in particular $W_i(C_i)$ holds.

The proof of Theorem 5.1 proceeds role by role: for each $i \in \{1, \dots, 6\}$ we exhibit an admissible tuple of the form in Definition 5.2. The six constructions are collected in §5.2.

5.2 Per-role witnesses

Each construction below specifies a raw feature cluster, threshold annotation, witness schema, level assignment, derived active projection, and explicit nonclaim record. The finite-description, closure, and audit/bookkeeping verifications are routine once these data are placed in a description built from the raw grammar of §4.1 and checked against the admissibility conditions of §4.2. In the P1 construction, $\sim$ is a relation on the source records and $\sim'$ a relation on the target records; descent asks whether $x \sim y$ implies $F(x) \sim' F(y)$. In the constructions, $\text{Ann}_\Theta[t]$ is a threshold annotation whose recorded label is t , and $v_i^{\text{fin}}, v_i^{\text{cl}}, v_i^{\text{aud}}$ are the checks that D_i satisfies Definitions 4.4, 4.5 and 4.6.

Annotations. In each construction, D_i includes a claim annotation naming C_i , W_i , the asserted check result, hypotheses and scope, and an audit annotation referencing it and the finite evidence used by the check, together with the threshold, level, witness-schema, collapse-case and nonclaim annotations as declared records; every endpoint or reference used in a construction is included as a declared record of that D_i , even where the abbreviated display names only its principal records. In each construction, the claim explicitly names P_i and C_i , and the threshold, witness-schema, collapse-case, level and nonclaim annotations each have a target field listing exactly C_i . The verification v_i^{aud} is the check that these entries are honest in the sense of Definition 4.6, not merely present.

Fields. Each D_i contains only the displayed witness records, their declared inputs, and the stated annotations. Every content record either contributes a required field to an aligned subcluster of C_i , carries a covered P_i seed, or is role-neutral data without an assertion of its own, such as a record named in a declared domain, codomain, source or target field of a required-field contributor, whether or not a check reads it. No content record makes an assertion independent of the displayed witness. Relation records other than Real in D_2 , and map, path, index and state records, carry only datum and reference fields; every law, invariant or comparison is a field of the displayed check, comparison or law-bearing record. Each threshold annotation $\text{Ann}_\Theta[t]$ carries only its typing, target and label t , and each level annotation carries only its typing, target and level; each claim annotation carries, besides its statement and hypothesis fields, only datum and reference fields; and each collapse-case, nonclaim, promotion and translation annotation carries only datum and reference fields.

Audits. In each D_i , the audit of the W_i claim lists the row's content certificates carried by members of C_i , each recomputed true, and a certificate $(W_i(C_i), \text{true})$; its conclusion is $W_i(C_i)$. The latter certificate is recomputed after the content checks and, for $i = 6$, after a_τ is established honest in $\text{Audit}_0(D_6)$. The conclusion thus follows from a signed premise, and the audit is honest.

Proposition 5.3 (P1 lower bound). *There exists $D_1 \in \text{FATCD}$ with a derived active projection $\pi_1(D_1)$ witnessing P1.*

Proof. Let D_1 carry the finite state set $S = \{x, y, x', y'\}$ with four distinct state records and a declared object record S listing x, y, x', y' , the source equivalence relation listing the classes $\{x, y\}$, $\{x'\}$, $\{y'\}$, and the target equality relation on S , listing the four singleton classes, both relation records of declared use quotient. Let $F : S \rightarrow S$ be a map record whose declared use is *update*, with $F(x) = x'$, $F(y) = y'$, $F(x') = x'$, and $F(y') = y'$. A comparison record checks $x \sim y$ but $F(x) \not\sim' F(y)$; hence the related source pair witnesses the descent obstruction. Attach a threshold annotation $\Theta_1 = \text{Ann}_\Theta[\text{descent obstruction}]$ to the cluster C_1 comprising these records, and record a witness-schema annotation naming P1 and the W_1 row; $W_1(C_1)$ holds by alternative

(b), since the comparison record names (x, y) and carries the certificate $(x \sim y \wedge ((x \in \text{dom}F \leftrightarrow y \notin \text{dom}F) \vee (x \in \text{dom}F \wedge y \in \text{dom}F \wedge F(x) \sim' F(x) \wedge F(y) \sim' F(y) \wedge \neg F(x) \sim' F(y))), \text{true})$, whose second disjunct recomputes true. Take Level_1 on the verification level, and let $\pi_1(D_1)$ be the derived active projection interpreting C_1 via $(\Theta_1, W_1, \text{Level}_1)$. The finiteness, closure, and audit verifications $v_1^{\text{fin}}, v_1^{\text{cl}}, v_1^{\text{aud}}$ then attach routinely: the claim has statement $W_1(C_1)$; its audit references the claim and every member of C_1 , lists the comparison certificate and $(W_1(C_1), \text{true})$, both recomputed true, and concludes $W_1(C_1)$. The descriptions D_2 – D_6 follow the same template; for D_6 , the event-or-trace audit is established honest in $\text{Audit}_0(D_6)$ before $(W_6(C_6), \text{true})$ is recomputed. The nonclaim record n_1 excludes any P5-packaging identification (Remark 2.2) and any claim of an unconditional descended map in the obstruction case. $\square$

In the running example, $\sim$ and $\sim'$ are both the grouping by q , $x = 0$ and $y = 1$: $q(0) = q(1)$, while $q(F0) = B \neq A = q(F1)$.

Proposition 5.4 (P2 lower bound). *There exists $D_2 \in \text{FATCD}$ with a derived active projection $\pi_2(D_2)$ witnessing P2.*

Proof. Let D_2 carry the state records 0, 1, 2, 3, the candidate class of realizers; an object record S_2 listing 0, 1, 2, 3, declared as the domain of q and as the candidate class of Real ; a quotient object record $\{A, B\}$ and a map record q of declared use packaging with $q(0) = q(1) = A$ and $q(2) = q(3) = B$, whose kernel E the comparison uses; a predicate record s declared as a macro-specification with formula $s(r) \equiv (r = 0 \vee r = 2)$; a relation record $\text{Real} = \{(0, s), (2, s)\}$, of declared use *realization*, with candidate class $K = \{0, 1, 2, 3\}$ and target s , whose datum fields record the formula's values true, false, true, false at 0, 1, 2, 3 and which carries the certificate $(s(0), \text{true})$ exhibiting the satisfying realizer 0; and a comparison record carrying the certificate $(q(0) = q(1) \wedge s(0) \wedge \neg s(1), \text{true})$, certifying that the E -class $\{0, 1\}$ meets both the realizer set $\{0, 2\}$ and its complement $\{1, 3\}$ in K , since $0 \in \{0, 2\}$ and $1 \notin \{0, 2\}$ although $q(0) = q(1)$, so that realizability is not reducible to quotient-class equality alone; the finite quotient-class comparison data are fields of this comparison record, and any separately declared class identifiers belong to C_2 . Attach $\Theta_2 = \text{Ann}_\Theta[\text{nontrivial representability}]$ to the cluster C_2 of the records 0, 1, 2, 3, S_2 , q , $\{A, B\}$, s , Real and the comparison record (the comparison reads no class-identifier record; the object records A, B , which occur only as values of q and are listed by $\{A, B\}$, are declared records of D_2 outside C_2 , have no assertion field and satisfy (R3)), and record a witness-schema annotation naming P2 and the W_2 row, and $W_2(C_2)$ holds by the satisfying branch: s has the single free variable r , Real records s at each member of K and exhibits the realizer 0, and the comparison certifies that the class $\{0, 1\}$ of $\ker q$, whose domain S_2 contains K , meets both $\{0, 2\}$ and $\{1, 3\}$. Take Level_2 on the verification level, and let $\pi_2(D_2)$ be the derived active projection determined by $(C_2, \Theta_2, W_2, \text{Level}_2)$. The finiteness, closure, and audit checks then follow from the construction. The nonclaim n_2 excludes collapse to P5 quotient equality alone and excludes arbitrary relation promotion. $\square$

The states and the map q are those of the running example.

Proposition 5.5 (P3 lower bound). *There exists $D_3 \in \text{FATCD}$ with a derived active projection $\pi_3(D_3)$ witnessing P3.*

Proof. Let D_3 carry a state record u and an object record U listing u ; generating moves a, b , recorded as map records of declared use *other* with declared source and target U and $a(u) = b(u) = u$; two path records $\rho_1 = ab$ and $\rho_2 = ba$, each listing the successive records (u, u, u) , both from u to u and each referencing the comparison record; and a comparison record referencing both ρ_1 and ρ_2 and

carrying the rewrite rule $aa \rightarrow a$ as a rule field (Definition 4.1) holding the pair of words (aa, a) , the finite monoid $N = \{e, \bar{a}, \bar{b}\}$, its elements recorded as the rational constants $e = 0$, $\bar{a} = 1$, $\bar{b} = 2$ in a list field and its multiplication as a finite table field, with $ey = y$ and $xy = x$ for $x \neq e$, the images $\varphi(a) = \bar{a}$ and $\varphi(b) = \bar{b}$, a move-list datum field $M = (a, b)$, and a certificate whose formula is the full prescribed $W_3(a)$ condition, reading R, N, e, m, φ, M from the comparison record and w_1, w_2 from the route records. The assignments $R = ((aa, a))$, $M = (a, b)$, $w_1 = ab$, $w_2 = ba$ specify their recorded values, not literal substitutions in the certificate. It checks that $e \in N$, that N is closed under m and associative with identity e , that $\varphi(a), \varphi(b) \in N$, that $\varphi(\lambda) = \varphi(\mu)$ for each pair (λ, μ) listed in that datum field (here $\varphi(aa) = \bar{a} = \varphi(a)$), and that $\varphi^{m,e}(w_{\rho_1}) \neq \varphi^{m,e}(w_{\rho_2})$, where w_{ρ_j} is the recorded word of ρ_j (here ab and ba) and m is the recorded multiplication table. On the displayed table, $\varphi^{m,e}(ab) = \bar{a}$ and $\varphi^{m,e}(ba) = \bar{b}$. Let $C_3 = \{u, U, a, b, \rho_1, \rho_2, \text{comparison}\}$. Attach $\Theta_3 = \text{Ann}_\Theta[\text{route mismatch}]$ to the cluster C_3 , and record a witness-schema annotation naming P3 and the W_3 row; $W_3(C_3)$ holds because the two generated same-source, same-target route records are distinguished by a finite-monoid non-reducibility certificate of the W_3 row. Take Level_3 on the verification level, and let $\pi_3(D_3)$ be the derived active projection determined by $(C_3, \Theta_3, W_3, \text{Level}_3)$. The routine verifications $v_3^{\text{fin}}, v_3^{\text{cl}}, v_3^{\text{aud}}$ then follow. The nonclaim n_3 excludes conflation with P1 route undefinedness and with direct cross-level comparison. Routes are compared in the category presented by the recorded moves and rules, not by equality of composite maps: here a and b have the same graph on $\{u\}$, and the witness certifies only that the recorded rules do not identify ab with ba . $\square$

The monoid is the first-move monoid of Section 2.

Proposition 5.6 (P4 lower bound). *There exists $D_4 \in \text{FATCD}$ with a derived active projection $\pi_4(D_4)$ witnessing P4.*

Proof. Let D_4 carry index records $0, \frac{1}{2}, 1, 2$; object records $\{0\}, \{1\}, \{2\}$ listing one index record each, and object records $I = \{0, 1, 2\}$ and $I' = \{0, \frac{1}{2}, 1, 2\}$; order records listing the pairs $(0, 1), (0, 2), (1, 2)$ on I and all six pairs of $0 < \frac{1}{2} < 1 < 2$ on I' ; transition records listing the pairs $(0, 1), (0, 2), (1, 2)$ on I and all six pairs of $0 < \frac{1}{2} < 1 < 2$ on I' ; transition records $t_{01} : \{0\} \rightarrow \{1\}$ with $t_{01}(0) = 1$ and $t_{12} : \{1\} \rightarrow \{2\}$ with $t_{12}(1) = 2$; and a refinement record for the inclusion $I \hookrightarrow I'$. Every endpoint and index reference resolves to a declared record. Each order, transition and refinement record carries its certificate as a law field: finite certificates check that both orders are irreflexive and transitive and each relates at least one pair, that each transition is defined on its declared domain with values in its declared codomain, and that the refinement maps I into I' and preserves order, each certificate being the prescribed law formula of Section 2 for its record. Attach $\Theta_4 = \text{Ann}_\Theta[\text{nontrivial stage/refinement}]$ to the cluster C_4 comprising these records, and record a witness-schema annotation naming P4 and the W_4 row; $W_4(C_4)$ holds because the records carry declared endpoints or indices and a law for them. Take Level_4 on the structural-categorical level, and let $\pi_4(D_4)$ be the derived active projection determined by $(C_4, \Theta_4, W_4, \text{Level}_4)$. The routine verifications $v_4^{\text{fin}}, v_4^{\text{cl}}, v_4^{\text{aud}}$ then follow. The nonclaim n_4 excludes bare labels and arbitrary relations or transitions without laws. $\square$

In the running example the transition record for F is present, but no threshold annotation is attached to it, so the P4 channel of D_{ex} is *below_threshold* (Section 7). Only the attachment data are missing there: the cluster formed by the transition record of F , S_0 and the four state records already satisfies W_4 . The description D_4 carries a threshold, a witness-schema and a level annotation on a larger law-bearing cluster, a stage tower $0 < 1 < 2$ with transitions and a refinement law.

Proposition 5.7 (P5 lower bound). *There exists $D_5 \in \text{FATCD}$ with a derived active projection $\pi_5(D_5)$ witnessing P5.*

Proof. Take two distinct declared object records x, y (the micro-elements), an object record listing them as the declared domain of q , a declared state record $*$ and a quotient object record Q listing it, the relation record E , of declared use quotient, relating both, and the map record $q : \{x, y\} \rightarrow Q$, with declared use *packaging*, sending both to $*$. Writing $X = \{x, y\}$ for the declared domain, the finite check record carries the certificate, recomputed true,

$$\begin{aligned} \forall u, v \in X (E(u, v) \leftrightarrow q(u) = q(v)) \wedge x \neq y \wedge q(x) = q(y) \\ \wedge \forall z \in Q \exists u \in X (q(u) = z) \wedge \forall u \in |E| (u \in X) \wedge \forall u \in X (q(u) \in Q). \end{aligned}$$

Thus $x \neq y$, $q(x) = q(y)$, and the complete quotient-validity condition in the W_5 row holds. Let C_5 comprise these declared records and this check, and let D_5 carry them. Attach $\Theta_5 = \text{Ann}_\Theta[\text{nontrivial quotient packaging}]$ to the cluster C_5 comprising these records, and record a witness-schema annotation naming P5 and the W_5 row; $W_5(C_5)$ holds because distinct raw objects are related and packaged together by a packaging map with quotient-validity evidence. Take Level_5 on the verification level, and let $\pi_5(D_5)$ be the derived active projection determined by $(C_5, \Theta_5, W_5, \text{Level}_5)$. The routine verifications $v_5^{\text{fin}}, v_5^{\text{cl}}, v_5^{\text{aud}}$ then follow. The nonclaim n_5 excludes identity packaging and witnesses whose identified records are not distinct. $\square$

Proposition 5.8 (P6 lower bound). *There exists $D_6 \in \text{FATCD}$ with a derived active projection $\pi_6(D_6)$ witnessing P6.*

Proof. Let D_6 consist of state records s_0, s_1, s_2 ; an event record e referencing s_0 ; a trace record τ listing (s_0, s_1, s_2) and referencing e ; a provenance record referencing e and τ ; a map record $t : \{s_0, s_1\} \rightarrow \{s_0, s_1, s_2\}$ of declared use other, with declared object records for its source and target, and $t(s_0) = s_1$, $t(s_1) = s_2$; and a check record referencing τ and t that carries the certificate $\tau_1 = t(\tau_0) \wedge \tau_2 = t(\tau_1)$, recomputed true; together with the annotations listed below. Let C_6 consist of s_0, s_1, s_2, e, τ , the provenance record, t , the source and target object records of t , and the check record. Record a claim annotation c_τ referencing e, τ and the check record, whose statement is the formula $\tau_1 = t(\tau_0) \wedge \tau_2 = t(\tau_1)$ of the check record's certificate (so c_τ satisfies the representation condition of Definition 4.2), with an audit a_τ referencing c_τ, e, τ , the provenance record, t and the check record and certifying precisely those equations from the check record; both are assigned to the active P6 projection; then record the claim naming P6 and C_6 with a second audit a_6 certifying $W_6(C_6)$, whose $\text{Audit}(D)$ conjunct is discharged by a_τ . Attach $\Theta_6 = \text{Ann}_\Theta[\text{nontrivial audit schema}]$ to the cluster C_6 , and record a witness-schema annotation naming P6 and the W_6 row; $W_6(C_6)$ holds because C_6 contains the event e , the trace τ , which references e , and the check record, which references τ and carries $\tau_1 = t(\tau_0) \wedge \tau_2 = t(\tau_1)$, exactly the conjunction of successive-entry replay equalities that the W_6 row requires; and $a_\tau \in \text{Audit}_0(D_6)$ refers to c_τ, e, τ and the check record and concludes that formula. Take Level_6 on the verification level, and let $\pi_6(D_6)$ be the derived active projection determined by $(C_6, \Theta_6, W_6, \text{Level}_6)$. The routine verifications $v_6^{\text{fin}}, v_6^{\text{cl}}, v_6^{\text{aud}}$ then follow. The nonclaim n_6 excludes passive logs, certificate-only content, and partial audit records. $\square$

Propositions 5.3–5.8 supply, for each $i \in \{1, \dots, 6\}$, an admissible witness tuple of the form in Definition 5.2. This proves Theorem 5.1.

The running example D_{ex} of Section 4 is an explicit description of this kind for three roles: its split pair witnesses P1, its quotient $\{A, B\}$ with packaging map q witnesses P5, and its trace with replay and check witnesses P6.

Remark 5.9 (What the lower bounds do and do not establish). Theorem 5.1 asserts that each of the six roles is instantiated somewhere in FATCD. It does not prove exact-six, does not rule out

candidate seventh-role threats, does not prove decomposition, and does not prove uniqueness of any later decomposition/exhaustion record. It also does not assert that any single description activates all six channels simultaneously; Proposition 7.5 (case $k = 6$) constructs one. Those questions are handled in Sections 6 and 7.

6 Upper-bound classification under the recognition hypothesis

The lower bounds of Section 5 rule out the trivialization in which some primitive role lacks an admissible witness in FATCD. The upper-bound direction addresses the complementary question: whether some additional, seventh closure role is needed in the covered scope. This section shows that, under the recognition hypothesis of Definition 6.7, every kind residual in the covered scope, every bookkeeping annotation, and every record residual of every well-formed decomposition record, is licensed only labels among (i)–(ix) of Definition 4.14, and that the recognized forms of probability, causality and agency receive such labels without it. The argument proceeds in three steps. First, we state the alignment rules that constrain how raw features may project to $P1, \dots, P6$. Second, we close the three high-priority candidate seventh-role threats: probability and uncertainty, residual causality, and residual agency. Third, we record the upper-bound classification theorem that collects these local classifications into a single covered-scope statement.

6.1 Role-projection alignment rules

The role meanings of Definition 2.1 carry alignment rules constraining which raw features may be interpreted as active projections of each role. These rules refine the alignments of Remark 2.2 and form the discipline under which the classification theorem is proved.

Proposition 6.1 (Role-projection alignment). *Under honest bookkeeping and the witness discipline of Definition 4.11, each role projection is constrained as follows.*

- *P1: admissible only when the candidate cluster carries a map or partial-map record of declared use update, source and target relation records of declared use quotient, and a certificate of alternative (a) or (b) of the W_1 row. In particular, a cluster with no map or partial-map record of declared use update, or no relation record of declared use quotient, does not satisfy W_1 .*
- *P2: admissible only when the candidate carries a macro specification, a realization relation, specification-satisfaction data, and evidence of certified unrealizability or failure of representability over the declared packaging kernel; when a satisfying realizer exists, the candidate must also carry a comparison with the classes of the kernel of a declared packaging map, whose declared domain contains the declared candidate class, certifying that some class of that kernel meets both the realizer set and its complement in the candidate class. Macro-level predicates or objectives count only insofar as they are treated as content to be represented. When no realizer exists, the candidate class must be the full declared domain of a map record of the cluster of declared use packaging or update, and the cluster must record a true evaluation of the specification at a declared same-kind record outside that class. Mere quotient equality and arbitrary relations without representability structure are not admissible as P2 projections.*
- *P3: admissible only when the candidate carries a route grammar (generating moves and all rewrite rules recorded in D), generated same-source, same-target route terms, and route-comparison data with a non-reducibility certificate of either alternative in the W_3 row. Generic transitions, generic causal arrows, generic path records, and route undefinedness arising from a partial update map are not admissible as P3 projections unless they are two defined same-source, same-target route records carrying a non-reducibility certificate of either alternative in the W_3 row.*

- P4: *admissible only when the candidate carries a stage, order, transition or refinement record (Definition 4.1) with a law in the sense of the witness table (for a transition record, a certificate that its table is defined on its declared domain with values in its declared codomain suffices). Bare labels, arbitrary relations without order, transition, or refinement law, quotient packaging, and audit bookkeeping are not admissible as P4 projections.*
- P5: *admissible only when the candidate carries a quotient relation, a packaging map, the identification of distinct micro-elements, an object record declared as the quotient object, and quotient-validity evidence. Generic structural paths, derivations, topologies or geometries merely because they have paths, and causal chains are not admissible as P5 projections unless quotient or packaging data is present.*
- P6: *admissible only when the cluster carries a declared event, a trace record, a record tied to the event that is that trace record or a provenance record also tied to that trace record, a replay or check record tied to the trace carrying a certificate of a replay or provenance property of the trace, and $\text{Audit}_0(D)$ contains an entry a , not necessarily a cluster member, referring to a claim annotation, to these event and trace records and to that replay or check record, whose conclusion is the formula of that certificate. Passive logs alone, arbitrary metadata, and certificate-only content without audit/provenance/replay/check structure are not admissible as P6 projections.*

Proof. By Definition 4.11, an admissible projection $\pi_i(D, C)$ requires $W_i(C)$, and W_i is defined by the row for P_i of the *Witness types* table in Section 2. Each constraint below follows by reading off the record kinds that row requires; the table is the definition and this proposition states its consequences for candidate features. We go through the roles in turn.

For P1, the fixed meaning is update-vs-quotient compatibility and descent obstruction. The witness type for that meaning is exactly a selected update together with source and target relation records of declared use quotient and a certificate of descent compatibility, descent obstruction, or update-vs-quotient mismatch; absent such a witness the candidate exhibits no descent datum to compare, so it cannot be an active P1 projection. In particular generic semantics, generic predicate meaning, generic selection criteria, and generic agency carry no descent witness and are inadmissible, which is the alignment of Remark 2.2.

For P2, the fixed meaning is macro-specification representability. Its witness type is a macro specification, a realization relation, specification-satisfaction data, and a certificate of unrealizability or of failure of representability over the declared packaging kernel; a macro-level predicate or objective enters only when treated as content to be represented, since only then does it instantiate the representability relation. Mere quotient equality is P5 data, an arbitrary relation lacking representability structure supplies no realization relation, and empirical observations lacking level and threshold data fail the witness discipline; none is an admissible P2 projection.

For P3, the fixed meaning is enriched route/coherence mismatch, and P3 is not a directionality certificate. Its witness type is a route grammar (generating moves and all rewrite rules recorded in D), generated same-source same-target route terms, and route-comparison data carrying a non-reducibility certificate of either alternative in the W_3 row. A candidate without such a route comparison has nothing of the P3 type to compare. Generic transitions and generic causal arrows are P4 data, generic path records are not route-comparison data, and route undefinedness arising from a partial update map is P1 content (Table 1, row P1/P3); none projects to P3 unless the route/coherence threshold is explicitly present and the comparison is specifically of the P3 type. This is the alignment that generic transition is not P3.

For P4, the fixed meaning is stage, order, transition, and refinement structure. Its witness type is a stage, order, transition or refinement record carrying a law beyond bare labels; staged dynamics

and regime ordering qualify when recorded in one of these kinds. Bare labels, arbitrary relations without an order, transition, or refinement law, quotient packaging (which is P5 data), and audit bookkeeping (which is P6 data) supply no such law and are inadmissible as P4 projections.

For P5, the fixed meaning is quotient packaging. Its witness type is a quotient relation, a packaging map, the identification of distinct micro-elements, an object record declared as the quotient object, and quotient-validity evidence. Generic structural paths, derivations, topologies or geometries that merely carry paths, and causal chains supply no quotient relation or packaging map, so they are inadmissible absent quotient or packaging data; this is the alignment that generic path or derivation is not P5.

For P6, the fixed meaning is audit, provenance, replay, and checkability. Its witness type is a declared event, a trace record tied to it directly or through a provenance record, a replay or check record carrying a certified replay or provenance property of the trace, and an entry of $\text{Audit}_0(D)$ concluding that property. Passive logs without check structure, arbitrary metadata, and certificate-only content lacking audit/provenance/replay/check structure supply no checkable audit datum and are inadmissible as P6 projections.

In each case the admissibility constraint is forced by the fixed role meaning through the witness discipline: a projection is admissible only with the witness type that the meaning of the role fixes. Formally, each bullet is the contrapositive of the row conjunct it names: W_1 requires an update map and quotient relations; W_2 a macro-specification and a realization relation; W_3 two defined parallel routes and a certificate; W_4 a law-bearing record; W_5 a quotient relation, a packaging map and a quotient object; and W_6 an event, a trace, a certified check and an Audit_0 entry. The remaining sentences gloss role meanings and are not used. $\square$

Remark 6.2 (Three excluded alignments, held fixed). Proposition 6.1 fixes three alignments that are held throughout: generic transition data does not project to P3, generic path or derivation data does not project to P5, and generic semantic or interpretation data does not project to P1. These are carried into every step of the classification theorem below.

6.2 Probability, causality and agency

Three classes of candidate seventh-role feature require particular scrutiny: probability and uncertainty, residual causality in the sense of the interventional and counterfactual hierarchy [26], and residual agency in the planning sense [7]. Each is classified in a proposition. In the titles of Propositions 6.3–6.5, *closure* of a threat means only that each listed form is licensed a label among (i)–(ix); it is unrelated to closure operations, to the condition $\text{Closed}(D)$, or to the closure content of Definition 6.7. In every case the candidate is classified under the alignment rules of Proposition 6.1 together with the residual scheme of Definition 4.14; no candidate is dismissed by definition. These three come first because they are the most natural candidates for a seventh role; the general classification theorem follows. A *probability or uncertainty feature* is a residual feature with a field of declared probability, weight, kernel, confidence or error type, or an empirical-bridge annotation whose identified bridge claim has such a field in its $\text{Audit}(D)$ entry; a *residual-causality feature* is one with a field of declared intervention, counterfactual or causal-dependence type; and a *residual-agency feature* is one with a field of declared goal, objective, plan, policy or choice type, or an empirical-bridge annotation whose identified bridge claim has such a field in its $\text{Audit}(D)$ entry. Value-type tags enter no label criterion (Definition 4.1), and the proofs below case only on content; hence each proposition below holds for every kind residual whose content takes one of its forms, whether or not it carries such a tag. Each recognized form is specified so that a criterion among (i)–(ix) applies to it; the propositions list which presentations of probabilistic, causal and agentic content the witness rows, seed clauses and annotation kinds already accommodate.

The P_i -form. In these propositions, a feature r has P_i -form when r is a record of a kind occurring in the W_i row of the witness table, contributes a required field to a finite cluster C satisfying $W_i(C)$, and every predicate, comparison, invariant, law or claim field of r is required for W_i at C or is an input to its check (Section 4.6); or when r carries a covered P_i seed and each of its predicate, comparison, invariant, law or claim fields is a named seed field of that clause. Such a feature satisfies criterion (i) when that criterion holds, and criterion (v) otherwise. In the bullets below, P_i -aligned content is labelled (i) or (v); a bullet does not assert that the decomposition record names an active projection for it. A seed alone is not an active projection.

Examples. For instance, a transition record whose fields are its endpoints, a kernel declared with values in the distributions on its state set, and the law that each row is such a distribution has P_4 -form: in an aligned cluster in which it is the only law-bearing record, deleting the row law leaves the cluster without a law, so W_4 fails. By contrast, the predicate N in the original D_{norm} of Remark 6.8 has no P_i -form; neither does a standalone transition record without a certified law or covered seed. For a composite form, the feature is carried by a finite set R_f of records, and each record of R_f has the P_i -form of its own component role (so, by the second sentence of this paragraph, it satisfies criterion (i) or (v), and also (ii) when it contributes required fields to two named active clusters and every predicate, comparison, invariant, law or claim field is required for, or an input to the check of, either cluster's witness predicate).

Composite forms. For a composite form, “classifies as a composite” is a derived reading of the finite record set R_f , not an additional residual label: Classify still labels each member of R_f . Label (ii) applies to a member only under the two-cluster condition just stated.

How the propositions proceed. Each proposition takes a kind residual of D whose content is of the named kind — probability or uncertainty; causation in the interventional and counterfactual sense; agency in the planning sense — and passes it through the residual scheme under the alignment rules. Its proof checks, for each recognized form, that the feature supplies the witness type of the role it is assigned to, or is an annotation or an outside-domain structure; the causality and agency propositions also show that one role is unavailable: P5 without quotient packaging, and P1 without an update-vs-quotient descent witness. A stochastic transition record carrying a certified row law, with no assertion fields beyond the named fields of its covered P_4 seed, has P_4 -form whether or not attachment data are present. Relative to a fixed well-formed $\text{Dec}(D)$, it satisfies criterion (i) if that criterion holds, and criterion (v) otherwise; an uncertainty record with provenance and replay data has P_6 -form when each of its predicate, comparison, invariant, law or claim fields is required for, or an input to, a W_6 check, or is a named field of its seed; an empirical-bridge annotation attaching a measurement error to a substrate observable is licensed label (vii) when it meets criterion (vii), the error record itself being classified by its own fields; and a confidence level recorded as the threshold predicate of a threshold annotation with no other field besides its typing, target and label fields, all of whose read records satisfy (R1), is an activation-threshold refinement. A normalization predicate with no free variable, such as the one in D_{norm} of Remark 6.8, is not a P2 seed merely because it states that finitely many weights sum to one. That remark shows that this specified predicate receives label (x), and discusses conservation laws and emergent obstructions separately. A record of P_i -form satisfies alternative (R1) of Definition 6.7, the threshold and bridge forms below satisfy (R2), and the outside-domain forms satisfy (R5); the propositions below are thus the cases, for these three classes, of the pointwise clause of Theorem 6.9, and assume nothing about the other features of D .

Proposition 6.3 (Closure of probability and uncertainty). (a) *Every kind residual (Definition 4.14), in particular every probability or uncertainty feature, of a description $D \in \text{FATCD}$ whose content takes one of the recognized forms below, where a form naming a P_i projection or unabsorbed P_i -*

aligned content requires the feature to have P_i -form (paragraph The P_i -form before Proposition 6.3), is licensed, relative to every well-formed decomposition record $\text{Dec}(D)$, at least one of the labels of

- P_2 -aligned content, label (i) or (v), when the feature has P_2 -form (for instance a macro-specification with one free variable and a realization relation whose recorded evaluations take both truth values);
- P_4 -aligned content, label (i) or (v), when the feature has P_4 -form (for instance a stochastic transition record carrying its row law);
- P_6 -aligned content, label (i) or (v), when the feature has P_6 -form (for instance an audited uncertainty-bookkeeping record);
- an activation-threshold refinement, when the feature is a threshold annotation targeting a cluster named for an active projection, with a confidence level recorded as its threshold predicate (Definition 4.9) as its only field besides typing, target and label, and that predicate reads at least one record, every record whose values it reads is listed in the annotation's target field, and each such record satisfies (R1); relative to a record that names its target cluster for no active projection, the annotation satisfies criterion (ix-b);
- an empirical-bridge annotation, when the feature is itself an empirical-bridge annotation identifying an admissible bridge claim (Definitions 3.1 and 3.11) whose $\text{Audit}(D)$ entry carries the five items of Definition 3.11, with no fields beyond its typing, targets and that identification;
- an outside-domain structure, label (viii), when an assertion field of the feature uses a declared predicate whose extension D does not record or quantifies over a sort whose carrier D does not record, the asserted property is true in some expansion of D and false in another, and every other assertion field of that record either also meets the outside-scope condition or is required for W_i at a cluster aligned with P_i , an input to its check at such a cluster, or named in a seed clause that the record carries (for instance the IsDist presentation of Remark 6.8);
- label (i), (ii) or (v), as a member of a composite of the foregoing projections, when the feature is one of a finite set of records each of which has the P_i -form of one of the roles above (so, by the paragraph The P_i -form, it satisfies criterion (i) or (v), and also (ii) when it contributes required fields to two named active clusters and every predicate, comparison, invariant, law or claim field is required for, or an input to the check of, either cluster's witness predicate);
- unabsorbed role-aligned content, label (v) of Definition 4.14, when it has P_i -form but criterion (i) does not hold for it, for instance because the cluster lacks the attachment data of Definition 4.9.

A content record targeted by such an empirical-bridge annotation is not thereby classified; it is classified by its own fields and is licensed label (ix-c) exactly when either branch of that criterion holds. Each case licenses a label among (i)–(ix) of Definition 4.14, so neither (x) nor (xi) is licensed for the feature. Probability or uncertainty content of any of these recognized forms therefore requires no seventh role; content of none of these forms is governed by the recognition hypothesis (Definition 6.7).

(b) Let a content record of D have as its only assertion field the formula $\bigwedge_k v(w_k) \geq 0 \wedge \sum_k v(w_k) = 1$ over value fields of pairwise distinct declared records $w_1, \dots, w_n$, none of which is the replaced record, suppose the recorded values $v(w_k)$ are nonnegative and sum to one (so the recomputed row check is true), and let D' be the description obtained from D by replacing that feature, with every incoming reference and affected annotation retargeted in a type-correct way and with the resulting description D' in **FATCD** (in particular, every realization relation still lists exactly the members of its candidate class at which its target's formula is true in D'), by a transition record $K : \{*\} \rightarrow \text{Dist}(\{w_1, \dots, w_n\})$ (here K is a transition record, not a candidate class) whose declared source is an object record $\{*\}$ listing a declared state record $*$, whose declared target is $\text{Dist}(X)$ for an object record X listing

$w_1, \dots, w_n$, recorded by a codomain reference to X and a datum field marking the codomain as distribution-valued (paragraph Laws), whose unique row is the list of terms $v(w_1), \dots, v(w_n)$, and whose law certificate is $(\bigwedge_k v(w_k) \geq 0 \wedge \sum_k v(w_k) = 1, \text{true})$; the records $*$, $\{*\}$ and X are added to D' . Then K is a covered P4 seed of D' and has P_4 -form, so relative to every well-formed decomposition record of D' it is licensed label (i) or (v), and neither (x) nor (xi) (its only assertion field is the law certificate, a named seed field of clause (4), so K has P_4 -form by the second branch of the paragraph The P_i -form); the added records $*$, $\{*\}$ and X have no assertion field and satisfy (R1) or (R3). The w_i and their values are unchanged, source and target declarations are added, and the claim and audit annotations of the replaced feature are retargeted to the row-law check of K ; in D itself the feature is not thereby classified.

Proof. Let r be a probability or uncertainty feature (a kind residual) of $D \in \text{FATCD}$, and let it pass through the residual scheme of Definition 4.14 under the alignment rules of Proposition 6.1. Role-aligned classifications below require the P_i -form specified in the proposition. Such a record satisfies criterion (i) when that criterion holds and criterion (v) otherwise. Composite forms are determined record by record; criterion (ii) applies under the two-cluster condition of the paragraph The P_i -form. We case on the content of r .

If the probabilistic content is a macro specification or a representability or non-representability condition on a description that contributes a required field to a cluster satisfying W_2 , with its realization relation and certificate, then it supplies the P2 witness, so it has P_2 -form and receives label (i) or (v) (paragraph The P_i -form). If instead the content is a stochastic transition, regime, or staged structure carrying a valid stage, order, transition, or refinement law and has P_4 -form, then it receives label (i) or (v); the row law of a kernel is such a law, as in the example before this proposition, and a normalization assertion that is not required for W_4 is carried by a separate predicate record, classified in its own right (Remark 6.8). If the content is an audited record of uncertainty bookkeeping — event reference, provenance, replay or check structure — it has P_6 -form. A P_6 -form record receives (i) when its named active cluster carries the certified trace check and honest audit required by W_6 ; a covered seed without that full witness receives (v). If the feature is a threshold annotation of a cluster named for an active projection whose only field besides typing, target and label is a threshold predicate recording a confidence level that decides when the projection is attempted, and that predicate reads at least one record, every record whose values it reads is listed in the annotation's target field, and each such record satisfies (R1), then it adds no role content and receives label (v-b): it is an activation-threshold refinement. Relative to a record that names its target cluster for no active projection, the annotation is a structural annotation whose fields are its typing, target and threshold predicate, with no independent role-theoretic assertion, and (v), (vi) and (vii) do not apply, so it satisfies criterion (ix-b). If it is an empirical-bridge annotation whose bridge claim is admissible and has an audit entry with the five items of Definition 3.11, and with no fields beyond its typing, targets and that identification, it receives (vii). A content record targeted by such an annotation is not thereby classified; it is classified by its own fields and is licensed label (ix-c) exactly when either branch of that criterion holds. Finally, if several role aspects occur together, take their finite record set R_f . By the composite-form hypothesis each member has its component role's P_i -form, so it satisfies criterion (i) or (v), and also (ii) under the two-cluster condition stated there; the set has the composite reading of those classifications. These are the recognized forms of probability and uncertainty content, and each reduces to the six roles as stated. A probability feature whose content has none of these forms is not classified by this proposition. In particular, the bare normalization predicate of Remark 6.8 receives label (x); other such constraints must be checked against the recognition alternatives of Definition 6.7. A feature whose probabilistic content lies outside the covered scope is classified as an outside-domain structure by Definition 4.14,

not absorbed by default: in the outside-domain case, the asserted property is true in some expansion of D and false in another, and together with the stated condition on the remaining assertion fields this satisfies criterion (viii). The closure is therefore scoped, as stated.

For (b), the row of K is a map from the declared one-point domain to the declared set $\text{Dist}(\{w_1, \dots, w_n\})$, and the finite check of nonnegativity and sum one certifies that its value lies in that codomain. This is a law in the sense of the witness table, so K is a law-bearing transition record, which is a covered P4 seed by Definition 7.1. Its declared source and target records are fixed by reference fields, and the value fields of the w_k are read by that check; none of these records carries an assertion of its own. Reference closure and honest bookkeeping of D' are hypotheses of the clause; the argument uses only the row of K and its law certificate. $\square$

Proposition 6.4 (Closure of residual causality). *Every kind residual (Definition 4.14), in particular every residual-causality feature, of a description $D \in \text{FATCD}$ whose content takes one of the recognized forms below, where a form naming a P_i projection or unabsorbed P_i -aligned content requires the feature to have P_i -form (paragraph The P_i -form before Proposition 6.3), is licensed under the residual scheme, relative to every well-formed decomposition record $\text{Dec}(D)$, at least one of the labels of*

- P_4 -aligned content, label (i) or (v), as the primary classification when the feature has P_4 -form (for instance a directed transition or refinement record with its law);
- P_3 -aligned content, label (i) or (v), only when the feature has P_3 -form and is specifically a route/coherence comparison between generated same-source, same-target routes;
- P_6 -aligned content, label (i) or (v), when the feature has P_6 -form;
- label (i), (ii) or (v), as a member of a composite of P4 and P6, when directed dependence and provenance audit are carried by a finite set of records containing the feature, each of which has P_4 -form or P_6 -form (so it satisfies criterion (i) or (v), and also (ii) when it contributes required fields to two named active clusters and every predicate, comparison, invariant, law or claim field is required for, or an input to the check of, either cluster's witness predicate);
- an outside-domain structure, when an assertion field of the feature (for instance a metaphysical-causation assertion) uses a declared predicate whose extension D does not record, the asserted property is true in some expansion of D and false in another, and every other assertion field of that record either also meets the outside-scope condition or is required for W_i at a cluster aligned with P_i , an input to its check at such a cluster, or named in a seed clause that the record carries;
- unabsorbed role-aligned content, label (v) of Definition 4.14, when it has P_i -form but criterion (i) does not hold for it, for instance because the cluster lacks the attachment data of Definition 4.9.

Classification as P5 is inadmissible in the absence of quotient packaging. Each case licenses a label among (i)–(ix) of Definition 4.14, so neither (x) nor (xi) is licensed for the feature. For these recognized forms, the covered content requires no seventh role; the outside-domain case receives label (viii) without a claim about roles beyond the covered scope. None of these classifications uses the recognition hypothesis (Definition 6.7); content of none of these forms is governed by it. An interventional mechanism is covered when recorded as a law-bearing transition or refinement record of P_4 -form, for instance one whose only assertion field is its law certificate (first bullet); a further assertion field serving no witness can leave it licensed only (x); a counterfactual assertion is covered only when it takes one of the listed forms, and otherwise falls under the recognition hypothesis.

Proof. Let r be a residual-causality feature (a kind residual) of $D \in \text{FATCD}$, causal content being taken from the hierarchy of Pearl [26], and pass r through the residual scheme of Definition 4.14 under Proposition 6.1. The forms of the proposition cover interventional mechanisms recorded

as transitions of P_4 -form; counterfactual assertions not of these forms fall under the recognition hypothesis. A causal feature presents a directed dependence, so we case on how that directedness is carried. Role-aligned and composite forms are labelled as in the paragraph *The P_i -form* before Proposition 6.3: each record receives (i) or (v), and (ii) under the two-cluster condition stated there.

If r has P_4 -form as a directed transition, refinement, or staged-dependence law, it receives label (i) or (v) by the paragraph *The P_i -form*; this is the primary classification. The record kind and its law alone do not establish the assertion-field guard defining P_4 -form. One case in which directedness routes elsewhere is when r is specifically a route/coherence comparison between generated same-source, same-target routes carrying a non-reducibility certificate of either alternative in the W_3 row; only then does it supply the P3 witness, have P_3 -form and receive label (i) or (v). We must not assign a generic causal arrow to P3, since by the alignment rules a generic transition is not a route comparison of the P3 type. If r is instead a provenance or dependency audit — event reference with a provenance or trace record — it has P_6 -form. A P_6 -form record receives (i) when its named active cluster carries the certified trace check and honest audit required by W_6 ; a covered seed without that full witness receives (v). When directed dependence and provenance audit occur together, take their finite record set R_f . Its P_4 -form and P_6 -form members are licensed (i) or (v) individually by their P_i -form; together they have the stated composite reading. Finally, if r records a metaphysical-causation assertion whose declared predicate has no recorded extension in D , its truth is not certified by the finite checks of D . In the outside-domain case, the asserted property is true in some expansion of D and false in another. Together with the stated condition on the remaining assertion fields, this satisfies criterion (viii).

Classification as a P5 projection is not available here: by Proposition 6.1 a P5 projection requires a quotient relation and packaging map, and a causal feature carrying no quotient packaging supplies none; the directedness of a causal chain is not packaging data. Hence the P5-inadmissibility caveat holds. The cases above are the recognized forms of residual-causality content; a causality feature whose content is of none of these forms is not classified here and falls under the recognition hypothesis (Definition 6.7). The closure is scoped and makes no scope-free claim. $\square$

Proposition 6.5 (Closure of residual agency). *Every kind residual (Definition 4.14), in particular every residual-agency feature, of a description $D \in \text{FATCD}$ whose content takes one of the recognized forms below, where a form naming a P_i projection or unabsorbed P_i -aligned content requires the feature to have P_i -form (paragraph *The P_i -form* before Proposition 6.3), is licensed under the residual scheme, relative to every well-formed decomposition record $\text{Dec}(D)$, at least one of the labels of*

- *label (i), as a constituent of an active P2, P4 or P6 projection, when the feature has the corresponding P_i -form and $\text{Dec}(D)$ names, for an active P_i projection, the cluster C of the paragraph *The P_i -form* (for the seed branch of that paragraph, any named cluster containing the feature);*
- *label (i), (ii) or (v), as a member of a composite of P2, P4, and P6, when the feature is one of an objective or specification record of P_2 -form, a transition, regime, or staged selection record of P_4 -form, and an audited decision-provenance record of P_6 -form, each licensed (i) or (v) by its P_i -form;*
- *an empirical-bridge annotation, when the feature is itself an empirical-bridge annotation identifying an admissible bridge claim (Definitions 3.1 and 3.11) whose $\text{Audit}(D)$ entry carries the five items of Definition 3.11, with no fields beyond its typing, targets and that identification;*
- *an outside-domain structure, when a proposed irreducible intentional or teleological predicate has no recorded extension in D , the asserted property is true in some expansion of D and false in another, and every other assertion field of that record either also meets the outside-scope condition*

or is required for W_i at a cluster aligned with P_i , an input to its check at such a cluster, or named in a seed clause that the record carries;

- unabsorbed role-aligned content, label (v) of Definition 4.14, when it has P_i -form but criterion (i) does not hold for it, for instance because the cluster lacks the attachment data of Definition 4.9.

A content record targeted by such an empirical-bridge annotation is not thereby classified; it is classified by its own fields and is licensed label (ix-c) exactly when either branch of that criterion holds. An update map receives (i) when it lies in a named active W_1 cluster and its assertion fields meet criterion (i); otherwise it receives (v) when it carries a covered P1 seed whose clause names each of its predicate, comparison, invariant, law or claim fields, or belongs to a failed W_1 offer each of whose such fields is an input to the recorded check; and (ix-c) when it supplies no witness field and has no such field; an update map of none of these three forms must be tested against the other recognized forms of this proposition; only when none applies is it not classified here and left to the recognition hypothesis. Each case licenses a label among (i)–(ix) of Definition 4.14, so neither (x) nor (xi) is licensed for the feature. For these recognized forms, the covered content requires no seventh role; the outside-domain case receives label (viii) without a claim about roles beyond the covered scope. None of these classifications uses the recognition hypothesis (Definition 6.7); content of none of these forms is governed by it.

Proof. Let r be a residual-agency feature (a kind residual) of $D \in \text{FATCD}$, agency being taken in the planning sense of Bratman [7], and pass it through the residual scheme of Definition 4.14 under Proposition 6.1. An agency feature in admissible form presents an objective pursued through selection and recorded for later inspection; we case on whether that content reduces to role witnesses or asserts more. Role-aligned and composite forms are labelled as in the paragraph *The P_i -form* before Proposition 6.3: each record receives (i) or (v), and (ii) under the two-cluster condition stated there.

If r has just one of the stated P_2 -, P_4 - or P_6 -forms and receives (i), criterion (i) supplies its named active cluster, W_i witness and attachments; hence it is the corresponding single-role projection. In the composite case, take the finite record set R_f carrying the objective or specification, staged selection, and decision provenance. By the stated P_2 -, P_4 - and P_6 -form hypotheses, each member receives (i), (ii) or (v) individually. Together those records have the asserted composite reading. If instead r is an empirical-bridge annotation whose bridge claim is admissible and has an audit entry with the five items of Definition 3.11, and with no fields beyond its typing, targets and that identification, it receives (vii). A content record targeted by such an annotation is not thereby classified; it is classified by its own fields and is licensed label (ix-c) exactly when either branch of that criterion holds. If r records a proposed irreducible intentional or teleological predicate whose extension D does not record, its truth cannot be checked over D . In the outside-domain case, the asserted property is true in some expansion of D and false in another. Together with the stated condition on the remaining assertion fields, this satisfies criterion (viii). A finite audited intentional-content claim with a recorded predicate extension and no recognized six-role disposition remains subject to the recognition hypothesis.

Classification as a P1 projection is not available here merely because agency selects an update: by Proposition 6.1 a P1 projection requires a selected update together with source and target relation records of declared use quotient and an explicit descent-compatibility, descent-obstruction, or update-vs-quotient mismatch witness. A selection event carrying no such witness supplies no P1 witness. Its update map receives (i) when it lies in a named active W_1 cluster and its assertion fields meet criterion (i); otherwise it receives (v) when it carries a covered P1 seed whose clause names each of its predicate, comparison, invariant, law or claim fields, or belongs to a failed W_1 offer each of whose such fields is an input to the recorded check; and (ix-c) when it supplies no

witness field and has no such field. It has P_4 -form only when recorded as a law-bearing transition record. This is the stated P1 caveat. The cases above are the recognized forms of residual-agency content; content of none of these forms falls under the recognition hypothesis (Definition 6.7). The closure is scoped and asserts nothing beyond it. $\square$

Remark 6.6 (Scope of the closures). Propositions 6.3–6.5 close the three threats within the covered scope; they do not make scope-free claims. A feature that falls outside the covered scope is classified as an outside-domain structure via Definition 4.14, not absorbed into a primitive role by default. Each named form is checked against its label criterion directly, without requiring recognition of every feature of D ; the propositions give the applicable label criterion for each named form and state when P5 and P1 projections are unavailable. Each recognized form in Propositions 6.3–6.5 is alternative (R1), (R2), (R3) or (R5) of Definition 6.7: each individual record in a recognized form satisfies (R1), (R2), (R3) or (R5), and composite forms are assessed member by member; the propositions verify those alternatives for the listed forms, and add that a P5 reading (causality) or a P1 reading (agency) is unavailable without quotient-packaging or descent data.

6.3 The recognition hypothesis and the classification theorem

Each alternative of Definition 6.7 is the decomposition-independent form of label criteria of Definition 4.14: (R1) of (i) or (v); (R2) of (v), (vi), (vii), (ix-a) and (ix-b); (R3) of (ix-c); (R4) of (iii) and (iv); and (R5) of (viii). Theorem 6.9 verifies that this matching does not depend on the chosen decomposition record.

The hypothesis below is imposed on three sets: the kind residuals of Definition 4.14 (content records and structural annotations, whose role reading is not fixed by their kind), all bookkeeping annotations, and the set $\text{Residual}(D)$ of each well-formed decomposition record (Definition 7.2). Stated for every record, it does not depend on a choice among them. Since $\text{Residual}(D)$ consists of kind residuals and bookkeeping annotations, the third clause below follows from the first two: D satisfies the recognition hypothesis if and only if every kind residual and every bookkeeping annotation of D is recognized, a condition that refers to no decomposition record.

Definition 6.7 (Recognition hypothesis). Terms used below: *aligned*, *contributes a required field*, *required for*, *input to the check*, *declared operation*, *independent assertion* and *outside the covered scope* are defined in the paragraph *Terms used in the label criteria* after Definition 4.14; *covered seed*, *named seed field* and *failed offer* in Definition 7.1; *honestly recorded* (of an offer: its audit is honest, Definition 4.6(b),(d)). The *closure content* of a feature r is the finite induced subdescription consisting of r , its fields, and all declared records reached by following its references in D or read by its formula-valued fields; it is used in Remark 6.8 (paragraph *Verification*). The alternatives below are evaluated in the whole of D , and a reference makes data available for checking without by itself certifying a role witness. A description $D \in \text{FATCD}$ satisfies the *recognition hypothesis* when every kind residual r of D (Definition 4.14), every bookkeeping annotation r of D (whether or not a decomposition record assigns it to a channel), and every element r of $\text{Residual}(D)$ for every well-formed decomposition record of D (Definition 7.2), is *recognized*, that is, satisfies at least one of the following:

- (R1) (first branch) r itself contributes a required field to a finite cluster aligned with one of the six witness types of the *Witness types* table of Section 2, or itself carries a covered seed for one of them (Definition 7.1), and every predicate, comparison, invariant, law or claim field φ of r is, for some index i_φ and some finite cluster $C_\varphi \ni r$ aligned with P_{i_φ} , required for W_{i_φ} at C_φ or an input to its check, or is named in a seed clause that r carries (so r carries no assertion beyond these fields; content beyond them must be recorded in a separate declared content

record, which is classified on its own); or (failed-offer branch) r is a member of the cluster of an honestly recorded failed offer for some P_i , and each of its predicate, comparison, invariant, law or claim fields is an input to the recorded check of W_i on that cluster; merely referencing a witness cluster does not suffice;

- (R2) *Structural annotations.* r is one of the six structural annotations of Definition 4.2; its fields contain only its declared operation, targets, and the finite typing, checking, preserved or suppressed data, losses and nonclaims required for that operation, and it makes no independent assertion (both terms as defined after Definition 4.14); in particular an instrument-visibility annotation whose fields other than its typing and target fields are those required by Definition 3.9, a lift annotation whose fields are those required by Definition 3.6, and an empirical-bridge annotation meeting criterion (vii), whose fields are only its typing, targets and the identification of its bridge claim and five items, satisfy this alternative, provided each of their formula-valued fields meets the condition of the next sentence. A formula-valued field of a structural annotation (a threshold predicate, a preserved or lost invariant) is part of its declared operation only when it reads at least one record, every record whose values it reads is listed in the annotation's target field, and each such record satisfies (R1); otherwise it is an independent assertion, and this alternative fails. *Bookkeeping annotations.* The alternative is also met when r is one of the seven bookkeeping annotations (for $D \in \text{FATCD}$ every condition below except the last, on field roles, holds automatically by Definitions 4.4–4.7) of Definition 4.2, it is finitely represented, has finite typed fields, and every reference it carries resolves to a declared finite record of D (Definitions 4.4 and 4.5), it is honest if it is an audit annotation, it carries the fields Definition 4.6(g) requires of its kind, and, if it is a claim annotation, it is referred to by an honest audit annotation and satisfies the representation condition (Definition 4.6(e)), and every field of r other than the statement and hypothesis fields of a claim annotation, the certificate field of an audit annotation and its conclusion field when its conclusion is true in every well-typed interpretation or is represented, in the sense of Definition 4.2, by fields of the content records named by the claim it audits, and the formula schema of a witness-schema annotation has field role datum or reference; a substantive assertion in a claim statement is represented by content records (Definition 4.2), which are assessed under (R1)–(R5);
- (R3) r is role-neutral data without an assertion of its own: r supplies no witness field and has no predicate, comparison, invariant, law or claim field. Typical instances are the inputs that a specified field of a feature satisfying (R1) uses in checking a required or seed field, and declared objects or states that no role datum uses; also a predicate record supplying no witness field whose only assertion field is a formula with at least one free variable whose truth value is not the same under all assignments respecting the declared sorts of its free variables, using only predicates whose extensions D records, and quantifying only over sorts whose carriers D records, each free variable ranging over a record kind or the rational sort, referenced by no claim annotation, and referenced by certificates only through the abbreviation $s(t)$ of the paragraph *Check language* (a definition, which has no truth value over D);
- (R4) r begins a finite chain $r = r_0, \dots, r_n$ of distinct declared records, with $n \geq 1$, such that, for each $j < n$, a field of r_j records a checked equality or isomorphism of presentation data with specified fields of r_{j+1} , or a checked level translation to r_{j+1} ; and, for each $j < n$, apart from one comparison field recording the equality, isomorphism or level translation with r_{j+1} , and the datum and reference fields it compares, r_j has no predicate, comparison, invariant, law or claim field. The terminal r_n satisfies (R1), (R2), (R3) or (R5). An equality or isomorphism of presentation data (a value-preserving relabelling of datum fields; no structural isomorphism is

meant) is a finite bijection β , recorded in the comparison field, from specified datum fields that are not term-valued (or the member list of an object record, a reference field) of r_j to datum fields that are not term-valued (or the member list of an object record, a reference field) of r_{j+1} , with a correct certificate, recorded true, that each recorded value equals that of its β -image; its formula is a conjunction of equalities between record identifiers, recorded field values and entries of recorded lists (written with indexing, as in the running example of Section 4). A checked level translation from r_j to r_{j+1} is a comparison field of r_j referencing a translation annotation carried in $\text{Audit}(D)$ whose source and target contexts are levels recorded by level annotations whose target fields list r_j and r_{j+1} , respectively, together with a finite bijection β of specified datum fields as above and a correct certificate, recorded true, that is such a conjunction of equalities, that each value equals its β -image;

- (R5) r lies outside the covered scope (Definition 4.14): a field of r asserts a property whose check is not finite over the records of D : it quantifies over a declared domain for which D supplies no finite checking range, or uses a declared predicate whose extension D does not record, and its truth value is not fixed by the recorded values of D (it is true in some expansion of D and false in another, in the sense of the paragraph *Certificates, assertion language and laws* of Section 2; for a formula with free variables: for some assignment α of its free variables and two expansions of D whose carriers contain the values α gives to variables of unlisted sorts, it is true under α in one and false under α in the other). In addition, every other predicate, comparison, invariant, law or claim field of r either has this property or is required for W_i at a cluster C aligned with P_i , an input to the check of W_i at such a cluster, or named in a seed clause that r carries; r is then recognized as outside-domain structure.

For $D \in \text{FATCD}$, every bookkeeping annotation whose fields other than the exempt fields listed in the second clause of (R2) are datum or reference fields satisfies the second clause of (R2); hence, when all bookkeeping annotations of D have this form, D satisfies the recognition hypothesis if and only if every kind residual of D is recognized. A feature satisfying (R1) is absorbed under label (i) when a decomposition record places it in an active projection under Proposition 6.1, and is covered under (v) when it remains unabsorbed. Remark 6.8 shows that some descriptions in FATCD do not satisfy this hypothesis.

Alternatives (R1)–(R5) are record-independent forms of label criteria: (R1) of (i) or (v); (R2) of (v), (vi), (vii), (ix-a) or (ix-b); (R3) of (ix-c); (R4) of (iii) or (iv); and (R5) of (viii) (the table in the proof of Theorem 6.9).

Remark 6.8 (The recognition hypothesis is a genuine restriction). The recognition hypothesis is an explicit assumption on D , not a consequence of the FATCD definitions, and some descriptions in FATCD do not satisfy it. The remark establishes five facts, each under the paragraph named: $D_{\text{norm}} \in \text{FATCD} \setminus \text{FATCD}_{\text{RH}}$, so $\text{ScopedExactSix}(\text{FATCD})$ fails (*The counterexample, Verification*); a transition-row presentation of the same weights lies in FATCD_{RH} (*Repair by a transition row*); every closed check-language formula without atomic W_j formulas has a covered P2-seed presentation, given two distinct declared records of one kind (or after adjoining them), whenever the completed description is again in FATCD (*Wrapping a closed formula as a P2 seed*; membership can fail when a correct certificate, or a realization relation's target formula, quantifies over all declared records of an adjoining kind); membership in FATCD_{RH} depends on presentation (*Scope depends on presentation*); and label (xi) is realized by D_{scr} (*A description passing the screen*). The paragraphs *One record suffices for recognition*, *Other unrecognized assertions* and *Open question* add a one-record failed-offer presentation, further unrecognized forms with the (R2) and (R3) presentation devices, and the open question. A record without an assertion of its own — one with no witness field and no predicate,

comparison, invariant, law or claim field, such as a declared object record — is recognized by (R3). The normalization predicate in the following description is not recognized.

The counterexample D_{norm} . Let D_{norm} , the description of Remark 4.8, consist of state records $w_1, \dots, w_n$ ($n \geq 1$), each with a nonnegative rational value field $v(w_k)$, the values summing to one (for instance $n = 2$ and $v(w_1) = v(w_2) = 1/2$); a predicate record N , declared as a macro-specification, whose formula asserts $\bigwedge_k v(w_k) \geq 0 \wedge \sum_k v(w_k) = 1$; a claim annotation naming N whose statement is the formula of N ; and an honest audit annotation listing the certificate $(\bigwedge_k v(w_k) \geq 0 \wedge \sum_k v(w_k) = 1, \text{true})$ and certifying that formula.

Verification. D_{norm} is finite, closed and audited, so $D_{\text{norm}} \in \text{FATCD}$. It contains no map, relation, comparison, path, derivation, order, transition, refinement, event, provenance, trace, replay or check record; in particular the closure content of N , namely N with $w_1, \dots, w_n$, contains none. So no cluster satisfies any W_i and no record is a covered seed (Definition 7.1). Hence N does not satisfy (R1); being a content record, not an annotation, it does not satisfy (R2); its predicate field contains a closed formula and it is referenced by a claim annotation, so neither branch of (R3) applies; it records no presentation or level translation, so not (R4); and N gives its predicate by the explicit formula $\bigwedge_k v(w_k) \geq 0 \wedge \sum_k v(w_k) = 1$ over recorded field values; this formula uses no declared predicate symbol and quantifies over no sort, so its check is finite over D_{norm} and (R5) fails. Although N is declared a macro-specification, it is not a covered seed: D_{norm} has no realization relation to a declared nonempty candidate class, and no realization relation can target N , since Definition 4.1 requires its target formula to have exactly one free variable. For the same reasons no criterion (i)–(ix) of Definition 4.14 holds for N : it is not in any cluster or composite, re-presents or restates nothing, is not an annotation, relates no instrument or substrate observable, is inside the covered scope, and carries an assertion. With no failed offer and no outside-scope declaration, every channel of D_{norm} is *absent_with_record*, so no active projection absorbs N , and N receives label (x) in every well-formed decomposition record. So ScopedExactSix(FATCD) fails, and the six roles are not *recognition-complete* for FATCD: some in-scope feature of some description in FATCD is placed by none of (i)–(ix). Label (x) records an unresolved candidate, not a seventh role: (xi) would further require the recorded witness of category (xi), itself a recorded screen rather than the identification of a new closure role.

Repair by a transition row. For distribution-valued weights the hypothesis constrains presentation, not content. For nonnegative rational weights summing to one, replace N by a transition record $K : \{*\} \rightarrow \text{Dist}(\{w_1, \dots, w_n\})$ whose unique row is the list of terms $v(w_1), \dots, v(w_n)$, so that K reads the value fields of the w_k and records no copies. Declare a state record $*$, a one-point object record $\{*\}$ listing it, and an object record X listing $w_1, \dots, w_n$, give K a codomain reference to X and a datum field marking it distribution-valued, and take as the law certificate of K the pair $(\bigwedge_k v(w_k) \geq 0 \wedge \sum_k v(w_k) = 1, \text{true})$, which expresses that the row lies in $\text{Dist}(X)$. Retarget the claim and its audit to K , replacing every reference to N by a reference to K . Set the claim's statement and the audit's conclusion to the law-certificate formula, and replace the audit's certificate list by that certificate, recomputed true, naming K and its declared inputs. Thus K represents the statement (Definition 4.2), and the audit is honest. Then K is a covered P4 seed satisfying (R1); $*$, $\{*\}$, X and the w_k carry no assertion field and satisfy (R1) or (R3); the claim and audit annotations satisfy (R2); and the resulting description lies in FATCD_{RH} .

Wrapping a closed formula as a P2 seed. In outline: the construction adjoins a predicate record s with one free variable over two records $c \neq c'$, a realization relation recording $s(c) = \text{true}$ and $s(c') = \text{false}$, and a one-class packaging map q with a comparison record; it chooses between two formulas for s according to the value of φ in the completed description D' . More generally, let φ be any closed check-language formula containing no atomic W_j formula and $c \neq c'$ declared records of one kind. Let D' be the final description, obtained once every record of this construction (including

the records $X_c, *, Q, q$ and the comparison record introduced under *Activation* below, the P2 claim and its audit, all attachment annotations and any fresh candidate records, when used) is adjoined and any removal below is made. The predicate record is $s(r) := \varphi \wedge r = c$ if φ holds in D' , and $s(r) := \varphi \vee r = c$ otherwise.

Choice of branch. Fix all additions, removals and retargetings, and all non-formula field values, before choosing the branch. The two branches differ only in the formula field of s , the retargeted claim's statement and its audit's conclusion. A check-language formula never reads a statement or conclusion (formula-valued fields are not terms) and reads a predicate record's formula only through an abbreviation $s(t)$ naming it; s is adjoined, so φ contains none. Hence φ has the same value in both candidates for D' . Its value agrees with its value in D provided every field it reads is unchanged and every quantifier has the same checking range. The construction applies when D' is again in FATCD. Adjoining records can falsify a correct certificate of D whose quantifier ranges over all declared records or over the declared records of an adjoined kind (for instance $(\forall x \in \text{State}(x = w_1 \vee x = w_2), \text{true})$, which the adjoined state record $*$ falsifies). A sufficient condition is that every retained realization relation still lists exactly its satisfying members and records correct evaluations, every retained certificate and law remains correct, every retained audit remains honest, and all additions, removals and retargetings preserve typing, reference closure, representation and the required annotation fields.

Seed and claim. This predicate record, declared a macro-specification whose variable r is declared to range over the common kind of c and c' , with a realization relation declaring $K = \{c, c'\}$, listing exactly c as satisfying, and recording both evaluations, is a covered P2 seed. Let the realization relation carry the evaluations as certificates $(s(c), \text{true})$ and $(s(c'), \text{false})$. The record that carried φ as an assertion field, if any (such as N), may be removed only if neither φ nor its named inputs reference it, it is neither c nor c' , and every incoming reference is a bookkeeping reference that can be retargeted to s and the realization relation while preserving typing, representation and audit honesty; otherwise that record is retained and classified separately; a claim whose statement is φ is retargeted to s and the realization relation, with statement φ in the first branch and $\neg\varphi$ in the second; its audit lists both certificates and concludes that statement. The statement is entailed by the signed certificates, so the claim satisfies the representation condition of Definition 4.2 and the audit is honest. Its predicate field is a named seed field, so it satisfies (R1). It receives (i) when criterion (i) holds and (v) otherwise.

Activation. Adjoining an object record X_c listing c, c' , a state record $*$ with a quotient object record Q listing it, a map record $q : X_c \rightarrow Q$ of declared use packaging with $q(c) = q(c') = *$, and a comparison record carrying $(q(c) = q(c') \wedge s(c) \wedge \neg s(c'), \text{true})$ gives the following. Let C contain c, c', s , the realization relation, $X_c, *, Q, q$ and the comparison record. Then $W_2(C)$ holds: the realization relation records both evaluations on $K = \{c, c'\}$, and the single kernel class of q meets the realizer set $\{c\}$ and its complement $\{c'\}$. Record a claim naming P2 and C with statement $W_2(C)$, and an honest audit listing the row's content certificates and $(W_2(C), \text{true})$ and concluding $W_2(C)$, and supply all attachments and level and lift conditions of Definition 4.11; fresh candidate records may be used to avoid pre-existing level, lift or forgetting annotations meeting C . The resulting projection is active. Since q is a covered P5 seed, in D' channel P2 is active and channel P5 is not *absent_with_record*; the transition-row repair likewise makes P4 at least *below_threshold*. Removing s from this cluster falsifies W_2 , and its sole predicate field is required for that witness, so criterion (i) applies in a decomposition record naming this projection.

What is recorded. This construction records the truth value of φ in D' in recognized seed content: in the first branch the recorded evaluation $s(c) = \text{true}$ entails φ , while in the second branch $s(c') = \text{false}$ entails $\neg\varphi$. Thus an assertion true in D' is recorded with its checked content; an assertion false in D' is represented by its checked negation. Agreement with its truth value in D

requires the preservation conditions above. The transition-row presentation above likewise preserves the checked normalization content for nonnegative weights of pairwise distinct records.

One record suffices for recognition. Let c be a declared record, $s(r) :\equiv \varphi \wedge r = c$, and let a realization relation with candidate class $K = \{c\}$ and target s carry $(s(c), \text{true})$ if φ holds in the completed description and $(\neg s(c), \text{true})$ otherwise. Fix all additions and fields other than that certificate's formula and the relation's list of satisfying members before selecting the branch. Assume φ reads neither changing field in either completion, including through quantified field lookups or predicate abbreviations. Then φ has the same value in both completions, so the branch choice is well defined. The construction applies when the completed description, with s , the relation, the claim and its audit adjoined, is again in FATCD. Record a claim naming P2 and $C = \{c, s, \text{Real}\}$ with an honest audit listing that certificate and $(W_2(C), \text{false})$. Since $K = \{c\}$, no kernel class can meet both the realizer set and its complement in K , and no evaluation outside K is recorded in C ; hence $W_2(C)$ is false in every well-typed interpretation; the formula of s and the certificate are inputs to the recorded check, so s and the relation satisfy the failed-offer branch of (R1) and receive $(v\text{-}c)$, and the certificate entails φ or $\neg\varphi$. Channel P2 is then *collapsed_boundary_case* unless active. The two-record proviso is needed only for the covered-seed and active presentations.

Scope depends on presentation. The covered scope also depends on presentation. Replace the formula of N by $\text{IsDist}(v(w_1), \dots, v(w_n))$, declare IsDist by a predicate record of arity n , each argument place declared rational-sorted, with no formula field and give no relation record for it, so that D does not record its extension (the declaring record has no assertion field and satisfies (R3); had the declaring record given an explicit formula in its n rational argument variables, such as $\bigwedge_k x_k \geq 0 \wedge x_1 + \dots + x_n = 1$, D would record the extension and N would again receive (x)), set the claim's statement to $\text{IsDist}(v(w_1), \dots, v(w_n))$, and let the audit list no certificate, conclude the valid formula $v(w_1) = v(w_1)$, and list the IsDist field of N in its non-certification field. This description remains honestly audited. Its records w_k and the declaring record of IsDist satisfy (R3), its claim and audit annotations satisfy the second clause of (R2), and N satisfies (R5) and receives (viii) rather than (x); so the description lies in FATCD_{RH} while D_{norm} does not, and no role is certified for N . Hence membership in FATCD_{RH} is not invariant under replacing an explicit formula by a declared predicate name whose extension D does not record, and Theorem 9.1 asserts nothing about content presented that way. More generally, replace at least one assertion field of an unrecognized kind residual by a declared predicate name whose extension the resulting description does not record. The changed record satisfies (R5) provided every other assertion field, evaluated in the resulting description, also lies outside the covered scope, is required for or an input to W_i at an aligned cluster containing the record, or is named in a seed clause the record carries. Unchanged records must be recognized separately. The result lies in FATCD_{RH} if it is again in FATCD (in particular, no replaced field is the formula of a realization relation's target), every remaining kind residual is recognized, and every bookkeeping annotation has only datum and reference fields besides the exempt fields of (R2). Theorem 9.1 certifies no role for the replaced fields.

Other unrecognized assertions. A standalone predicate record stating a conservation or balance law, or an assembled obstruction or holonomy, by a closed check-language formula over recorded inputs is not recognized as presented if it carries no covered seed, contributes no required witness field, is not an input to a prescribed witness check, and records no checked presentation or level translation. Under these conditions (R1)–(R5) fail as for N . An in-scope content record that contributes a required field to an aligned cluster or carries a covered seed, and whose assertion fields are each required at, or inputs to the check at, some aligned cluster containing it, possibly different clusters for different fields, satisfies (R1); one with no assertion field and no witness field satisfies (R3). Label (x) is also licensed for an unclaimed predicate record whose closed formula is true or false in D , such as $0 + 0 = 0$ or $0 + 0 = 1$ between rational constants: label (x) marks an assertion

placed by none of (i)–(ix), which need be neither true nor a candidate role. By contrast, the same record with formula $\theta \wedge r = r$ for a fresh variable r , still unclaimed and referenced by no certificate, does not satisfy the second branch of (R3), since its value does not depend on r , and still receives (x). Conversely, if c is a declared record of the kind over which r ranges and that kind has a second declared record, the unclaimed record with formula $\theta \wedge r = c$ (for θ true in D) or $\theta \vee r = c$ (for θ false in D), referenced by no certificate, satisfies the second branch of (R3), since its value at c differs from its value at the other records of that kind; for unclaimed closed assertions this branch constrains presentation, not content. The same holds for the structural-annotation branch of (R2): for a closed check-language formula θ that reads no record values outside a binary relation record E satisfying (R1), and a fixed declared record identifier u , a threshold annotation targeting $\{E\}$ with predicate $\theta \wedge (E(u, u) \vee \neg E(u, u))$ satisfies (R2) and receives (ix-b) (no cluster $\{E\}$ satisfies any W_i , so (v-b) cannot apply), whether or not $\text{Audit}(D)$ records the predicate checked true, provided the description with this annotation adjoined is again in FATCD. Since the W_6 certificate states a replay or provenance property of its trace, an arbitrary assertion about recorded values does not become a required W_6 field by adjoining a trace.

A description passing the screen. Let D_{scr} consist of index records 0, 1, an object record X listing them, an order record o for $0 < 1$ with its law certificate, a relation record r (the feature to be screened; not the free variable r of a macro-specification) of declared use other carrying a finite table field c with $c(0) = c(1) = 1$ and a datum field holding the list (id_X) , with certificates of extensivity, monotonicity, $c \circ c = c$, closure of the list under composition and $c \neq \text{id}_X$ (the prescribed conditions of criterion (xi)(d)–(e), with o as the order record and (id_X) as L), a witness-schema annotation naming a candidate outside $P_1, \dots, P_6$ with $\psi(C) :\equiv r \in C \wedge o \notin C$, where $r \in C$ abbreviates the finite disjunction $\bigvee_{d \in C} r = d$ over the listed identifiers of C and $o \notin C$ the conjunction $\bigwedge_{d \in C} o \neq d$, so that each instance $\psi(C')$ is a closed check-language formula, a check record carrying $\psi(\{r\})$, $\neg W_1(\{r\})$, $\dots$, $\neg W_6(\{r\})$, $\neg \psi(\{o\})$ and the finite conjunction over all $C' \subseteq \text{Raw}(D_{\text{scr}})$ of $(W_1(C') \vee \dots \vee W_6(C')) \rightarrow \neg \psi(C')$, a claim referencing r and that check record whose statement is these conditions, and an honest audit certifying it. The W_j atoms in these certificates are evaluated by the witness rows, which read no certificate containing a W_j atom and, for W_6 , only $\text{Audit}_0(D_{\text{scr}})$, which excludes this audit; so the certificates are well defined although the check record belongs to some clusters C' . Only o is law-bearing and no record supplies data for W_1, W_2, W_3, W_5, W_6 , so $W_i(C')$ holds only if $o \in C'$, where $\psi(C')$ is false; no map, partial-map, transition or refinement record is present, so the generated monoid is $\{\text{id}_X\}$. Recording c by a map record would place it in the list of criterion (xi)(e), and r would fail the screen. The assertion fields of r are required for no witness, r carries no seed, begins no (R4) chain and lies in the covered scope, so r satisfies (x) and (xi); the check record, which serves no witness and carries no seed, also receives (x) but not (xi), since it records no closure operation. Thus (xi) is realizable in FATCD, and its emptiness in Theorem 6.9 comes from the hypothesis.

Open question. Which assertion-bearing records admit a reduction in which the assertion is itself a required field of an active cluster (label (i)), without adjoining wrapper records such as the realization relation and one-class packaging map above, or a trace reference, is left open and is recorded with the endogenous-instrument meta-limit of Sections 11.3 and 12.1. The three threat closures of Propositions 6.3–6.5 label features of their recognized forms without the hypothesis; the classification theorem below assumes it in general.

Here and in the entry *Closed unconditionally* of the working vocabulary in Section 1.4, the phrase means that the stated recognized forms of Propositions 6.3–6.5 receive labels without assuming the recognition hypothesis. It does not assert that hypothesis for every feature in those three threat classes; Theorems 6.9 and 9.1 retain it for general residual exhaustion.

Three ingredients combine in the next theorem. The alignment rules of Proposition 6.1 fix which features each role can claim. The threat closures of Propositions 6.3–6.5 classify the recognized forms of probability, causality and agency. The residual scheme of Definition 4.14 supplies the labels. Under the recognition hypothesis, the theorem assigns every residual feature a label other than an unresolved threat or a seventh role: its proof matches each recognized disposition to one of labels (i)–(ix). The three preceding propositions classify their stated recognized forms without that hypothesis.

The proof sorts a recognized feature by its disposition, following the label table after Definition 4.14; a feature of any raw-grammar or annotation kind receives the label of the disposition its content has. The alternatives (R1)–(R5) of the recognition hypothesis are written to correspond to the label criteria, so the proof matches each alternative to a criterion and uses no substantive property of the particular witness rows (Remark 6.10); which features the hypothesis admits and which it excludes is described in Remark 6.8. Label (viii) and the channel status *inapplicable_outside_scope* of Definition 7.1 both concern content outside the covered scope: the label sorts a single feature, and the status marks a channel whose role content lies there.

Theorem 6.9 (Upper-bound classification). *Let $D \in \text{FATCD}$ satisfy the recognition hypothesis (Definition 6.7), let $\text{Dec}(D)$ be a well-formed decomposition record of D (Definition 7.2), and let r be a kind residual of D (Definition 4.14: any content record or structural annotation, including members of active clusters), a bookkeeping annotation of D (whether or not $\text{Dec}(D)$ assigns it to a channel), or a record residual, an element of $\text{Residual}(D)$. Then some criterion among (i)–(ix) holds for r relative to $\text{Dec}(D)$; hence neither (x) nor (xi) is licensed for r , and every label that the label criteria after Definition 4.14 license for r relative to $\text{Dec}(D)$ is one of*

- (i) a declared constituent of the cluster that $\text{Dec}(D)$ names for an active P1, P2, P3, P4, P5, or P6 projection, in the sense of criterion (i) of Definition 4.14;
- (ii) a shared constituent of two such projections;
- (iii) a presentation variant of an already classified feature;
- (iv) a level shift of an already classified feature;
- (v) unabsorbed role-aligned content, covered-seed content with no certified witness W_i , content checked in an honestly failed offer, or an activation-threshold refinement;
- (vi) an instrument-visibility artifact;
- (vii) an empirical-bridge annotation;
- (viii) an outside-domain structure;
- (ix) already-covered bookkeeping or annotation data, or role-neutral data without an assertion of its own.

In particular, categories (x) and (xi) of Definition 4.14, namely the unresolved-threat category and the recorded seventh-role screen, are empty: no candidate seventh-role feature in the covered scope remains as an unresolved threat, and no feature in the covered scope passes the recorded seventh-role screen of criterion (xi); here “no unresolved threat” means that some criterion among (i)–(ix) holds for r . Since seed clause (2) of Definition 7.1 accepts any one-variable macro-specification with a realization relation whose recorded evaluations take both truth values, this certifies the form in which r is recorded, not that its assertion concerns the role whose seed it carries (Remark 6.10). Record residuals. If r is an element of $\text{Residual}(D)$, every label licensed for it lies in (iii)–(ix). Pointwise form. The conclusion also holds pointwise, without the recognition hypothesis for the rest of D : for every $D \in \text{FATCD}$, every well-formed $\text{Dec}(D)$ and every r as above that satisfies

one of (R1)–(R5), every label licensed for r lies in (i)–(ix), and in (iii)–(ix) if $r \in \text{Residual}(D)$. For each fixed well-formed decomposition record, alternatives (R1)–(R5) supply, respectively, a criterion among (i)/(v), (v)/(vi)/(vii)/(ix-a)/(ix-b), (ix-c), (iii)/(iv), and (viii), as the proof below establishes.

Proof. Alternatives (R1)–(R5) of Definition 6.7 were written as record-independent forms of label criteria (i)–(ix) (paragraph before Definition 6.7); the proof checks, for a fixed well-formed record, that each alternative meets one of those criteria. Let $D \in \text{FATCD}$, fix a well-formed decomposition record $\text{Dec}(D)$, and let r be a kind residual of D (Definition 4.14), a bookkeeping annotation of D , or an element of $\text{Residual}(D)$ for that record. Fix any label L that the label criteria after Definition 4.14 license for r relative to $\text{Dec}(D)$. The label L is drawn from (i)–(xi); we show that, under the recognition hypothesis, it lies in (i)–(ix), so that (x) and (xi) are empty.

By the label table, L is (x) only if no criterion (i)–(ix) holds for r , and L is (xi) only if r has the property in (x).

Conclusion. Each recognition alternative of Definition 6.7 supplies a label criterion among (i)–(ix):

Alternative	Label criterion supplied
(R1)	(i) if criterion (i) holds; otherwise (v-a) for the first branch of (R1) and (v-c) for its failed-offer branch
(R2)	(vi) for visibility; (vii) for an admissible bridge claim (Definition 3.1) whose associated audit carries the five items of Definition 3.11; (v-b) for a threshold annotation targeting a cluster named for an active projection and carrying only a threshold predicate besides typing, target and label fields, which reads at least one record, every record it reads being listed in its target field and satisfying (R1); (ix-b) for remaining structural annotations satisfying (R2); (ix-a) for bookkeeping annotations satisfying the second clause of (R2)
(R3)	(ix-c)
(R4)	(iii) or (iv), the criterion of the terminal record being supplied by the rows for (R1), (R2), (R3) and (R5); no recursion occurs, since the terminal record satisfies (R1), (R2), (R3) or (R5)
(R5)	(viii)

Under the recognition hypothesis (Definition 6.7), every kind residual of D , every bookkeeping annotation of D and every element of $\text{Residual}(D)$ for every well-formed decomposition record of D , in particular r , satisfies one of (R1)–(R5), so by the table a criterion among (i)–(ix) holds for it; for (R4) the terminal record of the chain supplies it, and chains are finite. This uses recognition of r alone and, for (R4), the criterion that (R4) requires of the chain’s terminal record, so it holds pointwise in any $D \in \text{FATCD}$. Thus the criterion of (x), which requires that no criterion (i)–(ix) hold, fails; since (xi) requires (x), it fails as well. Hence L lies in (i)–(ix), and categories (x), the unresolved-threat category, and (xi), the recorded seventh-role screen, are empty. If $r \in \text{Residual}(D)$ for a decomposition record, L is neither (i) nor (ii): by Definition 7.2, that set omits the features already accounted for by an active projection, and (ii) requires $r \in C_i \cap C_j \subseteq A(\text{Dec}(D))$; so every label licensed for a record residual, relative to that record, lies in (iii)–(ix). The emptiness of (x) and (xi) is the content of the recognition hypothesis, not a consequence of the FATCD definitions: by Remark 6.8 a finite audited conservation law or an emergent composite obstruction, if unrecognized, would fall under (x), and under (xi) only if it also passes the recorded seventh-role screen; Remark 6.8 separately exhibits an unrecognized normalization predicate in FATCD. $\square$

Remark 6.10 (Scope of the theorem). Theorem 6.9 is conditional on the recognition hypothesis (Definition 6.7) and scoped to FATCD. Alternative (R1) yields criterion (i) or (v); (R2) yields (v),

(vi), (vii), (ix-a) or (ix-b); (R3) yields (ix-c); (R4) yields (iii) or (iv); and (R5) yields (viii). The alternatives are stated without reference to a decomposition record; the theorem records that these record-independent forms exclude (x). It does not derive that no finite audited feature requires a seventh role; its general conclusion assumes the recognition hypothesis. Propositions 6.3–6.5 classify only their stated recognized forms without that hypothesis. It does not claim that no seventh role can arise outside FATCD, that the alignment rules are exhaustive for richer domains, or that the recognition hypothesis holds in general. Nor does it prove exact-six on its own: the decomposition/exhaustion theorem of Section 7 remains necessary. Which further classes of assertion-bearing records are recognized, and broader questions, are deferred to Sections 11.3 and 12. For each fixed well-formed $\text{Dec}(D)$, the proof matches alternatives (R1)–(R5) to criteria (i)–(ix). This matching uses no substantive property of the particular witness rows or seed clauses. It remains valid for any finite family of check-language witness predicates and seed clauses, provided the recognition alternatives, label criteria, named seed fields, alignment and required-field conventions, and active-cluster assignments are changed consistently. The number six enters Theorem 9.1 through Definition 7.2, and the particular predicates through Theorem 5.1 and Propositions 6.3–6.5. The (R1) and (R4) cases use that record's active-cluster assignments and checked chains. The theorem proves that recognition excludes (x), using the table in its proof. The converse does not follow from the label criteria: bookkeeping fields may receive (ix) while substantive assertions require separate recognition, and criteria (iii) and (iv) accept a chain whose terminal record lies in $A(\text{Dec}(D)) \cup S(\text{Dec}(D))$, for instance a member of an active or failed-offer cluster with an assertion field serving no witness check, which need not satisfy (R1)–(R5). Criteria (vi) and (ix-b) impose the (R2) requirement that every record whose values a formula-valued prescribed field reads satisfy (R1), and each implies the structural-annotation branch of (R2). The upper bound therefore depends on the stated recognition hypothesis. Recognition is not necessary for the condition of Definition 4.18: adding to the nonclaim annotation of D_5 a claim field with formula $0 + 0 = 1$ gives $D \in \text{FATCD} \setminus \text{FATCD}_{\text{RH}}$ in which every well-formed record places that annotation (the only nonclaim annotation targeting C_5 , which the active P5 entry must assign) in $A(\text{Dec}(D))$, so no kind or record residual is licensed (x) or (xi). The proof uses recognition of kind residuals, bookkeeping annotations and record residuals. Since seed clause (2) accepts any one-variable macro-specification with a realization relation whose recorded evaluations take both truth values, (R1) certifies the form of a record, not that its assertion concerns the role whose seed it carries.

7 Decomposition and exhaustion

Under the recognition hypothesis every well-formed decomposition record meets the condition of Definition 4.18 (Theorem 6.9), so the exact-six question reduces to the existence of one well-formed record, which the theorem below proves for every description. Given $D \in \text{FATCD}$, is there a well-formed record that lists the six role channels, gives each a status, and accounts for every remaining raw feature through an inactive status record or a residual classification? This section shows that there is, and explains why a channel is not the same as an active role. The lower bounds show that each of the six channels is materially realizable in the domain; the upper-bound classification shows that, under the recognition hypothesis, no kind residual or record residual is licensed label (x) or (xi).

7.1 Decomposition records and channel statuses

Definition 7.1 (Channel status). For $D \in \text{FATCD}$, a decomposition record $\text{Dec}(D)$ (Definition 7.2) and $i \in \{1, \dots, 6\}$, the *role channel* $\text{Chan}_i(D)$ is the pair of i and the status entry a record $\text{Dec}(D)$ chooses for i ; the status value $\text{Status}_i(D)$ is fixed below from D alone, and the record only chooses

the cluster, offer, seed or declaration it names; the channel exists whether or not the status is active. A *channel status* for $\text{Chan}_i(D)$ is exactly one of the five values

- *active_projection* — an admissible $\pi_i(D)$ exists with $W_i(C)$ and all required attachments;
- *collapsed_boundary_case* — no admissible $\pi_i(D)$ exists, and an honestly recorded offer of a specified cluster has check result $W_i(C) = \text{false}$;
- *below_threshold* — neither preceding case applies, and a covered seed for P_i is present;
- *inapplicable_outside_scope* — none of the preceding cases applies, and an honest outside-scope declaration is present;
- *absent_with_record* — none of the preceding cases applies; the decomposition records that absence.

In prose these are called active, collapsed, below threshold, outside scope and absent. The status name *collapsed_boundary_case* refers to a failed offer; it is unrelated to the collapse-case annotations of Definition 4.9 and to the forbidden collapses of Table 1. The status $\text{Status}_i(D)$ depends only on D ; $\text{Chan}_i(D)$ and $\text{Residual}(D)$ are taken relative to a fixed record $\text{Dec}(D)$, and we write $\text{Chan}_i(\text{Dec}(D))$ and $\text{Residual}(\text{Dec}(D))$ when the record matters. Since $\text{Chan}_i(D)$ contains its status entry, the pair $(\text{Chan}_i(D), \text{Status}_i(D))$ in Definition 7.2 repeats the status as a separate component. A *covered seed* for P_i is, respectively:

- (1) a map or partial-map record whose declared use is *update*, together with a relation record of declared use quotient that it references as its source relation, or with the relation record serving as source relation in a cluster satisfying W_1 that contains it;
- (2) a predicate record whose predicate field is declared as a macro-specification whose formula has exactly one free variable r , occurring in it and ranging over records of the candidates' kind, and whose evaluations recorded, as in the W_2 row, in a field of the realization relation or of a comparison or check record of D , at declared records of that kind, are correct (each equals the value of the formula at that record in D) and include both truth values, together with a realization relation (a relation record of declared use realization) whose target field is this predicate record and whose declared candidate class is a nonempty set of declared records of that kind; the seed consists of these two records, and either may be named as carrying it; the named seed fields are the formula field of the predicate record and, on the realization-relation record, its candidate-class and target fields, its list of satisfying members, its recorded evaluations of that formula (as datum values or as certificates $(s(t), b)$ at declared records t of the candidates' kind), and its certificate exhibiting a satisfying realizer or checking that none exists; no other assertion field of either record is a named seed field;
- (3) a certificate, of one of the two types in the W_3 row and naming as inputs two path or derivation records with the same declared endpoints, that these are not related by the rewrite rules recorded in D , carried in a field of a content record of D ;
- (4) a law-bearing stage, order, transition or refinement record;
- (5) a relation record whose declared use is *quotient*, or a map record whose declared use is *packaging*;
- (6) an event, provenance or trace record with its declared event or trace identity and reference fields, or a replay or check record with its declared trace reference and a certificate whose formula has exactly one of the two forms prescribed for the W_6 row after the witness table, with that trace as τ (annotations are not seeds).

Unlike the no-realizer branch of the W_2 row, clause (2) does not require the candidate class to be the full declared domain of a packaging or update map. Every closed check-language formula without W_j

atoms admits the seed-(2) presentation of Remark 6.8, provided the completed wrapped description lies in FATCD and contains two distinct declared records of one kind (paragraph *Wrapping a closed formula as a P2 seed*). Such a seed supplies recognized P2 content but does not by itself certify $W_2(C)$ or an active P2 projection. In “covered seed”, “covered” names the seed clauses (1)–(6); it is not a reference to the covered scope.

The named seed fields are: in (1), the graph of the update map, any source-relation reference it carries, and the class data of the source relation; in (3), the fields of the comparison certificate; in (4), the law certificate and the endpoint or index fields it checks; in (5), the class data of the relation or the graph and quotient-object reference of the map; and in (2) and (6), the fields named there.

A record *carries* a seed when it is the map record in (1), the predicate or realization-relation record in (2), the record holding the certificate in (3), the law-bearing record in (4), the relation or map record in (5), or the event, provenance, trace, replay or check record in (6); a named seed field lying on another record is an input to that record’s seed. A comparison or check record whose recorded evaluations of a clause-(2) formula are used to meet that clause does not carry that seed. This does not exclude recognition through other fields or role data. Its classification must be checked against all the criteria of Definition 4.14; if none of (i)–(ix) holds, it is licensed (x). By the witness table, every cluster satisfying W_i contains a covered seed for P_i (for $i = 2$ the row, like seed clause (2), requires the recorded evaluations to be correct): the update map with its source relation data, the macro-specification with its realization relation, the comparison certificate of the two routes, the law-bearing record, the quotient relation or packaging map, and the event or trace record, respectively. A *failed offer* is an honestly recorded attempt to use a specified cluster C as a P_i witness for which the recorded check gives $W_i(C) = \text{false}$. An *outside-scope declaration* for P_i is a claim annotation naming P_i and a record r , with no witness-cluster field (so it is not an offer and its audit records no check of W_i), whose referenced record r lies outside the covered scope (the criterion after Definition 4.14) and whose kind occurs in the W_i row of the witness table; its honest audit lists the outside-scope field of r in its non-certification field. The status of $\text{Chan}_i(D)$ is the first that applies in the order: *active_projection* if some cluster supports an admissible derived projection $\pi_i(D)$ in the sense of Definition 4.11, including $W_i(C)$ and every applicable attachment of Definition 4.9; *collapsed_boundary_case* if there is a failed offer; *below_threshold* if a covered seed is present; *inapplicable_outside_scope* if there is an outside-scope declaration; and *absent_with_record* otherwise. An inactive status is recorded in $\text{Dec}(D)$; it is not a role witness. We write $\text{Status}_i(D)$ for the channel status of $\text{Chan}_i(D)$.

Statuses are assigned only by the ordered rule above. An admissible projection takes priority over every failed offer or incomplete seed. In its absence, a failed offer gives *collapsed_boundary_case*; otherwise a covered seed lacking a complete admissible projection gives *below_threshold*. Only then are an outside-scope declaration and the absent fallback considered. Proposition 2.6 uses the same two labels for local attempt verdicts, which need not equal channel statuses. In the running example, the P4 channel is *below_threshold*: its transition record is present, but no threshold annotation targets a cluster containing it.

Definition 7.2 (Decomposition/exhaustion record). A *decomposition/exhaustion record* for $D \in \text{FATCD}$ is a tuple

$$\text{Dec}(D) = (\text{Raw}(D), (\text{Chan}_i(D), \text{Status}_i(D))_{i=1}^6, \text{Residual}(D), \text{Classify}, \mathcal{L}, \mathcal{V}, \text{Audit}(D), \mathcal{N})$$

where

- the six role channels together with their statuses form the six-channel status sequence, each $\text{Status}_i(D)$ being the value that the ordered rule of Definition 7.1 assigns. If $\text{Status}_i(D) =$

active_projection, the channel carries an active projection $\pi_i(D)$ with its full attachment data. If $\text{Status}_i(D)$ is inactive, the channel carries the corresponding below-threshold, collapsed-boundary, absent-with-record, or outside-scope status record;

- $\text{Residual}(D) \subseteq \text{Raw}(D)$ is the residual feature set, defined as follows. Let $A(\text{Dec}(D))$ be the union of the content clusters of the active projections and the attachment annotations (the threshold, witness-schema, level and collapse-case annotations attached to the cluster, the claim annotation naming P_i and the cluster, the audit annotations referring to that claim, the nonclaim annotations targeting it, the translation and lift or forgetting annotations that Definition 4.11 requires for a level crossing of the projection, and, for P_6 , the event-or-trace claim and honest audit used to establish the W_6 audit-entry conjunct) that the channel entry assigns to its projection (an active entry assigns at least one annotation of each kind that Definition 4.11 requires for admissibility and may leave further annotations targeting the cluster unassigned). Let $S(\text{Dec}(D))$ be the union of the clusters and offers explicitly named by the inactive status entries: the records of the failed offered cluster with the offer's claim annotation and the audits referring to it, for *collapsed_boundary_case*; all records of the covered seed it names (for clauses (1) and (2), both records the clause names), for *below_threshold*; and the outside-scope declaration with the audits referring to it, for *inapplicable_outside_scope*; *absent_with_record* accounts for no feature, and a status entry accounts only for records it names. Then $\text{Residual}(D) = \text{Raw}(D) \setminus (A(\text{Dec}(D)) \cup S(\text{Dec}(D)))$, relative to the chosen record, and Classify assigns to each $r \in \text{Residual}(D)$, including any bookkeeping annotation not assigned to a channel, a label in the scheme of Definition 4.14; already covered bookkeeping may receive (ix);
- $\mathcal{L}$ collects the loss and lift records for any level crossings involved in projections or classifications;
- $\mathcal{V}$ collects instrument-visibility and empirical-bridge annotations where these are asserted;
- $\text{Audit}(D)$ is the honest bookkeeping subrecord;
- $\mathcal{N}$ is the explicit nonclaim record.

A decomposition/exhaustion record is *well-formed* when each $\text{Status}_i(D)$ is drawn from the five values of Definition 7.1, each active $\pi_i(D)$ is admissible in the sense of Definitions 4.11 and 4.9, each inactive status record names a failed offer, covered seed or outside-scope declaration of D for that channel, as Definition 7.1 requires (none for *absent_with_record*), and every raw feature r lies in $A(\text{Dec}(D))$, or in $S(\text{Dec}(D))$, or in $\text{Residual}(D)$ and receives a label, and, for each $r \in \text{Residual}(D)$, $\text{Classify}(r)$ is a label whose criterion in the table after Definition 4.14 holds for r relative to $\text{Dec}(D)$. The records in $\mathcal{L}$ and $\mathcal{V}$ are auxiliary annotations on this coverage, not additional role channels. The status entries and the collections $\mathcal{L}$, $\mathcal{V}$ and $\mathcal{N}$ are data of $\text{Dec}(D)$. Each cluster, seed, offer or declaration named by a status is selected from, or supported by, the records of $\text{Raw}(D)$ and $\text{Audit}(D)$ required above; the selection and the status entries need not themselves be raw records. Forming $\text{Dec}(D)$ does not change D .

By the ordered rule of Definition 7.1, the status $\text{Status}_i(D)$ depends only on D , not on the decomposition record; the cluster, offer, seed or declaration that a record names, and hence $A(\text{Dec}(D))$, $S(\text{Dec}(D))$ and $\text{Residual}(D)$, may depend on the record. Because $\text{Residual}(D)$ leaves out every feature already accounted for by the active projection named for its channel or by an inactive status record, labels (i) and (ii) of Definition 4.14 do not occur for a residual feature of $\text{Dec}(D)$: both require the record to lie in a cluster named for an active projection. This is why the results below list labels (iii)–(ix).

Two residual sets. A residual feature in the sense of Definition 4.14 is also called a *kind residual*, in statements and elsewhere, and an element of $\text{Residual}(D)$ a *record residual*. Membership in the first depends on the raw-grammar kind of a feature, and membership in the second on whether the record $\text{Dec}(D)$ accounts for it through a channel. In particular, a bookkeeping annotation is not a kind residual but may be a record residual if the chosen channel assignments do not account for it. A kind residual absorbed into an active projection is not a record residual, which is why label (i) does not occur among record residuals. The recognition hypothesis and Theorem 6.9 cover both sets.

7.2 Existence, six-channel exactness, and residual exhaustion

Theorem 7.3 (Decomposition existence). *Every description $D \in \text{FATCD}$ admits a well-formed decomposition/exhaustion record $\text{Dec}(D)$ in the sense of Definition 7.2.*

Proof. For each $i \in \{1, \dots, 6\}$, range over the finitely many clusters $C \subseteq \text{Raw}(D)$. If some C satisfies $W_i(C)$ together with every item of Definitions 4.9 and 4.11, name one such C , record $\text{Status}_i(D) = \text{active_projection}$ and the active projection $\pi_i(D)$ it determines; otherwise record the inactive status that the ordered rule of Definition 7.1 assigns, and name one failed offer, covered seed or outside-scope declaration for the first case of the rule that applies. Six status entries result, one per role. The raw features absorbed by these channel records are removed from the residual stage: the statuses, their supporting clusters and their assigned annotations are selected first, and $\text{Residual}(D)$ is then formed by the set difference of Definition 7.2. For each r in it, choose a licensed label among (i)–(ix) if one applies. If none applies, then r is not outside the covered scope (otherwise (viii) would apply), D is finite, and every claim concerning r is referred to by an honest audit annotation; so the criterion of (x) holds. Thus Classify is total, and the record is well-formed irrespective of which labels occur. An unassigned bookkeeping annotation is assessed by the same criteria, ordinarily (ix). Relative to the chosen record, (i) cannot apply to a record residual, since absorbed content was placed in $A(\text{Dec}(D))$; under the recognition hypothesis (Definition 6.7), Theorem 6.9 further places every label in (iii)–(ix), with (x) and (xi) empty. The records $\mathcal{L}$ and $\mathcal{V}$ are read from $\text{Raw}(D)$ and $\text{Audit}(D)$; the bookkeeping subrecord is $\text{Audit}(D)$; and $\mathcal{N}$ records the nonclaims carried forward from the witness discipline. The resulting tuple is a decomposition/exhaustion record, well-formed by construction. Non-vacuity of the six channels follows from Theorem 5.1, which supplies, for each channel, a description in which that channel is active. $\square$

Theorem 7.4 (Six-channel exactness and residual exhaustion). *For every $D \in \text{FATCD}$ and every well-formed $\text{Dec}(D)$ in the sense of Definition 7.2:*

- (i) *the channel sequence $(\text{Chan}_i(D))_{i=1}^6$ has length exactly six, one per primitive role P1, ..., P6;*
- (ii) *every raw feature of D is accounted for at the channel/exhaustion level by an active projection $\pi_i(D)$, a channel-status record for an inactive channel, or a residual classification under the scheme of Definition 4.14, possibly with label (x) or (xi); since label (x) applies to in-scope features for which no criterion (i)–(ix) holds, this clause alone does not exclude an unresolved candidate (see (iii)); the records in $\mathcal{L}$ and $\mathcal{V}$ are auxiliary annotations on this coverage;*
- (iii) *under the recognition hypothesis (Definition 6.7) and within the covered scope, no raw feature is classified as an unresolved seventh-role threat or as passing the recorded seventh-role screen (Theorem 6.9); the three principal threat classes are closed without the recognition hypothesis, in the recognized forms of Propositions 6.3–6.5;*
- (iv) *the status sequence $(\text{Status}_i(D))_{i=1}^6$ is the same for every well-formed $\text{Dec}(D)$.*

Proof. Exactness of the channel sequence is built into the definition of $\text{Dec}(D)$; there are six role channels by Definition 7.2. Coverage of raw features is the well-formedness condition of Definition 7.2 together with totality of Classify on $\text{Residual}(D)$, which holds by the case argument in the proof of Theorem 7.3. Emptiness of categories (x) and (xi), under the recognition hypothesis (Definition 6.7) and within the covered scope, is Theorem 6.9 directly. The last assertion of (iii) is Propositions 6.3, 6.4 and 6.5, which hold for every $D \in \text{FATCD}$ and every well-formed $\text{Dec}(D)$ without the recognition hypothesis. For (iv), Definition 7.2 records each status by the ordered rule of Definition 7.1, whose clauses refer only to D . $\square$

7.3 Channel-versus-activation discipline

Theorem 7.4 asserts six channels in every decomposition/exhaustion record. It does not assert six active projections in every record, nor does it assert that a well-formed record is unique. The distinction between channel presence and channel activation prevents six-channel exactness from being read as an all-six-active claim.

Membership of $D_1, \dots, D_6$ in FATCD_{RH} is verified record by record in the proof of Theorem 9.1; that verification uses only Theorem 5.1 and the definitions. In brief: every content record of D_i contributes a required field to C_i or carries a covered seed, each of its assertion fields being required or a named seed field (R1), or it has no assertion field (R1 or R3); every structural annotation of D_i carries only its typing, target and label or level field, and every bookkeeping annotation only datum and reference fields besides its exempt fields (R2).

Proposition 7.5 (Channel-versus-activation discipline). *Let $D \in \text{FATCD}$ and let $\text{Dec}(D)$ be a well-formed decomposition/exhaustion record. The following are distinct statements:*

- (i) Six channels: the channel sequence $(\text{Chan}_i(D))_{i=1}^6$ has length six with the fixed role names $P_1, \dots, P_6$.
- (ii) Six active projections: for each i , $\text{Status}_i(D) = \text{active_projection}$ and the channel carries an active $\pi_i(D)$.

Theorem 7.4 establishes (i) for every $D \in \text{FATCD}$. It does not establish (ii). In particular, an admissible $D \in \text{FATCD}$ may have any number of active channels between zero and six; inactive channels are recorded with status *below_threshold*, *collapsed_boundary_case*, *absent_with_record*, or *inapplicable_outside_scope* and are not active witnesses for the corresponding roles. For each $k \in \{0, \dots, 6\}$ some $D \in \text{FATCD}_{\text{RH}}$ has exactly k active channels, and some single description in FATCD_{RH} has channels of all five statuses.

Proof. Statements (i) and (ii) differ whenever some channel is inactive, which the last clause provides for every $k < 6$.

Exactly k active channels. For $k = 0$, let D_0 consist of a single object record: it is finite and closed, it has no claims, so it is audited, and it has no cluster satisfying any W_i , no failed offer, no covered seed and no outside-scope declaration, so the status rule of Definition 7.1 gives every channel *absent_with_record*. Its object record has no witness field and no predicate, comparison, invariant, law or claim field, so it satisfies (R3) of Definition 6.7 and $D_0 \in \text{FATCD}_{\text{RH}}$. For $1 \leq k \leq 6$, choose k distinct indices S and let D_S be the disjoint union of the descriptions D_i , $i \in S$, of Theorem 5.1, with record identifiers made disjoint by an injective renaming applied recursively to every record-identifier occurrence in reference, datum, term-valued and formula-valued fields, including identifiers inside finite lists, tuples, relation classes and map tables, and the argument lists of W_j atoms. Rational constants and other non-identifier values are unchanged. A numeral in record-sorted position (paragraph *Sorts* of Section 2), such as 0 and 2 in the formula of s in D_2 , is a record-identifier occurrence and is renamed; a numeral in rational-sorted position, such as an

element of the monoid list of D_3 , is not. Each condition of Definition 4.7 concerns a record, an annotation or a claim together with the records it refers to, which lie in one component, except that certificate correctness and the evaluation of W_j atoms may read description-wide data (all declared records or those of one kind, the rule list R , $\text{Audit}_0(D)$); the next sentences check that this data leaves every recomputed value unchanged. The certificates' quantifiers are bounded by listed object records or finite field lists, except $\forall u \in |E|$, which is guarded by $E(u, u)$; this is false at every record of another component, since renaming keeps the classes of E inside its component. So the content checks retain their values in D_S . The W_1, W_2, W_4, W_5 checks remain within their components. The W_3 row also requires the rules of its cluster to be all rewrite rules recorded in the description; since the indices in S are distinct and D_3 is the only component recording rewrite rules, this clause keeps its value in D_S . For $W_6(C_6)$, the honest event-or-trace audit entry of D_6 remains in $\text{Audit}_0(D_S)$, so its separate audit conjunct also remains true. Thus the audits remain honest and $D_S \in \text{FATCD}$. For $i \in S$, the cluster C_i with its attachments still satisfies W_i , so channel P_i is *active_projection*. For $j \notin S$, no component has a witness-schema annotation naming P_j , together with a threshold and an honest audit of $W_j(C)$ for a cluster C , and making record identifiers disjoint creates no such annotation or cross-component reference. So no admissible $\pi_j(D_S, C)$ in the sense of Definition 4.11 exists, and channel P_j is not active. Hence D_S has exactly k active channels. Each of these descriptions lies in FATCD_{RH} . The inventory in the proof of the last clause of Theorem 9.1 (Section 9, p. 88), which uses Theorem 5.1 and the definitions but not this proposition, establishes (R1) or (R3) for every content record of each D_i using records and checks within that component, and (R2) for its structural and bookkeeping annotations. Under this renaming each certificate, claim statement and audit conclusion of D_i becomes its renamed image, which recomputes to the same value in D_S . They apply to every kind residual and every possible record residual of D_S , independently of the chosen decomposition record. The case D_0 was checked above.

All five statuses. For the five statuses, D_S supplies *active_projection* and D_0 supplies *absent_with_record*. Let o be the object record of D_0 . Let D_c consist of state records $x \neq y$, an object record X_c listing x and y , a relation record E on X_c of declared use quotient with the single class $\{x, y\}$, a claim naming P5 and $C = \{x, y, E\}$, and an honest audit listing $(W_5(C), \text{false})$; no packaging map is present, so this failed offer lies on a cluster carrying a covered P5 seed, and channel P5 is *collapsed_boundary_case*. Here x, y and X_c satisfy (R3), E carries a covered seed whose class data are named seed fields and satisfies (R1), and the annotations satisfy (R2). Let D_o be D_0 together with a predicate record Q (here Q is a predicate record, unrelated to the quotient objects Q of the W_5 row and Remark 6.8, and o is the object record of D_0 , not the order record of D_{scr}) of arity one, its argument place declared to range over object records, with no formula field and no relation record presenting it, a predicate record s whose only assertion field is $Q(o)$, a claim naming P2 and s with statement $Q(o)$, and an honest audit that lists no certificate, records no check of W_2 (so the claim is not an offer), concludes $o = o$ and lists the field of s in its non-certification field. This claim is an outside-scope declaration, so channel P2 of D_o is *inapplicable_outside_scope*. The declaration is not an admissible claim in the sense of Definition 3.1, since no audit certifies $Q(o)$; Definition 7.1 requires only that its audit be honest, and the conclusion $o = o$ makes the audit honest while certifying nothing about Q . In D_o o and Q satisfy (R3), s satisfies (R5), and the annotations satisfy (R2). The description D_{ex} of Section 4 supplies *below_threshold* for its P4 channel. It lies in FATCD_{RH} . The records F, E , the split-pair comparison, $q, \{A, B\}$, the quotient-validity check, the event, the trace and the check record each contribute a required field to C_1, C_5 or C_6 , and their assertion fields (the split-pair, validity and replay certificates) are required for W_1, W_5 and W_6 respectively, so these records satisfy (R1). The provenance record carries a covered P6 seed, and the transition record carries a covered P4 seed whose law field is a named seed field; both satisfy (R1). The state records, S_0 , the object records A, B and the replay record have

no assertion field and satisfy (R1) or (R3); the index record satisfies (R4) by the one-step chain to E , which satisfies (R1); and its annotations satisfy (R2).

All five in one description. Let D_{all} be the renamed disjoint union of D_6 , D_c , D_o and D_T , where D_T has a state record z , an object record Z listing it, and a transition record $T : Z \rightarrow Z$, $T(z) = z$, whose law field carries the certificate, recomputed true, that T is defined on Z with values in Z . The argument for D_S gives $D_{\text{all}} \in \text{FATCD}$; T carries a covered P4 seed with named law field and satisfies (R1), z and Z satisfy (R1) or (R3), and the other components are inventoried above, so $D_{\text{all}} \in \text{FATCD}_{\text{RH}}$. No component supplies a P1, P2 or P3 seed, an active projection other than that of D_6 , or a failed offer other than that of D_c ; so by the ordered rule P6 is *active_projection*, P5 *collapsed_boundary_case*, P4 *below_threshold*, P2 *inapplicable_outside_scope*, and P1 and P3 *absent_with_record*. $\square$

Proposition 7.6 (Decomposition records are not unique). *A description $D \in \text{FATCD}$ may admit more than one well-formed decomposition/exhaustion record in the sense of Definition 7.2. In particular, distinct well-formed records may name different witness clusters for one active channel, and then have different sets $A(\text{Dec}(D))$, $\text{Residual}(D)$ and residual labels; Definition 7.2 also leaves the choice among level annotations attached to the named cluster and the placement of loss/lift and visibility records to the record. Such a description can be chosen in FATCD_{RH} . Theorems 7.3 and 7.4 assert existence, not uniqueness.*

Proof. Let D be the disjoint union of two copies of the description D_5 of Theorem 5.1 (with the same renaming), as in the proof of Proposition 7.5; then $D \in \text{FATCD}$, and each copy carries a cluster, C_5 and C'_5 , that satisfies W_5 with its attachments. The construction of Theorem 7.3 may name either cluster for the P5 channel. In the record naming C_5 , the content records of C'_5 that contribute a required field to C'_5 , a six-role-aligned cluster that the chosen projection does not absorb, receive label (v), and any others, having no predicate, comparison, invariant, law or claim field and supplying no witness field, receive (ix-c); in the record naming C'_5 , the same holds with the copies exchanged. The W_5 assertion of the unchosen copy is represented by its check record and other required-field records, labelled (v); its bookkeeping annotations receive (ix-a) and its threshold and level annotations (ix-b). The two records name different clusters, so they differ, and both are well-formed. The description D satisfies the recognition hypothesis: each content record of either copy supplies a witness field to that copy's cluster satisfying W_5 , or is an input without assertion fields, and so satisfies (R1) or (R3) as in the inventory in the proof of Theorem 9.1 (p. 88); its annotations satisfy (R2). Hence $D \in \text{FATCD}_{\text{RH}}$. $\square$

The running example, decomposed. For the description D_{ex} of Section 4, a well-formed decomposition record assigns the six channels the following statuses; the records named are those of the table in Section 4. The channels P1, P5 and P6 have status *active_projection*: the split pair $(0, 1)$ witnesses the descent obstruction, the quotient object $\{A, B\}$ with its packaging map witnesses the packaging, and the replayed and checked trace witnesses the audit. The channel P4 has status *below_threshold*: the transition record of F is present, but no threshold annotation targets a cluster containing it. The channels P2 and P3 have status *absent_with_record*: the description records no macro-specification and no path or derivation record. The index record copying the classes of E is a residual feature, classified as a presentation variant of the relation record. So D_{ex} carries six channels and three active projections, as Proposition 7.5 allows.

Remark 7.7 (What the decomposition theorems do and do not prove). Theorems 7.3 and 7.4 establish the structural side of the scoped exact-six claim: every $D \in \text{FATCD}$ admits a six-channel decomposition/exhaustion record, with residual exhaustion in the covered scope under the recognition

hypothesis (Definition 6.7). Together with the lower-bound theorem (Section 5) and the upper-bound classification (Section 6), they prepare the scoped exact-six theorem but do not replace the integrated-stress check that follows. They do not assert that every description activates all six roles, that the decomposition is canonical or unique, or that the scoped exact-six program extends beyond FATCD; those remain nonclaims.

8 Checks against misreadings

In this section a misreading is *excluded* when it is not licensed by the results of Sections 4–7. The section is a checklist. Each of its six remarks lists misreadings of the results of Sections 5–7, for example asserting exact six without the lower bounds, or reading a role channel as an active role, and names the earlier result that rules each misreading out; “stress” means a check against such a list, and the chain *satisfies* a stress family when the result named for each misreading in its list rules that misreading out. Proposition 8.7 checks that the earlier results block the misreadings listed in the six remarks and combine into the terminal statement. The section adds no new mathematics; it collects in one place what the results do not permit.

Misreadings of	Blocked by
dependency (Remark 8.1)	the FATCD domain and the order of Theorems 5.1, 6.9, 7.3 and 7.4
circularity (Remark 8.2)	Proposition 4.16 and the direct proof of Theorem 7.3
channel versus activation (Remark 8.3)	Proposition 7.5 and Definition 7.1
role alignment (Remark 8.4)	Proposition 6.1
upper-bound residual (Remark 8.5)	Propositions 6.3–6.5 and Theorem 6.9
overclaim (Remark 8.6)	Remark 2.9 and Propositions 3.12 and 7.6

8.1 Dependency stress

Remark 8.1 (Dependency stress). The scoped exact-six target $\text{ScopedExactSix}(\text{FATCD}_{\text{RH}})$ depends on the FATCD domain of Section 4, the upper-bound classification of Theorem 6.9, and the decomposition and exhaustion theorems of Section 7; the last clause of Theorem 9.1 also depends on the lower-bound theorem of Section 5. Accordingly, the following are excluded:

- asserting the target without the FATCD domain;
- asserting the target without the upper-bound classification;
- asserting the target without the decomposition and exhaustion theorems;
- asserting that every channel is active somewhere in FATCD_{RH} without the lower-bound theorem.

The integrated stress check of Proposition 8.7 is a separate consistency check; the proof of Theorem 9.1 does not use it.

8.2 Circularity stress

Remark 8.2 (Circularity stress). Three circularity patterns are excluded under the theorem chain:

- redefining FATCD so that it already contains exactly six role slots by construction;
- declaring the absence of a seventh role by a membership condition of FATCD rather than by an explicitly stated hypothesis;
- justifying decomposition existence by assuming scoped exact-six.

The non-circularity schema of Proposition 4.16 and the direct proof of Theorem 7.3 block each pattern.

8.3 Channel-versus-activation stress

Remark 8.3 (Channel-versus-activation stress). The following conflations are excluded:

- requiring that every $D \in \text{FATCD}$ activates all six role channels;
- treating an inactive channel of status *below_threshold* as an active witness for its role;
- treating an inactive channel of status *collapsed_boundary_case* as an active witness;
- treating an inactive channel of status *absent_with_record* as an active witness.

Proposition 7.5 enforces each distinction. The same inactive-status check also applies to *inapplicable_outside_scope*.

8.4 Role-alignment stress

Remark 8.4 (Role-alignment stress). The three forbidden assignments of Remark 6.2 are excluded: generic transition data does not project to P3, generic path or derivation data does not project to P5, and generic semantic or interpretation data does not project to P1. The alignment rules of Proposition 6.1 enforce each prohibition.

8.5 Upper-bound residual stress

Remark 8.5 (Upper-bound residual stress). Under Theorem 6.9 and within the covered scope, the following misreadings are excluded:

- probability or uncertainty features of the recognized forms of Proposition 6.3 reopening as unresolved threats;
- causality features of the recognized forms of Proposition 6.4 reopening as unresolved threats;
- agency features of the recognized forms of Proposition 6.5 reopening as unresolved threats;
- any residual feature in the covered scope of a description satisfying the recognition hypothesis remaining unclassified under Definition 4.14;
- any residual feature in the covered scope of a description satisfying the recognition hypothesis passing the recorded seventh-role screen (without that hypothesis such a feature exists: the relation record r of D_{scr} , Remark 6.8).

Propositions 6.3–6.5 settle the three threat closures, and Theorem 6.9, under the recognition hypothesis, delivers the emptiness of categories (x) and (xi).

8.6 Overclaim stress

Remark 8.6 (Overclaim stress). The following restatements of the scoped exact-six target are excluded:

- restatement as a full six-primitives algebra;
- restatement as a global primitive-independence theorem;
- restatement as unique or canonical decomposition;
- restatement as empirical realization;
- restatement as a claim about all emergence phenomena, or any domain broader than FATCD.

Remark 2.9, Proposition 3.12, and Proposition 7.6 carry the corresponding nonclaims forward through the theorem chain. In particular, nonclaim closure survives the exact-six assembly.

8.7 The integrated stress certification

Proposition 8.7 (Integrated stress certification). *Clauses (a)–(d) are statements about descriptions; clause (e) records the dependency structure of the proof of Theorem 9.1. The following hold: (a) every $D \in \text{FATCD}$, including $D_{\text{norm}} \notin \text{FATCD}_{\text{RH}}$, admits a well-formed decomposition record, and every well-formed record of such D satisfies clauses (i)–(ii) of Theorem 7.4; (b) for each $k \in \{0, \dots, 6\}$ some description in FATCD_{RH} has exactly k active channels (Proposition 7.5); (c) some description has two distinct well-formed decomposition records (Proposition 7.6); (d) the description D_{norm} of Remark 6.8 lies in FATCD but not in FATCD_{RH} ; and (e) Theorem 9.1 follows from Theorems 5.1, 6.9, 7.3 and 7.4, Propositions 6.3–6.5, and the inventory of the records of $D_1, \dots, D_6$ given in its proof.*

Proof. We verify that each of the six stress families is discharged by the earlier result it rests on, so that none of the listed misreadings can arise in the combined chain.

Dependency. The components named in Remark 8.1 are present and ordered: the FATCD domain is fixed in Section 4, the lower-bound theorem in Section 5 (Theorem 5.1), the upper-bound classification in Section 6 (Theorem 6.9), and the decomposition and exhaustion theorems in Section 7 (Theorems 7.3 and 7.4). Theorem 9.1 uses Theorems 7.3 and 7.4; clause (iii) of Theorem 7.4 uses Theorem 6.9; and the last clause of Theorem 9.1 uses Theorem 5.1. Theorem 6.9 does not use Theorem 5.1. The existence argument of Theorem 7.3 does not use it either; that proof cites it only for the separate non-vacuity observation at its end. So the target cannot be asserted while skipping one of these uses, and no link consumes its own output.

Circularity. The three circularity patterns of Remark 8.2 are blocked by the non-circularity schema of Proposition 4.16, which defines FATCD without reference to a six-slot count, and by the direct proof of Theorem 7.3, which produces a decomposition without assuming scoped exact-six. The absence of labels (x) and (xi) is not a membership condition of FATCD : it follows, through the label criteria of Definition 4.14, from the recognition hypothesis, which is assumed and fails for the description of Remark 6.8.

Channel versus activation. The confluents of Remark 8.3 are excluded by Proposition 7.5 together with the five-status taxonomy of Definition 7.1: a channel carrying status *below_threshold*, *collapsed_boundary_case*, *absent_with_record*, or *inapplicable_outside_scope* is an inactive channel and is never read as an active projection, so no description is required to activate all six channels.

Role alignment. The three forbidden assignments of Remark 8.4 are ruled out by the alignment rules of Proposition 6.1, which refuse generic transition data as P3, generic path or derivation data as P5, and generic semantic or interpretation data as P1, in accordance with the alignments of Remark 6.2.

Upper-bound residual. The misreadings of Remark 8.5 are closed within the covered scope by Propositions 6.3–6.5, which settle the probability, causality, and agency threat closures, and by Theorem 6.9, which — under the recognition hypothesis (Definition 6.7) — classifies every covered residual feature and delivers the emptiness of categories (x) and (xi); hence, under that hypothesis, no covered residual reopens as an unresolved threat or passes the recorded seventh-role screen, while the three principal threat classes are closed without that hypothesis, in the recognized forms of Propositions 6.3–6.5.

Overclaim. The excluded restatements of Remark 8.6 are blocked because the nonclaims of Remark 2.9 are carried forward by Proposition 3.12, and non-uniqueness of the decomposition is recorded by Proposition 7.6; thus nonclaim closure survives the exact-six assembly, and no full-algebra, global-independence, unique-decomposition, empirical-realization, or broadened-domain reading is admitted.

Thus each misreading listed in the stress remarks is excluded by the result cited for it; dependencies cohere, the channel-versus-activation distinction holds and nonclaim closure holds, so the results combine into the terminal statement of Section 9.

For the concrete clauses: (a) holds because Theorem 7.3 gives a well-formed decomposition record for every $D \in \text{FATCD}$, and its existence construction and clauses (i)–(ii) of Theorem 7.4 use only Definitions 7.1 and 7.2 and the label criteria after Definition 4.14, and clause (iii) alone cites Theorem 6.9; (b) is Proposition 7.5; (c) is Proposition 7.6; (d) is shown in Remark 6.8; and (e) holds because the proof of Theorem 9.1 in Section 9 uses exactly the results listed in (e) and not this proposition. $\square$

In the sense of the checklist of this section, the results named in its table rule out each listed misreading: dependency, circularity, channel-versus-activation, role-alignment, upper-bound residual and overclaim misreadings.

9 The scoped exact-six theorem

The proof assembles Theorems 5.1, 6.9, 7.3 and 7.4 over the descriptions in FATCD that satisfy the recognition hypothesis, together with Propositions 6.3–6.5 and an inventory of the records of the lower-bound descriptions. The theorem statement also records the three threat closures of Propositions 6.3–6.5, and its last clause, from Theorem 5.1, says that none of the six channels is idle on that class.

In brief: for every $D \in \text{FATCD}_{\text{RH}}$ a well-formed $\text{Dec}(D)$ exists; every well-formed $\text{Dec}(D)$ has the channels $P_1, \dots, P_6$, each with the status of Definition 7.1; every label licensed for a kind residual or a bookkeeping annotation lies in (i)–(ix), and every label licensed for $r \in \text{Residual}(D)$ in (iii)–(ix); and P_i is active for D_i ($i = 1, \dots, 6$).

Theorem 9.1 (Scoped exact-six). *Let FATCD_{RH} be the class of descriptions $D \in \text{FATCD}$ that satisfy the recognition hypothesis (Definition 6.7), and let $D \in \text{FATCD}_{\text{RH}}$. (a) Then D admits a well-formed decomposition/exhaustion record (Theorem 7.3), and every well-formed record $\text{Dec}(D)$ of D has exactly six typed closure-role channels:*

- P1: *update-vs-quotient compatibility / descent obstruction;*
- P2: *macro-specification representability (when active, witnessed by a certified absence of realizers in a candidate class that is the full declared domain of a packaging or update map, with a recorded true evaluation at a declared same-kind record outside that class, or a realizer set such that some class of a declared packaging kernel whose domain contains the candidate class meets both it and its complement in the candidate class);*
- P3: *enriched route/coherence mismatch;*
- P4: *stage/order/transition/refinement structure;*
- P5: *quotient packaging / quotienting;*
- P6: *audit/bookkeeping/provenance/replay/checkability.*

(b) *Each channel carries exactly one recorded status from the five statuses active_projection, collapsed_boundary_case, below_threshold, inapplicable_outside_scope and absent_with_record: the first of these, in the order listed, whose case in Definition 7.1 applies. Every active projection on a cluster C carries threshold, witness-schema, collapse-case, level and nonclaim annotations targeting exactly C , and an honest audit certifying $W_i(C)$ (Definition 4.11).*

(c) *Let $\text{Dec}(D)$ be any well-formed record of D . Every label that the criteria license for a kind residual of D (Definition 4.14) or a bookkeeping annotation of D relative to $\text{Dec}(D)$ lies among (i)–(ix).*

Every element of $\text{Residual}(D)$ for that record is classified, under the upper-bound classification of Section 6, as one of:

- (iii) presentation variant;
- (iv) level shift;
- (v) unabsorbed role-aligned content, covered-seed content with no certified witness W_i , content checked in an honestly failed offer, or activation-threshold refinement;
- (vi) instrument-visibility artifact;
- (vii) empirical-bridge annotation;
- (viii) outside-domain structure;
- (ix) already-covered bookkeeping or annotation data, or role-neutral data without an assertion of its own.

Under the recognition hypothesis, no kind residual and no element of $\text{Residual}(D)$ remains as an unresolved candidate seventh role or passes the recorded seventh-role screen. Since seed clause (2) of Definition 7.1 accepts any one-variable macro-specification with a realization relation whose recorded evaluations take both truth values, this clause certifies the form in which each feature is recorded, not that its content concerns the role whose seed or witness it carries (Remark 6.10).

(d) Moreover, for every $D' \in \text{FATCD}$, whether or not it satisfies the recognition hypothesis, any kind residual of D' whose content takes one of the recognized forms of Propositions 6.3–6.5 is licensed, relative to every well-formed $\text{Dec}(D')$, at least one of the labels those propositions list, and neither (x) nor (xi).

(e) The conclusions stated for $D \in \text{FATCD}_{\text{RH}}$ hold for every such D ; that is, the statement $\text{ScopedExactSix}(\text{FATCD}_{\text{RH}})$ holds. Moreover, for each $i \in \{1, \dots, 6\}$ the description D_i of Theorem 5.1 lies in FATCD_{RH} and has a well-formed record, and in every well-formed record of D_i the channel P_i is active_projection (Theorem 7.4(iv)); so each of the six channels is active somewhere in FATCD_{RH} .

The recognition hypothesis restricts how assertions are presented. An explicit finite normalization formula can fail it; replacing an assertion by a declared predicate whose extension D does not record places the feature outside the covered scope when the resulting record meets criterion (viii), while presenting nonnegative weights as the row of a transition kernel makes them a covered P4 seed (Remark 6.8, paragraphs *Scope depends on presentation* and *Repair by a transition row*). Every closed check-language formula φ containing no atomic W_j formula can be wrapped, provided the completed description lies in FATCD , given two distinct declared records of one kind, in the covered P2 seed described in Remark 6.8 (paragraph *Wrapping a closed formula as a P2 seed*). This recognizes the predicate and realization-relation records and records φ 's truth value in the resulting description; a false formula is represented by its checked negation. If the wrapped description is again in FATCD , it satisfies the recognition hypothesis exactly when all its other kind residuals, all its bookkeeping annotations and all its record residuals are recognized; for the bookkeeping annotations this holds in particular when each has only datum and reference fields besides its exempt fields (paragraph after Definition 6.7).

The proof assembles the earlier results. Proposition 8.7 checks separately that they combine without the misreadings of Section 8; the proof does not use it.

Proof. Fix $D \in \text{FATCD}$ satisfying the recognition hypothesis. By Theorem 7.3, D admits a well-formed decomposition/exhaustion record. Let $\text{Dec}(D)$ be any well-formed record of D in the sense of Definition 7.2. By Theorem 7.4, the channel sequence of $\text{Dec}(D)$ has length six, each channel

carries one of the five statuses of Definition 7.1, and every raw feature lies in $A(\text{Dec}(D))$, $S(\text{Dec}(D))$ or $\text{Residual}(D)$; features in the last set receive a residual classification, and $\mathcal{L}$ and $\mathcal{V}$ annotate this coverage rather than forming a fourth coverage case. By Theorem 6.9, which applies because D satisfies the recognition hypothesis, every label licensed for a kind residual or a bookkeeping annotation of D lies among (i)–(ix) and every label licensed for an element of $\text{Residual}(D)$ among (iii)–(ix) (both clauses of Theorem 6.9), so categories (x) and (xi) are empty, which is the condition of Definition 4.18. Hence $\text{ScopedExactSix}(\text{FATCD}_{\text{RH}})$. The clause on active projections is the admissibility clause of Definition 4.11, which Definition 7.2 requires of every active channel of a well-formed record. The clause on D' is Propositions 6.3–6.5, which hold for every $D' \in \text{FATCD}$ and every well-formed $\text{Dec}(D')$ without the recognition hypothesis.

For the last clause, fix i and D_i . Its content records satisfy (R1) or (R3). In D_1 , the update, relation and comparison records supply the descent-obstruction witness; the four state records and the object record S are named inputs or declared-domain records of those checks and carry no assertion field. In D_2 , the specification and realization records supply the representability data, the comparison record supplies the required comparison in its satisfying-realizer instance, and candidate and S_2 records are direct checking inputs; the packaging map q carries a covered P5 seed with no assertion field, so it satisfies (R1), and the object record $\{A, B\}$, the codomain of q , supplies no witness field and has no assertion field, so it satisfies (R3). In D_3 , the route and comparison records supply the generated-route and separation data; their endpoints, and the object record U listing them, are direct inputs, and the move records a, b contribute required fields (their graphs make ρ_1, ρ_2 defined) and, having only datum fields, satisfy (R1). The rewrite rule $aa \rightarrow a$ is a datum field of the comparison record, which its monoid certificate reads by quantifying over the listed rules; it carries no assertion, and the comparison record satisfies (R1). In D_4 , each certified order, transition and refinement record is a covered P4 seed, while its declared indices and the object records listing them are inputs to its law check. In D_5 , the quotient relation, packaging map and validity check supply the quotient witness; the micro-objects and quotient object are declared inputs to those checks. In D_6 , the event, trace, provenance and check records supply required witness data or covered P6 seeds. Every declared input carries no predicate, comparison, invariant, law or claim field, so the assertion-field condition of (R1) holds vacuously for it; it therefore satisfies (R1) if it supplies a witness field, as the micro-object records x, y of D_5 do, and (R3) otherwise. The related-pair comparison in D_1 belongs to its (R1) comparison record. The structural and bookkeeping annotations satisfy (R2); their recorded assertion $W_i(C_i)$ is represented by the content records. These alternatives cover every kind residual and every possible element of $\text{Residual}(D_i)$, so $D_i \in \text{FATCD}_{\text{RH}}$. Finally $W_i(C_i)$ and its recorded attachments give P_i status *active_projection* by Definition 7.1. $\square$

For the running example, the record of Section 7 illustrates the six-channel and status clauses of the theorem: six channels, each with one status (three *active_projection*, one *below_threshold*, two *absent_with_record*), with the index record classified as a presentation variant.

The following table traces the clauses of the theorem to their sources. Residual exhaustion uses the recognition hypothesis; the integrated-stress check carries that assumption through its upper-bound residual case.

Clause	Source
a well-formed record $\text{Dec}(D)$ exists	Theorem 7.3
six channels, one status each, every raw feature accounted for	Definition 7.2, Theorem 7.4
each active projection has threshold, witness, level and bookkeeping data	Definition 7.2
record residuals receive labels in (iii)–(ix), kind residuals in (i)–(ix); (x) and (xi) are empty	Theorem 6.9, under the recognition hypothesis
probability, causality and agency, in their recognized forms	Propositions 6.3–6.5
each channel is active for some description in FATCD_{RH}	Theorem 5.1

9.1 What scoped exact-six means and does not mean

Theorem 9.1 has a reading that should be preserved throughout what follows.

- *Six channels, not six active witnesses.* Theorems 7.3 and 7.4 give six role channels in $\text{Dec}(D)$ for every $D \in \text{FATCD}$, and the theorem restates this for $D \in \text{FATCD}_{\text{RH}}$. It does not assert that every D activates all six roles, that every D carries six active projections $\pi_1(D), \dots, \pi_6(D)$, or that a single D realizes all six witness schemas of Section 5; such a description exists in FATCD_{RH} (the case $k = 6$ of Proposition 7.5), but it is not a clause of the theorem.
- *Existential, not unique, decomposition.* The theorem asserts existence of $\text{Dec}(D)$; it does not assert uniqueness or canonicity. Distinct well-formed records of the same D may exist (Proposition 7.6).
- *Conditional residual exhaustion.* The residual exhaustion clause is conditional on the recognition hypothesis (Definition 6.7) and restricted to the covered scope: the probability, causality and agency threats are closed without that hypothesis, in the recognized forms of Propositions 6.3–6.5, while not every finite audited feature is recognized: the normalization description of Remark 6.8 has a residual feature labelled (x). Features whose assertions use a predicate whose extension D does not record, or quantify over a sort whose carrier D does not record, are classified as outside-domain structures under category (viii) of Definition 4.14, not absorbed into primitive roles, when each such assertion is true in some expansion of D and false in another and the other assertion fields meet that criterion; an assertion of this kind with the same truth value in every expansion, such as $Q(o) \vee \neg Q(o)$ for an unrecorded Q , lies in the covered scope and is subject to the recognition hypothesis.
- *No broader claim.* Theorem 9.1 is a statement about FATCD ; the claims it does not make are listed in Section 11.

10 Discussion

Theorem 9.1 establishes $\text{ScopedExactSix}(\text{FATCD}_{\text{RH}})$; its six-channel decomposition clauses hold for all $D \in \text{FATCD}$. This section discusses what the statement earns and what it leaves unstated, how the three parts of Sections 2, 3, and 4–8 combine into one theorem stack, and how the result sits next to earlier Six Birds literature.

10.1 The shape of the scoped claim

Each of the three scope features — domain restriction, existence rather than uniqueness, residual exhaustion within a covered scope — is structural rather than cosmetic. An unrestricted exact-six theorem would require a domain-free argument for which the present paper does not claim to have the instruments. A uniqueness or canonicity claim would require a theory of translation records identifying distinct well-formed decompositions, which lies outside the present admissibility regime.

Scope-free residual exhaustion would require removing the recognition hypothesis, which fails in FATCD (Remark 6.8); the three threat closures of Section 6 are already stated without it for their recognized forms. Stating the theorem at the scope the proof supports keeps the endogenous-instrument meta-limit (Section 11.3) separate from the current claim instead of folded into it. The posture aligns with the broader view that scientific laws form a patchwork of domain-local regularities rather than a single global theory [9].

10.2 The three parts as one theorem stack

The three parts introduced in Section 1.1 play complementary roles. The primitive role architecture (Section 2) fixes the vocabulary: six role meanings, the behavioral, verification, and structural-categorical level pattern, and the local boundary distinctions. This vocabulary alone does not certify any claim, since it does not specify which role talk is admissible. The admissibility and meta-theory regime (Section 3) supplies that certification through honest bookkeeping, typed non-collapse, the forgetting and selected-lift architecture, activation thresholds, instrument-relative visibility, and empirical-bridge admissibility. Admissibility alone, however, yields no terminal claim; it only regulates what may be asserted. The exact-six theorem chain (Sections 4–8) supplies the quantification: the FATCD domain, the six lower-bound witnesses, the upper-bound classification and the decomposition/exhaustion theorem together deliver Theorem 9.1, and the integrated stress certification is a separate consistency check that its proof does not use. Vocabulary without admissibility would be undisciplined; admissibility without a theorem chain would be inconclusive; a theorem chain without the first two would lack a subject.

10.3 Channel versus activation

The channel-versus-activation discipline of Proposition 7.5, enforced throughout Sections 4–9, separates two statements that are easily conflated. The structural statement is that every admissible description carries six typed closure-role slots. The activation statement is that every admissible description realizes all six slots with active projections. The activation statement is too strong: it fails on descriptions that carry no route grammar and therefore have no active P3 projection, and on descriptions whose raw material carries no P6-aligned event/provenance/replay/check witness cluster and therefore have no active P6 projection. The structural statement, which is what we prove, is compatible with any activation profile. Inactive channels carry one of the statuses *below_threshold*, *collapsed_boundary_case*, *absent_with_record*, or *inapplicable_outside_scope*, describing the way a role fails to activate. Separating the two statements is what lets the six-channel clauses (Theorems 7.3 and 7.4) hold of every $D \in \text{FATCD}$, and Theorem 9.1 of every $D \in \text{FATCD}_{\text{RH}}$, without requiring activation of all six roles in every description; it also lets the lower-bound theorem of Section 5 be stated existentially over the domain rather than uniformly. An informal reading that elides the distinction silently recovers the stronger activation claim, which we do not prove.

10.4 Instrument-relative methodology

A methodological choice underlying Section 3 deserves explicit statement as a contribution. Instruments are treated as mathematical probes rather than as neutral tools, in the spirit of ontic structural realism [18] and the sheaf-theoretic treatment of contextuality [1]. The technical form is Definition 3.9; the associated nonclaim is that no result here rests on an instrument-free truth.

The motivation is reflexive. The subject matter — finite audited typed closure descriptions and the six typed closure roles that project them — is the same descriptive machinery from which any instrument for detecting closure structure must itself be built. Operator-presentation, quotient/descent, coherence/route, constructive/proof-theoretic, and information/regime instruments

are not vantage points outside the domain: each is a closure structure probing another closure structure. A claim asserted under one such instrument while silently suppressing what another would have revealed would privilege that instrument without bookkeeping for the privilege — the overhead that Definition 3.9 blocks.

The consequence is epistemic. Every admissible visibility claim carries an explicit visibility record: its visible content, its suppressed content, the instruments under which visibility and suppression are asserted, and the resulting instrument-relative claim strength (Definition 3.9). Triangulating a role projection across several instruments is a disciplined way to strengthen an assertion; reading a projection off a single instrument and declaring its content canonical is not admissible. The discipline is one of epistemic hygiene, in the spirit of Bohr’s complementarity doctrine [6]: we do not claim that instruments are incompatible in principle, only that they are not interchangeable without explicit visibility bookkeeping and, where contexts shift, explicit translation records (Definitions 3.9 and 3.5).

This posture also explains why the endogenous-instrument meta-limit (Sections 11.3 and 12.1) is deferred to future work. Taking instrument-relativity seriously from the outset lets the meta-limit be stated as its own question — whether the accepted internal instrument family can certify its own soundness and completeness — a question made statable, but not settled, by the stance adopted here.

10.5 Relation to earlier Six Birds results

Six Birds Foundations I [28] introduced the theory-package formalism $\mathcal{T} = (Z, f, \Sigma_f, E, \mathcal{A})$ and derived the P1–P6 vocabulary under minimal hypotheses of composability with limited interface access. The present paper does not re-derive that vocabulary; it takes the six roles as given and asks a different question: whether, within a specific admissibility regime and over a specific class of admissible descriptions, the six roles are exactly enough. The Foundations theorem gives the canonical unavailability side of the picture; the terminal theorem here gives the scoped exhaustion side. The two statements address different questions and address them at different levels; combining them does not yield a stronger claim, and neither theorem follows from the other.

The theorem chain also draws on a nearby result. Tsiokos [30], complementing the no-go analysis of Tsiokos [29], established that protocol holonomy is not a directionality certificate: noncommutativity of update sequences can be nonzero while path-reversal asymmetry remains zero, so route mismatch is a diagnostic, not an arrow of time. That observation is why Definition 2.1 records that P3 is not a directionality certificate, and why Remark 2.3 keeps P3 route/coherence mismatch distinct from the drive specialization of P6. The next section records the scope boundaries and nonclaims of the present theorem stack.

11 Limitations and nonclaims

This section collects the scope restrictions of Theorem 9.1 and the global nonclaims preserved throughout.

11.1 Scope of the terminal theorem

Theorem 9.1 establishes $\text{ScopedExactSix}(\text{FATCD}_{\text{RH}})$; its six-channel decomposition clauses hold for all $D \in \text{FATCD}$, the finite audited typed closure description domain of Definition 4.7. The theorem asserts:

- existence of a decomposition/exhaustion record $\text{Dec}(D)$ with six role channels and one status per channel;

- residual-feature exhaustion under the upper-bound classification, conditional on the recognition hypothesis (Definition 6.7) and within the covered scope, so that no seventh irreducible role is required there (the three principal threat classes are closed without the hypothesis, in the recognized forms of Propositions 6.3–6.5);
- activation of any $\pi_i(D)$ only under the threshold, witness-schema, collapse-case, level, nonclaim and audit data of Definitions 4.9 and 4.11.

It does not assert:

- decomposition uniqueness or canonicity;
- six active projections, or six active nontrivial witnesses, in every description;
- scope-free residual exhaustion, or any claim about features outside the covered scope;
- that every finite audited feature of a description in FATCD satisfies (R1)–(R5); Remark 6.8 exhibits a feature with label (x) and a feature with label (xi);
- unrestricted exact-six, or coverage of all emergence phenomena;
- that broader domains require no further work.

11.2 Global nonclaims

Each nonclaim below is preserved throughout the paper and is enforced locally by the theorem-chain result that would otherwise overread it.

- *No full six-primitives algebra.* Scoped exact-six is proved over FATCD; it is not claimed that P1–P6 generate a global algebra of closure operations (Remark 2.9, Remark 8.6).
- *No global primitive-independence theorem.* Typed non-collapse (Proposition 3.3) is the only claim made at that level; no global independence notion is defined or claimed.
- *No empirical realization.* Empirical-bridge admissibility (Definition 3.11) admits threshold-dependent, level-indexed, instrument-visible substrate evidence; it does not assert that any $D \in \text{FATCD}$ is itself a substrate observation.
- *No unsupported unique selected lift.* The displayed atlas lifts are selected and, except where uniqueness is proved and recorded (as for a descended map), non-unique; a local uniqueness assertion is admissible only with a recorded proof (Definition 3.6).
- *No unsupported lower-to-higher determinacy.* Behavioral or verification content is not claimed to determine structural-categorical content without a recorded proof, as for a descended map.
- *No instrument-free formulation.* Admissible visibility claims carry an instrument-relative visibility record (Definition 3.9).
- *No internal instrument-family completeness.* The accepted instrument family is not asserted to be sufficient for every exact-six judgment formulable in the same regime.
- *No unrestricted all-emergence coverage.* Theorem 9.1 is scoped to FATCD.
- *No uniqueness of primitive presentations.* Distinct admissible presentations of the same role may coexist.
- *Cross-level comparison requires translation.* Direct comparison of claims at different levels is inadmissible without a translation record (Definition 3.5).

11.3 The endogenous-instrument meta-limit

An adjacent meta-theoretic question, deferred to future work, is whether the accepted internal instrument family can itself certify both its soundness and its completeness for exact-six judgments formulable inside the same regime. The notions of an internal argument and of the soundness and completeness of an instrument family are not defined in this paper; what follows is a heuristic target, not a mathematical statement. A plausible sharp formulation is that no internal argument should be expected to accomplish all three of the following at once:

1. establish the scoped exact-six claim;
2. certify the soundness of the accepted instrument family;
3. certify the completeness of that family for exact-six judgments expressible inside the same regime.

Such self-certification limits are of a piece with the classical Gödel–Löb tradition on internal consistency and provability statements [12, 20]. This question is not part of the proof of Theorem 9.1 and does not weaken or retract the theorem’s scoped conclusion. It is taken up as a follow-on direction in Section 12.

12 Future work

Theorem 9.1 and the limitations of Section 11 suggest several follow-on research directions. Each subsection below states a direction, not a commitment.

12.1 Endogenous-instrument meta-limit

A formalization of the meta-limit described in Section 11.3 would first define internal arguments and the soundness and completeness of an instrument family, and then target a precise form of the heuristic of Section 11.3: that no internal argument can simultaneously establish the scoped exact-six claim, certify the soundness of the accepted instrument family, and certify its completeness for exact-six judgments formulable inside the same regime. Theorem 9.1 would remain intact under such a result; the additional content would be a precise account of self-certification boundaries. A closely related open question concerns recognition. Not every finite audited feature of a description in FATCD is recognized: the bare normalization predicate of Remark 6.8 receives the unresolved label (x). What remains open is which classes of assertion-bearing records — normalization constraints not presented as the law of a transition record (for instance for signed weights), finite conservation laws, emergent composite obstructions — admit a reduction in which the assertion is itself a required field of an active cluster (label (i)) without adjoining wrapper records such as a realization relation and a one-class packaging map (Remark 6.8), or a trace reference, and whether any of them carries the recorded witness of category (xi) of Definition 4.14, which is a recorded screen rather than the identification of a new closure role.

12.2 Broader-domain exact-six

Candidate extensions beyond FATCD include admissibility regimes that weaken finiteness, closure, or audit (Section 4), and regimes that admit annotation kinds beyond the six Ann_* categories of Definition 4.2. Any such extension would require fresh upper-bound and decomposition/exhaustion analysis, since the alignment rules and threat closures are not stated at scope-free strength.

12.3 Uniqueness or canonical decomposition

Theorem 9.1 is existential. A uniqueness study would investigate when distinct well-formed decomposition/exhaustion records of the same $D \in \text{FATCD}$ are equivalent under explicit translation

records, or when a canonical representative can be selected. The channel-versus-activation discipline (Proposition 7.5) is compatible with such a study; no uniqueness claim is part of the present theorem.

12.4 Full six-primitives algebra

The generator-and-relation structure of P1–P6 beyond typed non-collapse remains open. A full algebra would require scope-free independence data, not just the local boundary distinctions of Table 1, and would likely require a categorical apparatus richer than the raw grammar of Section 4.1.

12.5 Empirical substrate mapping

The empirical-bridge admissibility regime (Definition 3.11) can be connected to concrete substrate-level observables. The present paper treats empirical bridges as threshold-dependent, level-indexed, and instrument-visible annotations; later work can couple this to substrate-specific measurement theory while preserving the nonclaim of empirical realization.

12.6 Mechanization

A Lean 4 development accompanies the present paper; Appendix B records, claim by claim, how far each definition and theorem is currently mechanized. Strengthening that mechanization toward full proofs — in particular a Lean account of the boundary distinctions, the decomposition-existence construction, and the recognition hypothesis underlying the upper-bound classification — is a natural direction, building on the closure-related infrastructure already mechanized in the Foundations program [11, 28].

12.7 Cross-instrument comparison

Decomposition/exhaustion records of the same $D \in \text{FATCD}$ can be compared under different admissible instrument assignments. Definition 3.9 admits visibility records that suppress different portions of a claim under different instruments; a cross-instrument study could make the instrument-dependence of $\text{Dec}(D)$ explicit and sharpen the notion of instrument-relative claim strength.

A Provenance and evidence hygiene

The development of the theorem chain of Sections 4–8 was assisted by automated drafting and proof-search tooling, the Aristotle system [2]. We record this for transparency and to fix the division of evidential labor precisely. Such tooling produces *candidate* definitions, statements, and arguments; it carries no evidential weight of its own, and nothing in this paper rests on it.

The mathematical content is established solely by the proofs given in the body. The machine-*checked* component of the work — which declarations are realized in Lean, and at what fidelity — is documented separately in Appendix B; the automated tooling named above is distinct from that mechanization and assisted drafting only. No raw output of that tooling is offered as mathematical evidence anywhere in the manuscript: where a section draws on machine assistance, the load-bearing content is the section’s own proof, and the recorded mechanization is exactly as described in Appendix B.

A.1 Code availability

The L^AT_EX source of this manuscript and the accompanying Lean mechanization artifacts are available at <https://github.com/ioannist/six-birds-foundations-ii>.

B Mechanization: declarations, coverage, and trust base

The theorem chain of this paper is accompanied by a Lean 4 development, the integrated `SixBirds` project, available at the code repository of Appendix A.1. This appendix is the canonical disclosure venue for that development. For every numbered definition and claim it records the Lean declaration that realizes it and the *fidelity* of that realization, and Subsection B.5 gives a note on each result. The mapping is maintained in the repository and reproduced here.

B.1 Fidelity of the correspondence

Each theorem and proposition has a note in Subsection B.5 naming its Lean declaration and its fidelity. We distinguish six fidelities:

- *verified* — a faithful Lean derivation of the statement from non-trivial hypotheses;
- *narrowed / parametric* — a genuine Lean proof, but over a single concrete witness, a parametric record, or a narrowed input space rather than the full generality of the paper statement;
- *schema (projection)* — the Lean declaration reuses supplied hypotheses or encoded definitions, or constructs, projects or assembles records and their proofs, without formalizing the paper’s full mathematical content; it is therefore *not* described as a verified proof of the corresponding paper statement;
- *trivial* — the Lean statement reduces to `True` by construction, or ranges over a hand-built total classifier; it records a bookkeeping fact, not a derivation;
- *typed mirror* — for definitions, a Lean structure or inductive mirroring the paper definition;
- *none* — the declaration does not establish the statement for any instance.

We state plainly that **no claim in the present development is a faithful derivation in the first sense**. The Lean track is a typed harness: it fixes the objects as typed records, exposes the channel, status, and residual machinery, and checks that malformed and misaligned cases are rejected (the semantic regression tests of the `SixBirds.SemanticTests` module); but the load-bearing classification and closure statements are projections, schemas, or trivially-true bookkeeping over those records. The notes are worded accordingly. A reader who wants the mathematics should read the body proofs; the Lean track certifies typed well-formedness and case rejection, not the theorems.

B.2 The recognition hypothesis and the upper-bound classifier

The honest status of the central no-seventh-role result (Theorem 6.9) is legible in the Lean. The classifier `SixBirds.classifyResidual` is a total function on the finite raw-record kinds, every branch of which lands in a covered class; the declarations `upper_bound_classification` and the three closure lemmas are `True` by construction over it. This is exactly why the body states the result conditionally, under the recognition hypothesis (Definition 6.7): the Lean classifier imposes covered outputs by construction; it neither represents nor proves the paper-side recognition hypothesis or the quantification over every licensed label. The paper-side recognition hypothesis has no independent Lean correspondent, and the failure of recognition-completeness for FATCD (Remark 6.8) lies precisely in the gap between the classifier’s by-construction coverage and a derived exhaustiveness.

B.3 Declaration and coverage table

The table maps each result to its Lean declaration in the `SixBirds` namespace and its audited coverage. All declarations are `sorry`-free and axiom-clean relative to the trust base of §B.4; the coverage column records fidelity in the sense of §B.1, not mere presence.

§	result	Lean declaration	coverage
§2	Definition 2.1	<code>SixBirds.Role</code>	typed mirror (def.)
§2	Definition 2.4	<code>SixBirds.Level</code>	typed mirror (def.)
§2	Definition 2.8	<code>SixBirds.SelectedLiftAtlas</code>	typed mirror (def.)
§2	Proposition 2.6	<code>SixBirds.boundary_distinctions</code>	schema (projection)
§3	Definition 3.1	<code>SixBirds.honest_bookkeeping_</code> <code>admissibility</code>	schema (projection)
§3	Proposition 3.3	<code>SixBirds.typed_non_collapse</code>	schema (projection)
§3	Definition 3.5	<code>SixBirds.level_profile</code>	trivial
§3	Definition 3.6	<code>SixBirds.forgetting_lifts</code>	schema (projection)
§3	Remark 3.8	<code>SixBirds.activation_thresholds</code>	schema (projection)
§3	Definition 3.9	<code>SixBirds.visibility</code>	schema (projection)
§3	Definition 3.11	<code>SixBirds.empirical_bridge</code>	schema (projection)
§3	Proposition 3.12	<code>SixBirds.nonclaim_closure</code>	schema (projection)
§4	Definition 4.1	<code>SixBirds.RawRecord</code>	typed mirror (def.)
§4	Definition 4.2	<code>SixBirds.AnnotationKind</code>	typed mirror (def.)
§4	Definition 4.4	<code>SixBirds.FiniteDescription</code>	typed mirror (def.)
§4	Definition 4.5	<code>SixBirds.ClosedDescription</code>	typed mirror (def.)
§4	Definition 4.6	<code>SixBirds.AuditedDescription</code>	typed mirror (def.)
§4	Definition 4.7	<code>SixBirds.FATCD</code>	typed mirror (def.)
§4	Definition 4.9	<code>SixBirds.ThresholdAttachment</code>	typed mirror (def.)
§4	Definition 4.11	<code>SixBirds.DerivedRoleProjection</code>	typed mirror (def.)
§4	Definition 4.14	<code>SixBirds.</code> <code>ResidualClassificationScheme</code>	typed mirror (def.)
§4	Definition 4.18	<code>SixBirds.ScopedExactSixQuestion</code>	typed mirror (def.)
§4	Proposition 4.16	<code>SixBirds.non_circularity</code>	schema (projection)
§4	Proposition 4.17	<code>SixBirds.non_overbreadth</code>	schema (projection)
§5	Definition 5.2	<code>SixBirds.CommonWitnessData</code>	typed mirror (def.)
§5	Theorem 5.1	<code>SixBirds.six_role_lower_bounds</code>	schema (projection)
§5	Proposition 5.3	<code>SixBirds.p1_witness</code>	narrowed / parametric
§5	Proposition 5.4	<code>SixBirds.p2_witness</code>	narrowed / parametric
§5	Proposition 5.5	<code>SixBirds.p3_witness</code>	narrowed / parametric
§5	Proposition 5.6	<code>SixBirds.p4_witness</code>	narrowed / parametric
§5	Proposition 5.7	<code>SixBirds.p5_witness</code>	narrowed / parametric
§5	Proposition 5.8	<code>SixBirds.p6_witness</code>	narrowed / parametric
§6	Proposition 6.1	<code>SixBirds.alignment_rules</code>	schema (projection)
§6	Proposition 6.3	<code>SixBirds.probability_closure</code>	trivial
§6	Proposition 6.4	<code>SixBirds.causality_closure</code>	trivial
§6	Proposition 6.5	<code>SixBirds.agency_closure</code>	trivial
§6	Theorem 6.9	<code>SixBirds.upper_bound_</code> <code>classification</code>	trivial
§7	Definition 7.1	<code>SixBirds.ChannelStatus</code>	typed mirror (def.)
§7	Definition 7.2	<code>SixBirds.DecompositionRecord</code>	typed mirror (def.)
§7	Theorem 7.3	<code>SixBirds.decomposition_existence</code>	schema (projection)
§7	Theorem 7.4	<code>SixBirds.six_channel_exactness</code>	schema (projection)
§7	Proposition 7.5	<code>SixBirds.channel_vs_activation</code>	schema (projection)
§7	Proposition 7.6	<code>SixBirds.non_uniqueness</code>	none
§8	Remark 8.1	<code>SixBirds.dependency_stress</code>	schema (projection)
§8	Remark 8.2	<code>SixBirds.circularity_stress</code>	schema (projection)
§8	Remark 8.3	<code>SixBirds.channel_activation_</code> <code>stress</code>	narrowed / parametric
§8	Remark 8.4	<code>SixBirds.role_alignment_stress</code>	narrowed / parametric
§8	Remark 8.5	<code>SixBirds.residual_stress</code>	trivial
§8	Remark 8.6	<code>SixBirds.overclaim_stress</code>	schema (projection)
§8	Proposition 8.7	<code>SixBirds.integrated_stress</code>	schema (projection)
§9	Theorem 9.1	<code>SixBirds.scoped_exact_six</code>	schema (projection)

B.4 Trust base and axiom audit

The dependency audit accepts only the standard Lean and Mathlib trust base: `propext`, `Classical.choice` and `Quot.sound` (function extensionality is a theorem derived from `Quot.sound`). No project-local axiom is admitted for any declaration recorded as a theorem. The validation harness (`scripts/validate_lean.sh`) rejects project-local axioms and `sorry`, and runs a semantic audit that rejects `:= True` and `:= False` definition stubs and all-purpose residual classifiers. The trust-base check certifies that the typed harness is axiom-clean relative to that base; it does not, and is not intended to, upgrade a schema or a trivially-true statement to a derivation. The fidelity column of §B.3 is the governing record of what each declaration establishes.

B.5 Notes on individual results

For each result, the note below records the Lean declaration that corresponds to it and the extent of that correspondence.

Proposition 2.6. The Lean declaration `SixBirds.boundary_distinctions` has fidelity schema (projection) of the legend above: its encoded hypotheses, definitions or record data supply its conclusion; it does not derive the corresponding paper statement.

Proposition 3.3. The Lean declaration `SixBirds.typed_non_collapse` has fidelity schema (projection) of the legend above: its encoded hypotheses, definitions or record data supply its conclusion; it does not derive the corresponding paper statement.

Proposition 3.12. The Lean declaration `SixBirds.nonclaim_closure` has fidelity schema (projection) of the legend above: its encoded hypotheses, definitions or record data supply its conclusion; it does not derive the corresponding paper statement.

Proposition 4.16. The Lean declaration `SixBirds.non_circularity` has fidelity schema (projection) of the legend above: its encoded hypotheses, definitions or record data supply its conclusion; it does not derive the corresponding paper statement.

Proposition 4.17. The Lean declaration `SixBirds.non_overbreadth` has fidelity schema (projection) of the legend above: its encoded hypotheses, definitions or record data supply its conclusion; it does not derive the corresponding paper statement. The Lean declaration is identical in statement and proof to `SixBirds.non_circularity` and re-extracts the admissibility fields of the record rather than establishing the structural-restriction content of the proposition.

Theorem 5.1. The Lean declaration `SixBirds.six_role_lower_bounds` has fidelity schema (projection) of the legend above: its encoded hypotheses, definitions or record data supply its conclusion; it does not derive the corresponding paper statement.

Proposition 5.3. The Lean development establishes this for one concrete description `witnessDescription Role.P1` under parametric hypotheses (alignment, threshold attachment, and finite/closed/audited verifications of that fixed witness); the narrowing is recorded in the table of this appendix (`SixBirds.p1_witness`).

Proposition 5.4. The Lean development establishes this for one concrete description `witnessDescription Role.P2` under parametric hypotheses (alignment, threshold attachment, and finite/closed/audited verifications of that fixed witness); the narrowing is recorded in the table of this appendix (`SixBirds.p2_witness`).

Proposition 5.5. The Lean development establishes this for one concrete description `witnessDescription Role.P3` under parametric hypotheses (alignment, threshold attachment, and finite/closed/audited verifications of that fixed witness); the narrowing is recorded in the table of this appendix (`SixBirds.p3_witness`).

Proposition 5.6. The Lean development establishes this for one concrete description `witnessDescription Role.P4` under parametric hypotheses (alignment, threshold attachment, and finite/closed/audited verifications of that fixed witness; the attached level is fixed to the verification level rather than the structural-categorical level of the prose); the narrowing is recorded in the table of this appendix (`SixBirds.p4_witness`).

Proposition 5.7. The Lean development establishes this for one concrete description `witnessDescription Role.P5` under parametric hypotheses (alignment over the same raw-kind cluster used for P1, threshold attachment, and finite/closed/audited verifications of that fixed witness); the narrowing is recorded in the table of this appendix (`SixBirds.p5_witness`).

Proposition 5.8. The Lean development establishes this for one concrete description `witnessDescription Role.P6` under parametric hypotheses (alignment over event/provenance/replay/check records, threshold attachment, and finite/closed/audited verifications of that fixed witness); the narrowing is recorded in the table of this appendix (`SixBirds.p6_witness`).

Theorem 6.9. The Lean declaration is a check of the classifier: every `RawKind` branch of `classifyResidual` lands in a class whose `CoveredResidualClass` branch is `True`, and the uncovered constructors `unresolvedThreat/genuineSeventhRole` are never produced; “no seventh role” holds by construction rather than by derivation from feature content. (`SixBirds.upper_bound_classification`).

Proposition 6.1. The Lean declaration `SixBirds.alignment_rules` has fidelity schema (projection) of the legend above: its encoded hypotheses, definitions or record data supply its conclusion; it does not derive the corresponding paper statement.

Proposition 6.3. The Lean declaration is a single-record check: `classifyResidual` maps `RawKind.probability` to `composite`, whose `CoveredResidualClass` branch is `True`; it does not establish the paper’s universal probability/uncertainty closure. (`SixBirds.probability_closure`).

Proposition 6.4. The Lean declaration is a single-record check: `classifyResidual` maps `RawKind.causality` to `composite`, whose `CoveredResidualClass` branch is `True`; the P5-inadmissibility content of the proposition is absent and no universal residual-causality closure is established. (`SixBirds.causality_closure`).

Proposition 6.5. The Lean declaration is a single-record check: `classifyResidual` maps `RawKind.agency` to `empiricalBridgeAnnotation`, whose `CoveredResidualClass` branch is `True`; the P1-inadmissibility content is absent and no universal residual-agency closure is established. (`SixBirds.agency_closure`).

Theorem 7.3. The Lean declaration `SixBirds.decomposition_existence` has fidelity schema (projection) of the legend above: its encoded hypotheses, definitions or record data supply its conclusion; it does not derive the corresponding paper statement.

Theorem 7.4. The Lean declaration `SixBirds.six_channel_exactness` has fidelity schema (projection) of the legend above: its encoded hypotheses, definitions or record data supply its conclusion; it does not derive the corresponding paper statement.

Proposition 7.5. The Lean declaration `SixBirds.channel_vs_activation` has fidelity schema (projection) of the legend above: its encoded hypotheses, definitions or record data supply its conclusion; it does not derive the corresponding paper statement.

Proposition 7.6. The Lean development exhibits two concrete decomposition records of a single description that differ only in their P1 channel status; since the status of a channel is fixed by the description, at most one of these records is well-formed, so the Lean declaration does not witness the proposition, and the further sources of non-uniqueness in the prose (level assignments, choice of witness cluster or representation of the witness predicate, residual labels, loss/lift placement) are not mechanized. The Lean declaration is `SixBirds.non_uniqueness`.

Proposition 8.7. The Lean declaration `SixBirds.integrated_stress` has fidelity schema (projection) of the legend above, bundling the six stress conjuncts: its encoded hypotheses, definitions or record data supply its conclusion; it does not derive the corresponding paper statement.

Theorem 9.1. The Lean declaration `SixBirds.scoped_exact_six` has fidelity schema (projection) of the legend above: its encoded hypotheses, definitions or record data supply its conclusion; it does not derive the corresponding paper statement.

The running example. The module `SixBirds.Examples.RunningExample` constructs a typed proxy for D_{ex} as a finite, closed and audited FATCD, proves the split pair, the replay and check of the trace and the equivalence of the lens classes by computation, constructs the P1, P5 and P6 projections, and gives a well-formed decomposition record with the statuses stated in Section 7, classifying the index record as a presentation variant. For each active status the record checks that the raw record kinds its role requires are present in the description; the three projections are constructed separately and are not linked to the status field. The absent statuses of P2 and P3 and the below-threshold status of P4 are assigned, not derived from the content, and the record does not track a residual set. The typed layer has no annotation kind for transitions; the transition is a separate raw record referring to the map record of F , and no threshold annotation targets it. The paper’s expanded D_{ex} explicitly includes a q -map, a quotient-validity check and claim and audit contents absent from this Lean raw-record list. Lean checks kind presence and assigned statuses; it does not verify the honesty predicate of Definition 4.6, the paper’s W_5 check, or the singleton residual calculation.

C Version history

Version 1 appeared on Zenodo on 20 April 2026, Version 2 on Qeios on 9 June 2026, and Version 3 on 2 October 2026. Each subsection lists the differences from the preceding version.

C.1 Version 3

Version 3 keeps every numbered result of Version 2 with its number, label and type, in the same order. The titles of Remarks 2.2 and 6.2 (“corrected” became “excluded” alignments) and of Remark 6.8 changed.

Corrected statements. Definition 2.4 qualifies lossy forgetting, and Definition 2.8 qualifies non-uniqueness and lossiness. Definition 3.6 makes the loss-record and uniqueness-proof requirements explicit; Version 2 already admitted lossless forgetting and unique lifts when proved and recorded. Proposition 2.6 is stated as the failure of W_i for a cluster built from the bare boundary datum. Proposition 4.17 no longer claims that unaudited material cannot enter FATCD, which fails for

unclaimed records. Propositions 6.3–6.5 concluded that a feature of a recognized form classifies as a P_i projection, which needs attachment data; they now license label (i) or (v), or another listed label, for every well-formed decomposition record. Definition 7.1 replaces five conditions that were not shown to be exclusive and exhaustive by an ordered rule. Remark 6.8 left open whether the recognition hypothesis holds on FATCD ; Version 3 proves that it does not ($D_{\text{norm}} \in \text{FATCD} \setminus \text{FATCD}_{\text{RH}}$), and Theorem 9.1, whose conclusion Version 2 named $\text{ScopedExactSix}(\text{FATCD})$, now concludes $\text{ScopedExactSix}(\text{FATCD}_{\text{RH}})$; Definition 4.18 records that the unrestricted statement fails. Proposition 8.7 replaces the readiness claim of Version 2 by clauses (a)–(e). The scope statements of Remarks 3.4, 3.8, 6.10, 8.1 and 8.5 were corrected in the same way.

Added hypotheses and strengthened results. Definition 4.7 requires realization relations to list exactly the satisfying candidates, and Definition 4.9 requires nonclaim, collapse-case and witness-schema annotations targeting the cluster. Theorem 6.9 holds for every well-formed decomposition record and adds a pointwise form; Theorem 7.4 adds clause (iv), independence of the status sequence from the decomposition record; Proposition 7.5 realizes every active-channel count from 0 to 6 in FATCD_{RH} , and all five statuses in one description; Proposition 7.6 exhibits distinct well-formed records in FATCD_{RH} ; Proposition 3.3 covers every W_k ; and Theorem 9.1 adds clauses (d) and (e).

Definitions and scope. Version 3 specifies the witness predicates through the *Witness types* table, audit honesty through Definition 4.6, and licensed labels through Definition 4.14. It distinguishes kind residuals from record residuals and extends recognition to bookkeeping annotations, with alternatives (R1)–(R5). Category (xi), called a genuine seventh irreducible typed role in Version 2, is a recorded screen, not proof of a new role. Remark 6.8 adds the screen example and conditional closed-formula wrapping constructions, showing that recognition depends on presentation.

Additions and removals. Version 3 adds Subsection 1.4 with its vocabulary, symbols and example index; a running example; figures of the section dependencies and of the three levels; the label-criteria table; and collects the Lean-counterpart notes, formerly footnotes to the results, in Appendix B.5. The nonclaim “no surrogate terminal evidence” and the statement that $P6_{\text{drive}}$ is a face of $P6$ were removed. “Regressions” and “guardrails” are called misreadings and checks, and Section 8 is titled accordingly.

C.2 Version 2

Version 2 made the upper bound conditional on an explicit recognition hypothesis: it added Definition 6.7 and Remark 6.8, so that the upper-bound classification of Version 1 (Theorem 6.7) became Theorem 6.9 and its scope remark became Remark 6.10, and the absence of a seventh role in the terminal theorem was stated under that hypothesis. Statements of Section 3 that fix admissibility conditions rather than assert consequences were stated as definitions (Definitions 3.1, 3.5, 3.6, 3.9 and 3.11) or as a remark (Remark 3.8), the stress checks of Section 8 as Remarks 8.1–8.6, and the integrated stress theorem as Proposition 8.7. The proof of Proposition 2.6 was completed over all fifteen rows of the boundary matrix, corrections from external review were incorporated, and notes on the Lean counterpart of each result were added, with an appendix describing the Lean development.

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